

The Microstructure of Ferrous Alloys

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Constituents in Steels

Products of diffusion-controlled transformations.

Products of diffusionless athermal transformations.

Products of precipitation before or during solidification due to limited solid solubility, or pick-up from external sources.

Grain Boundaries

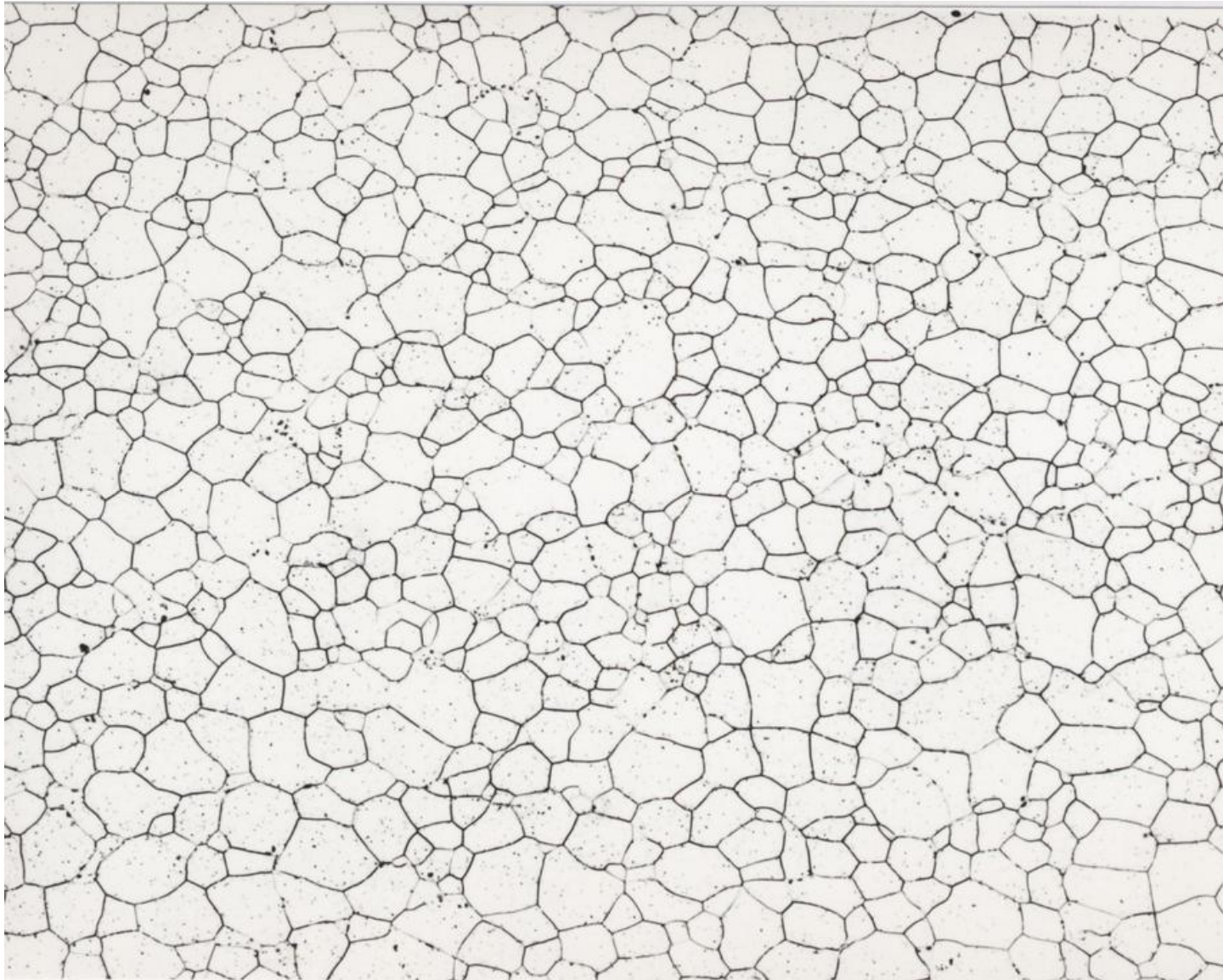
- The interface between two grains where the crystal lattice changes from that of one grain to that of the other grain.
- There are three types of grain boundaries in Fe-based alloys: ferrite, austenite and prior-austenite. The prior-austenite grain boundaries are those of the parent (austenite) phase before transformation. Ferrite and austenite grain boundaries are those of the product phase, although there are compositions where allotropic transformations do not occur.

ALPHA IRON

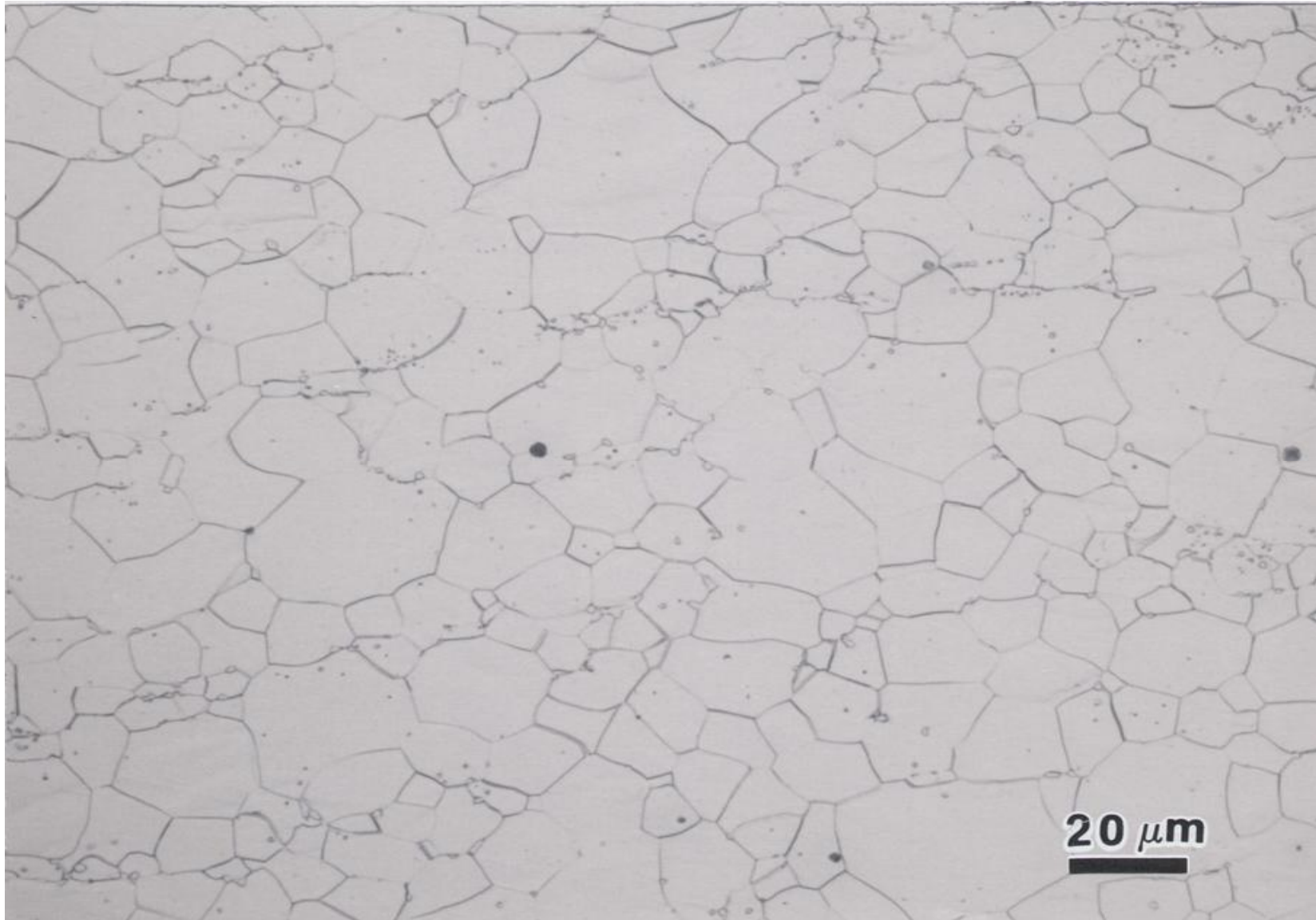
The bcc form of pure iron stable below 912 °C
(1674 °F).

FERRITE

Solid solution of one or more elements in body-centered cubic iron.

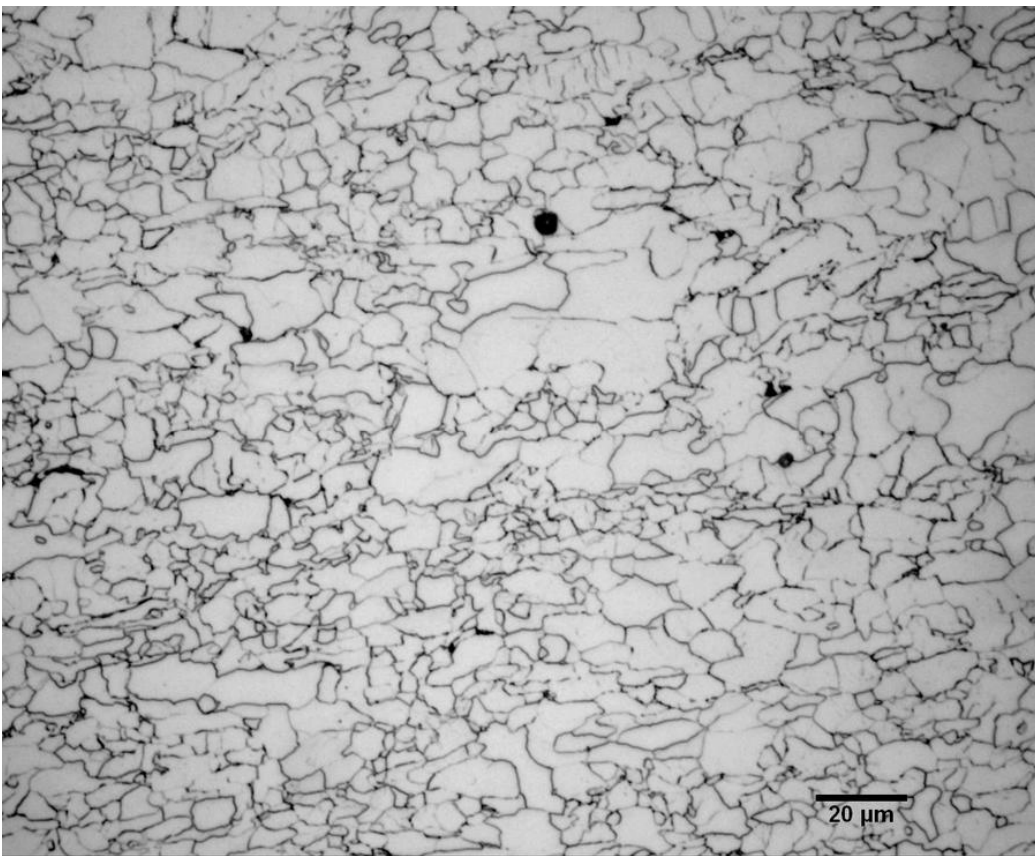


Ferrite grain boundaries in an interstitial-free sheet steel. Etched with Marshall's Reagent + HF. Original at 200X.

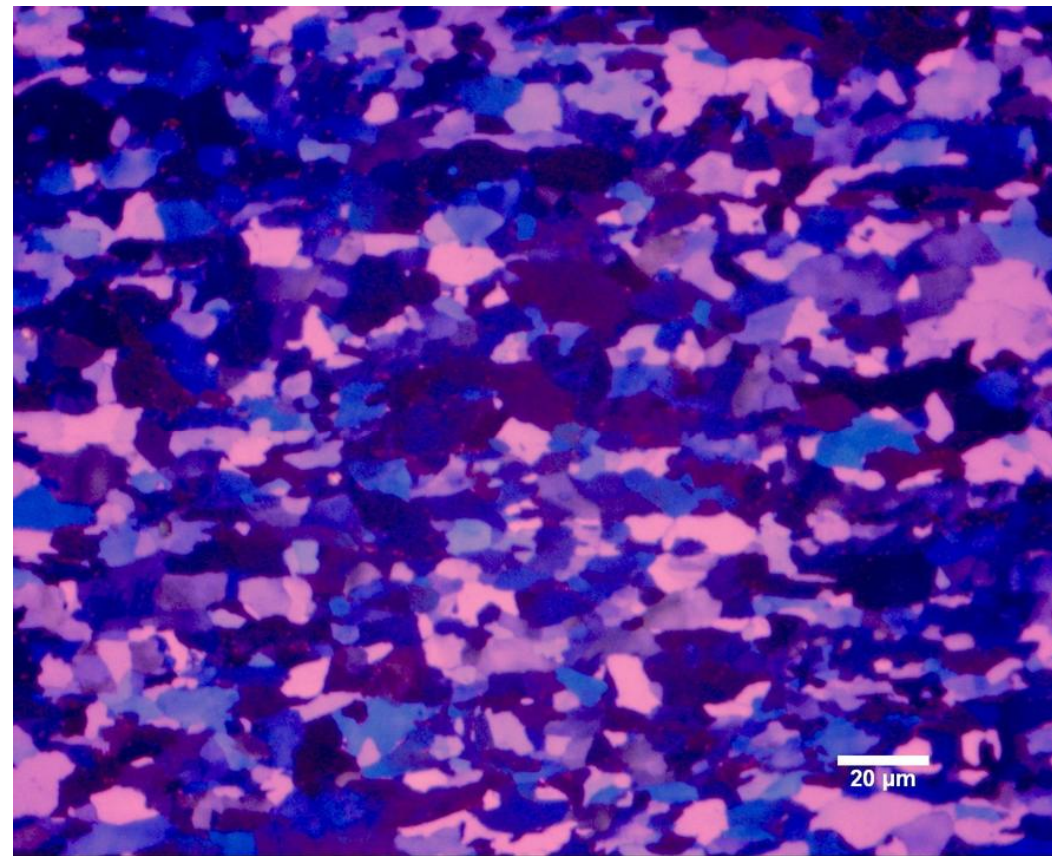


Electrical iron ($<0.02\%$ C) etched with 2% nital revealing a ferrite grain structure.

X-65 Line Pipe Steel

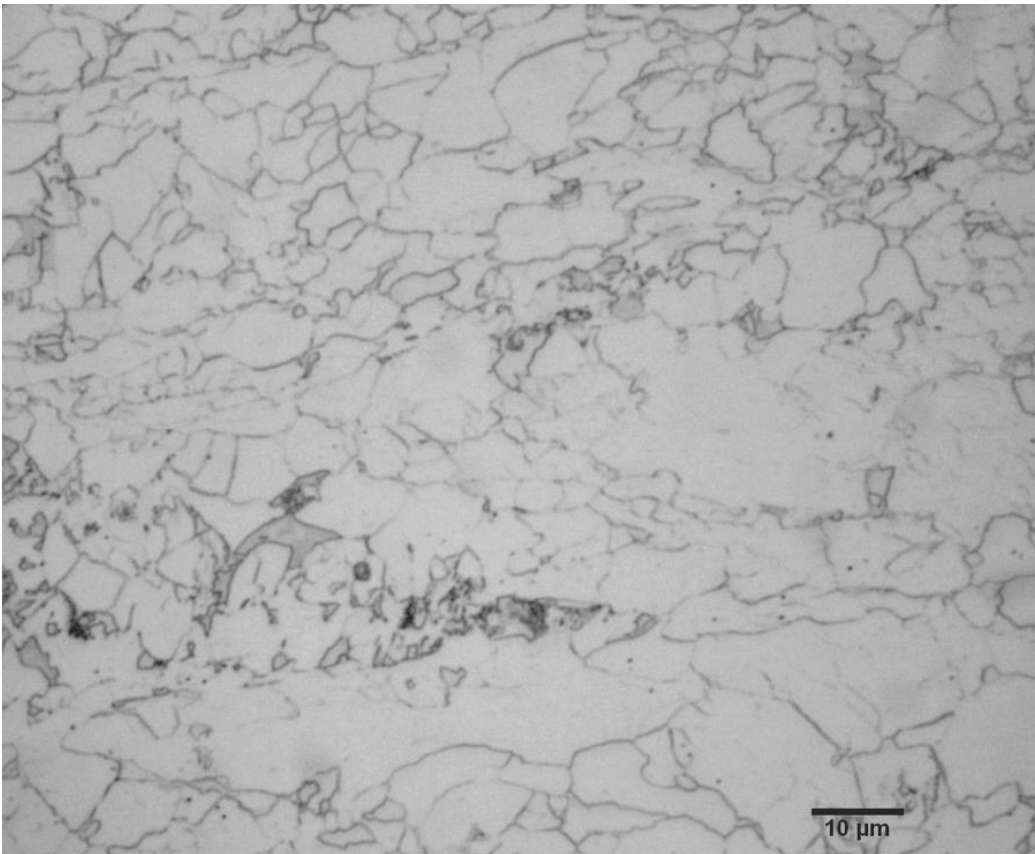


2% Nital

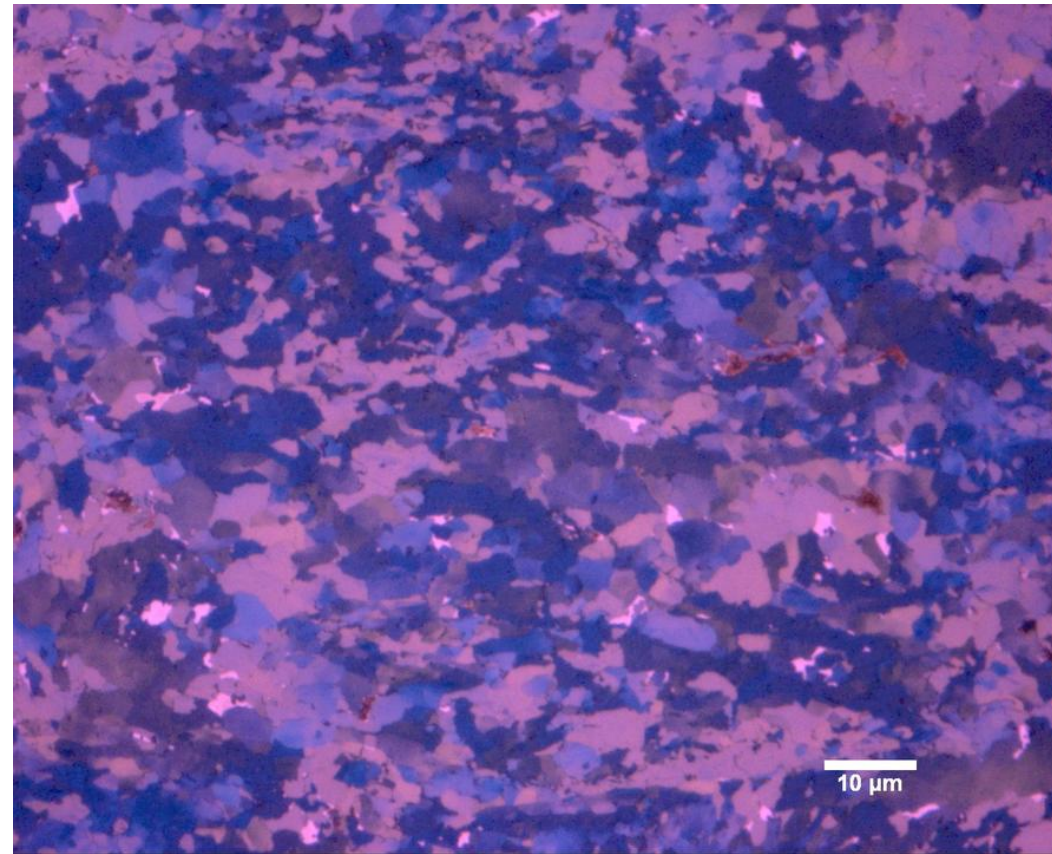


Klemm's I

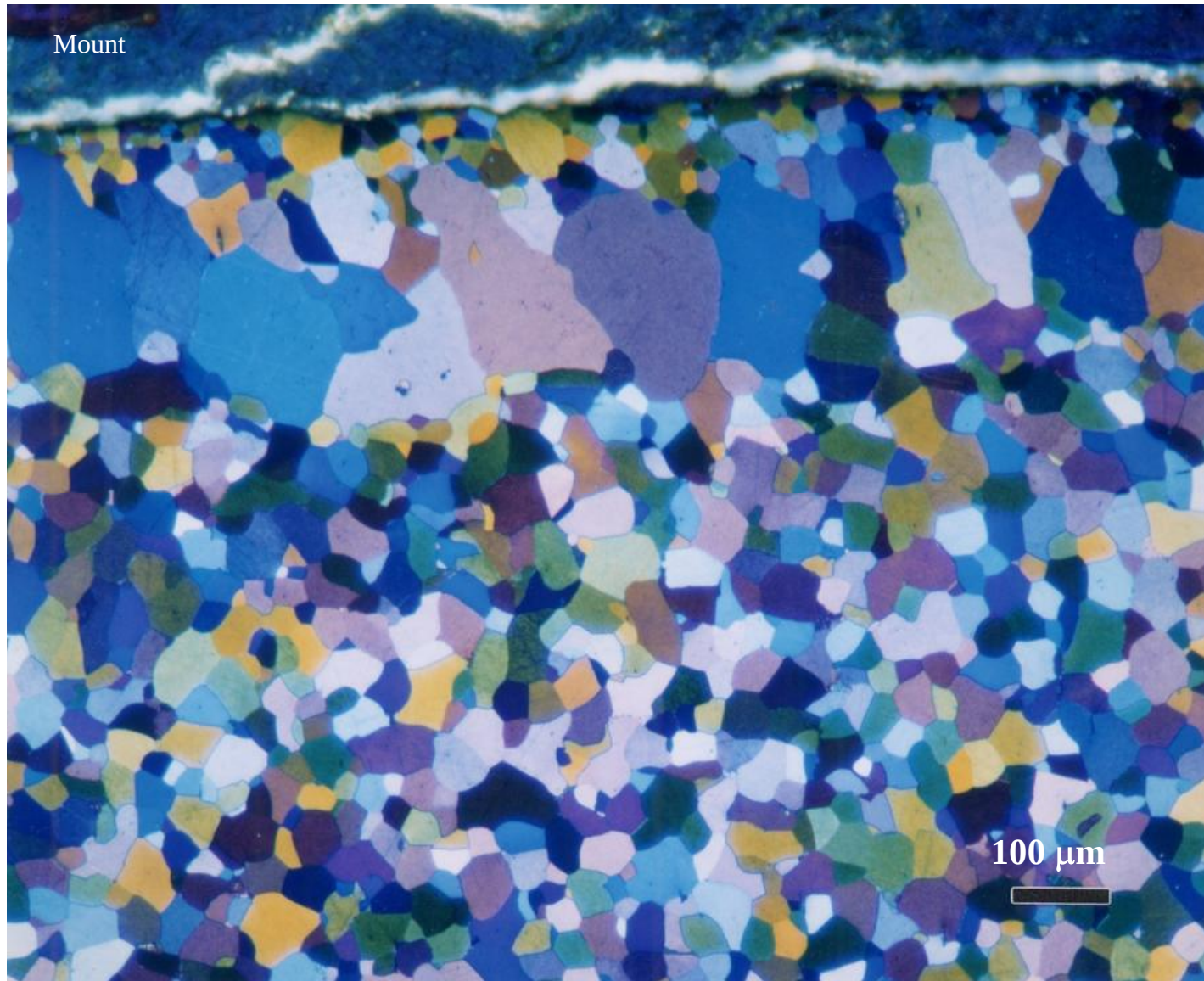
X-80 Line Pipe Steel



2% Nital



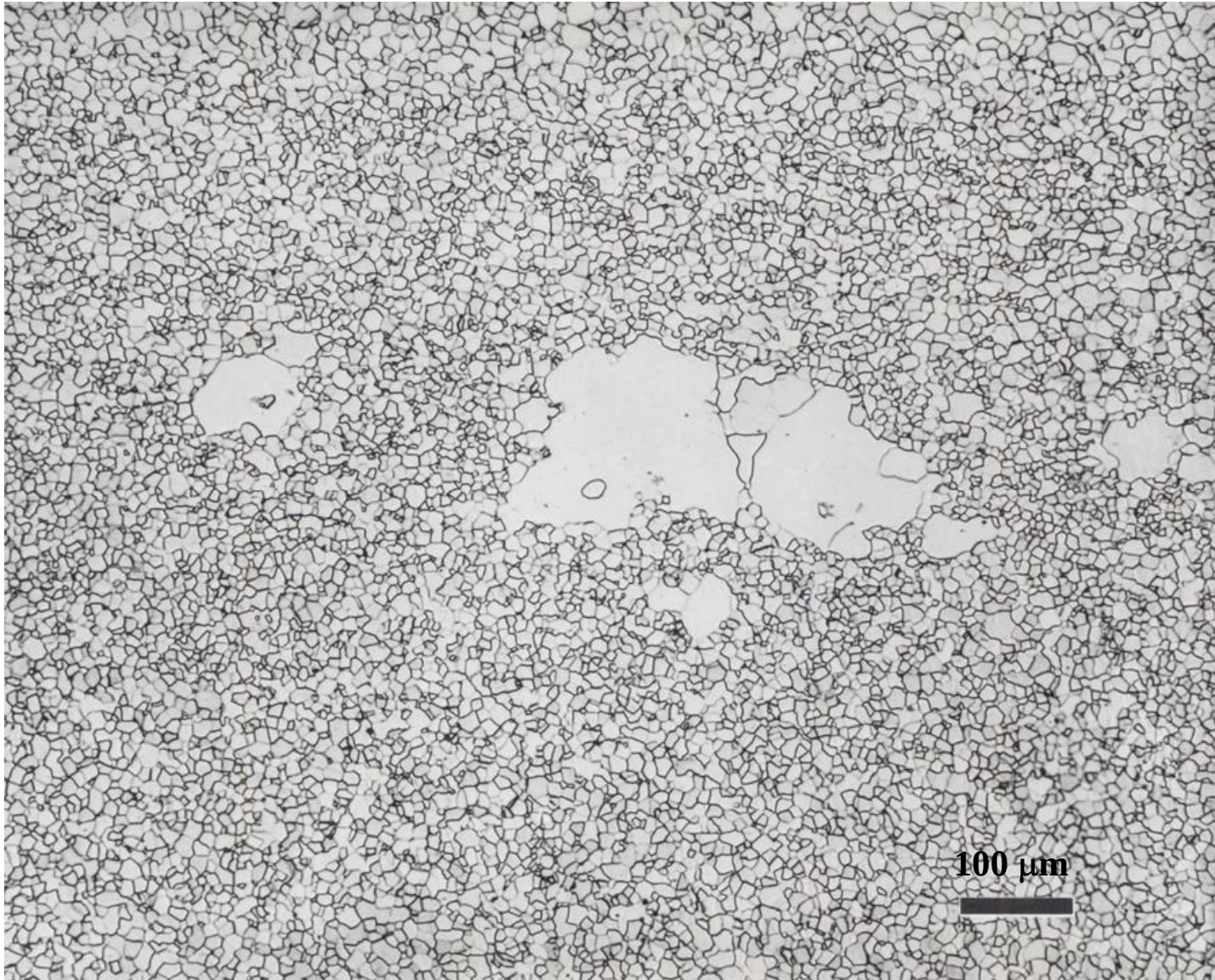
Klemm's I



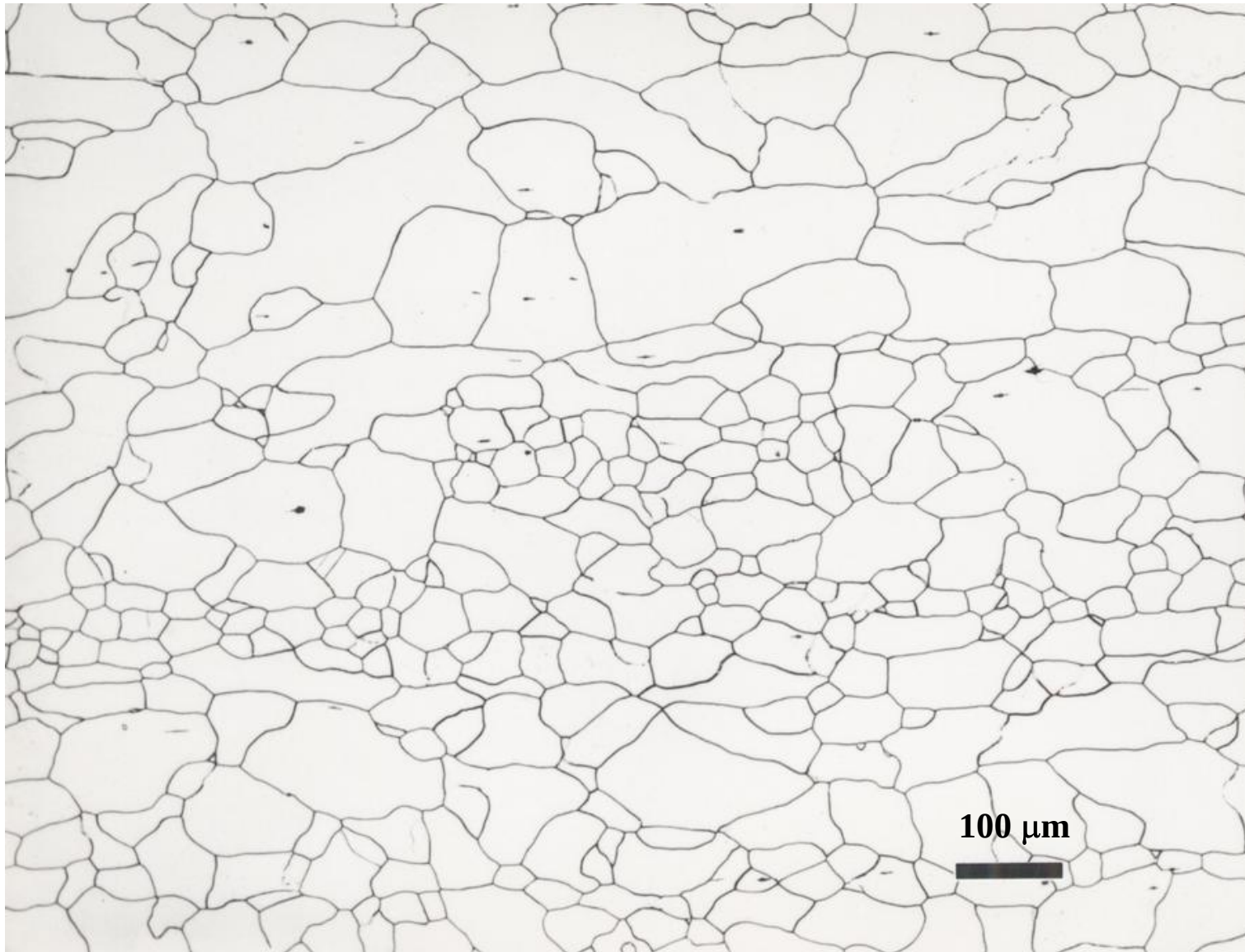
Ferrite grains in lamination sheet steel revealed using Klemm's I tint etch. This is a duplex condition where there are much larger grains near the surface. Viewed with polarized light plus sensitive tint.



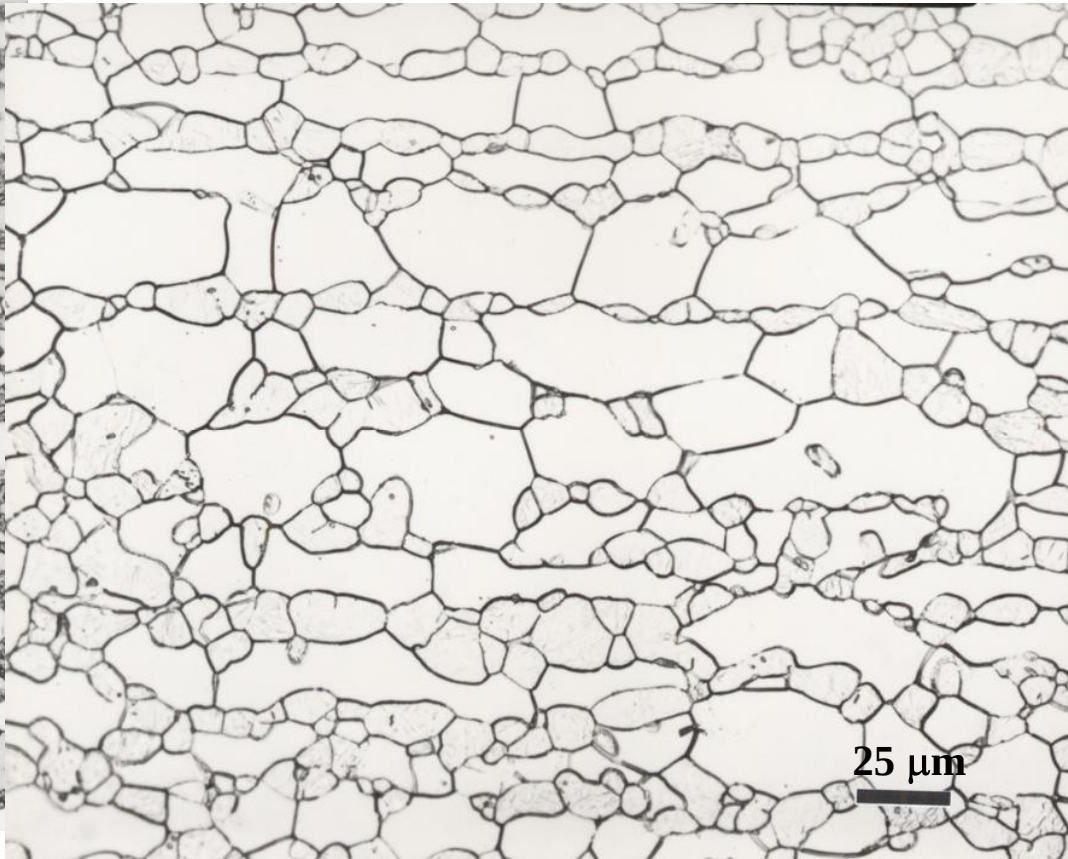
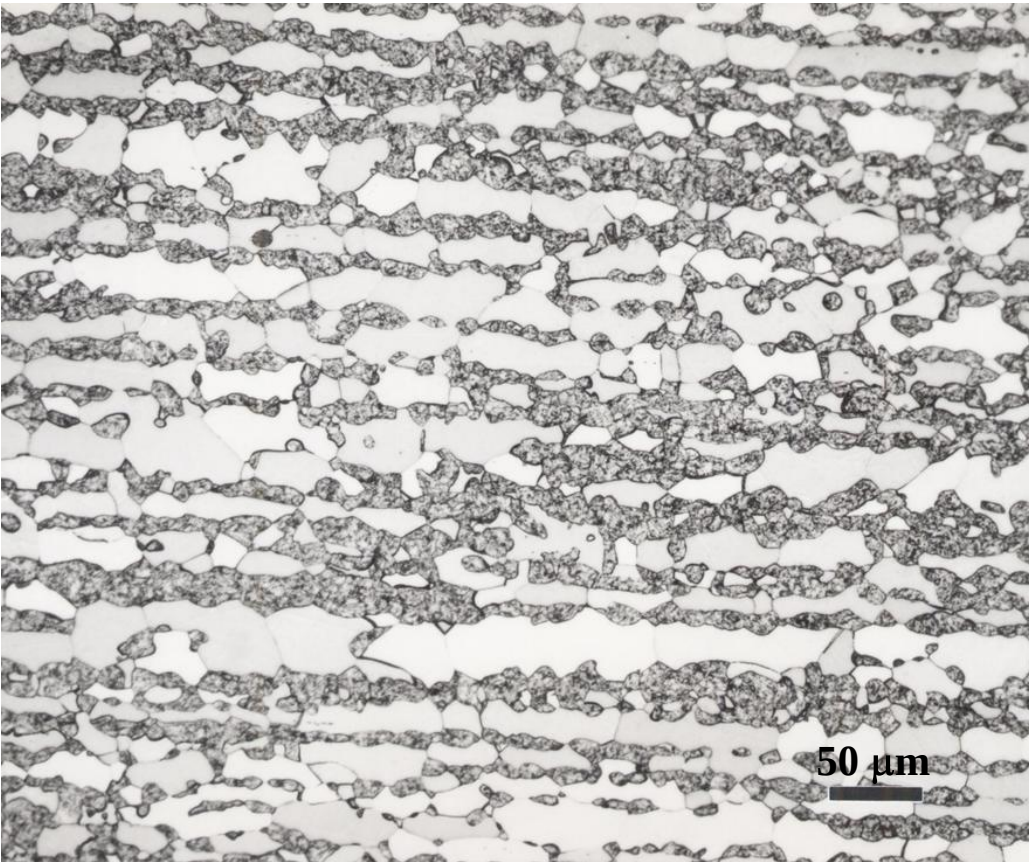
Ferrite grains (in the interior of the lamination sheet steel previously shown) revealed using Klemm's I tint etch. This is a normal unimodal grain size distribution. Original magnification was 100X. Viewed with polarized light plus sensitive tint.



Duplex grain size condition in a low-carbon sheet steel. This is a case where there are only a few grains that are far larger than the rest of the grains present. Etched with 2% nital.



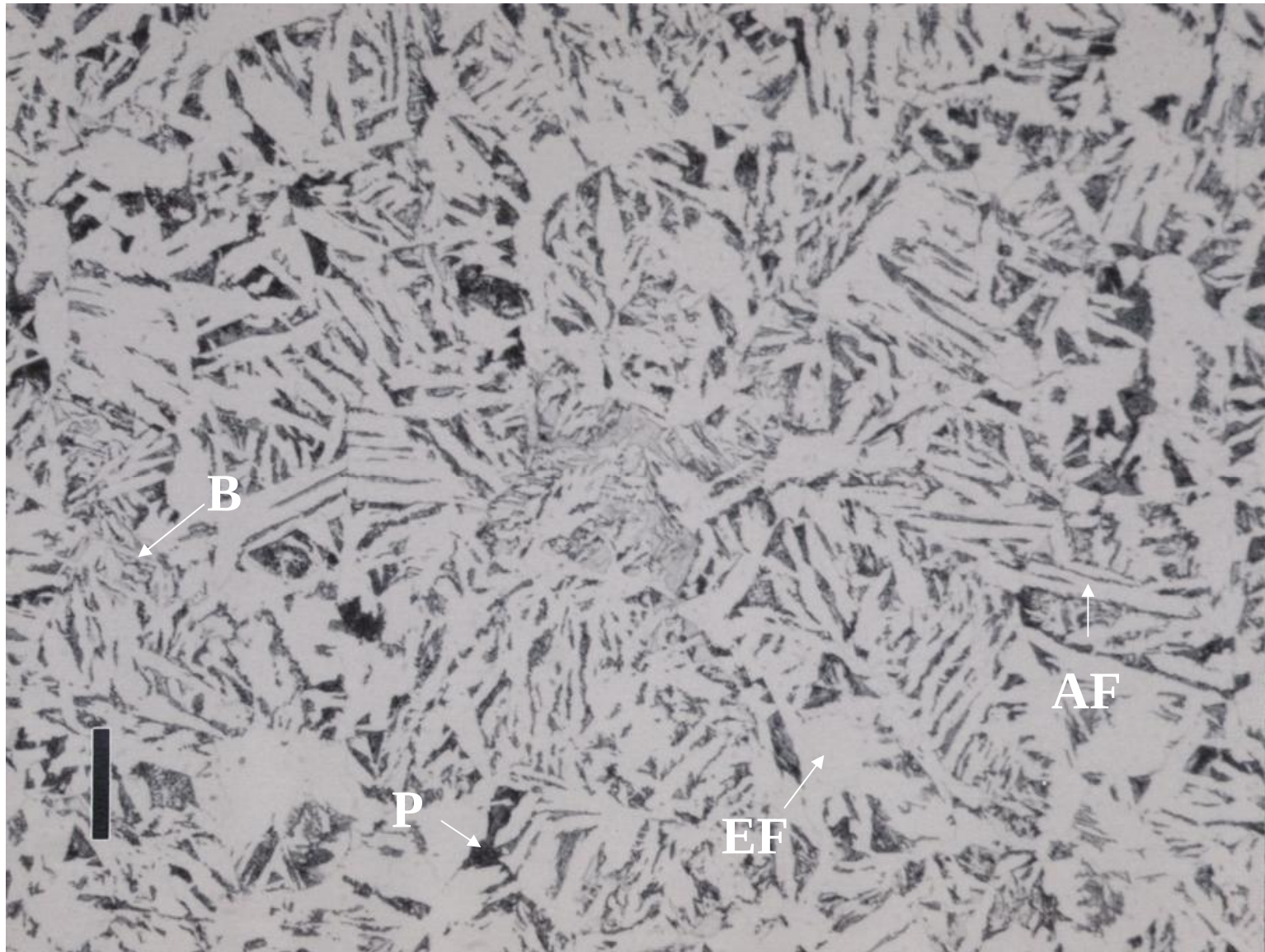
Ferrite grains in E-Brite (Fe - <0.01% C – 26% Cr – 1.1% Mo - <0.015% N) ferritic stainless steel in the hot rolled and annealed condition (6-mm thick plate). Etched with aqueous 60% HNO₃, 1.2 V dc, Pt cathode, 15 s (longitudinal plane). Note that a duplex grain size condition is present.



High-carbon versions of 430 ferritic steels are produced, by the microstructure consists of martensite and ferrite. The micrographs show a longitudinal view of such a modified grade after etching with Beraha's reagent (aqueous 10% HCl plus 1% potassium metabisulfite) to reveal the phases (left) and electrolytically with aqueous 60% nitric acid, 1 V dc, 40 s, to bring out the grain structure (right). Ferrite-ferrite, martensite-ferrite and martensite-martensite grain boundaries are revealed.

ACICULAR FERRITE

A highly sub-structured non-equiaxed ferrite that forms during cooling by a mixed diffusion and shear mode transformation beginning above the temperature range for upper bainite formation.



Acicular ferrite (AF) in as-rolled 4027 alloy steel (Fe – 0.27% C – 0.8% Mn – 0.25% Si – 0.25% Mo) revealed by etching with 2% nital. There are also patches of equiaxed ferrite (EF), occasional patches of pearlite (P) and considerable bainite (B) in the matrix. The magnification bar is 20 μm in length.



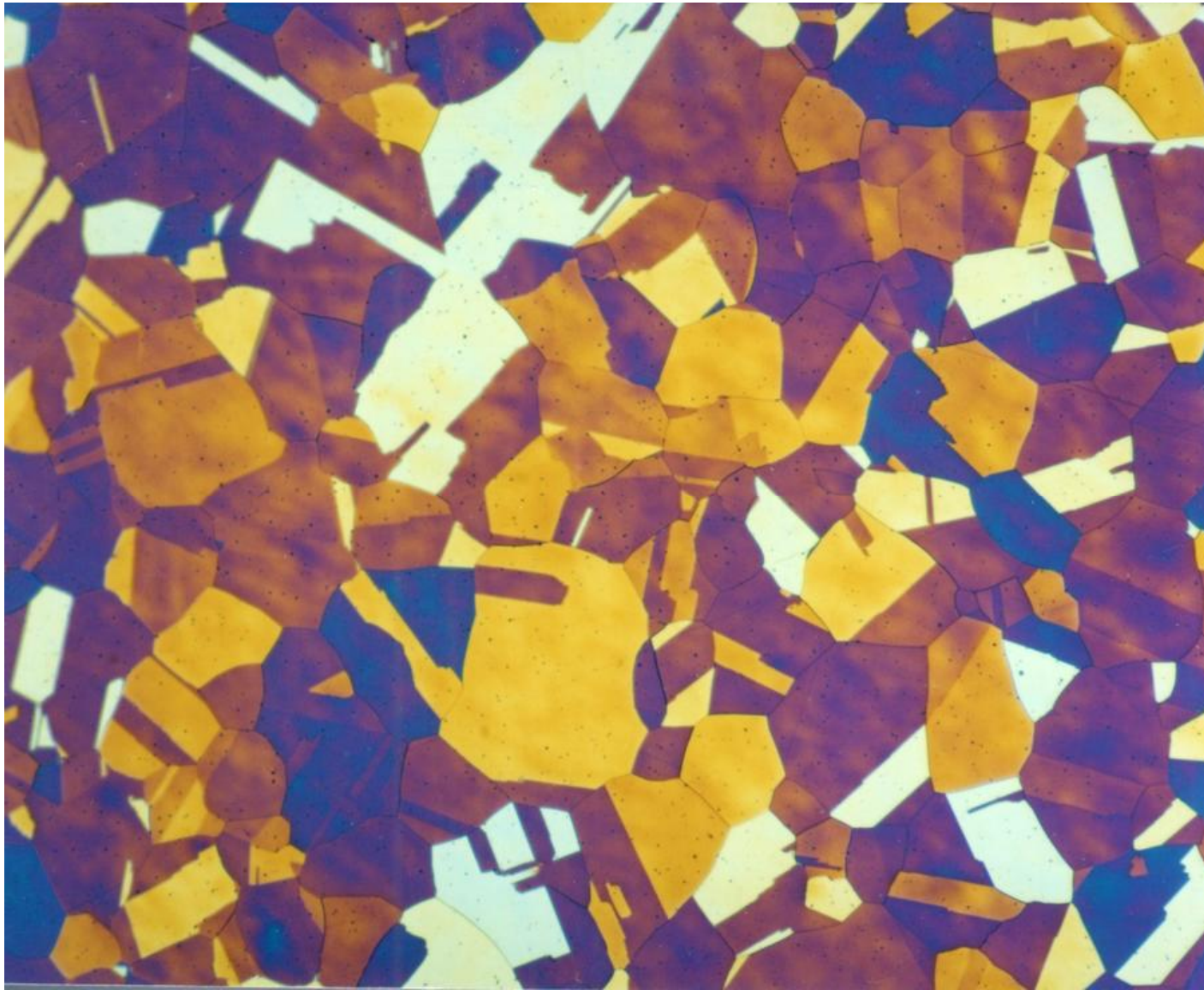
Acicular ferrite (arrows) in an as-cast low-carbon steel (Fe – 0.09% C – 0.39% Mn – 0.04% Si – 0.068% Al) revealed by Klemm's I reagent (viewed with crossed polarized light plus sensitive tint). Original at 100X magnification.

GAMMA IRON

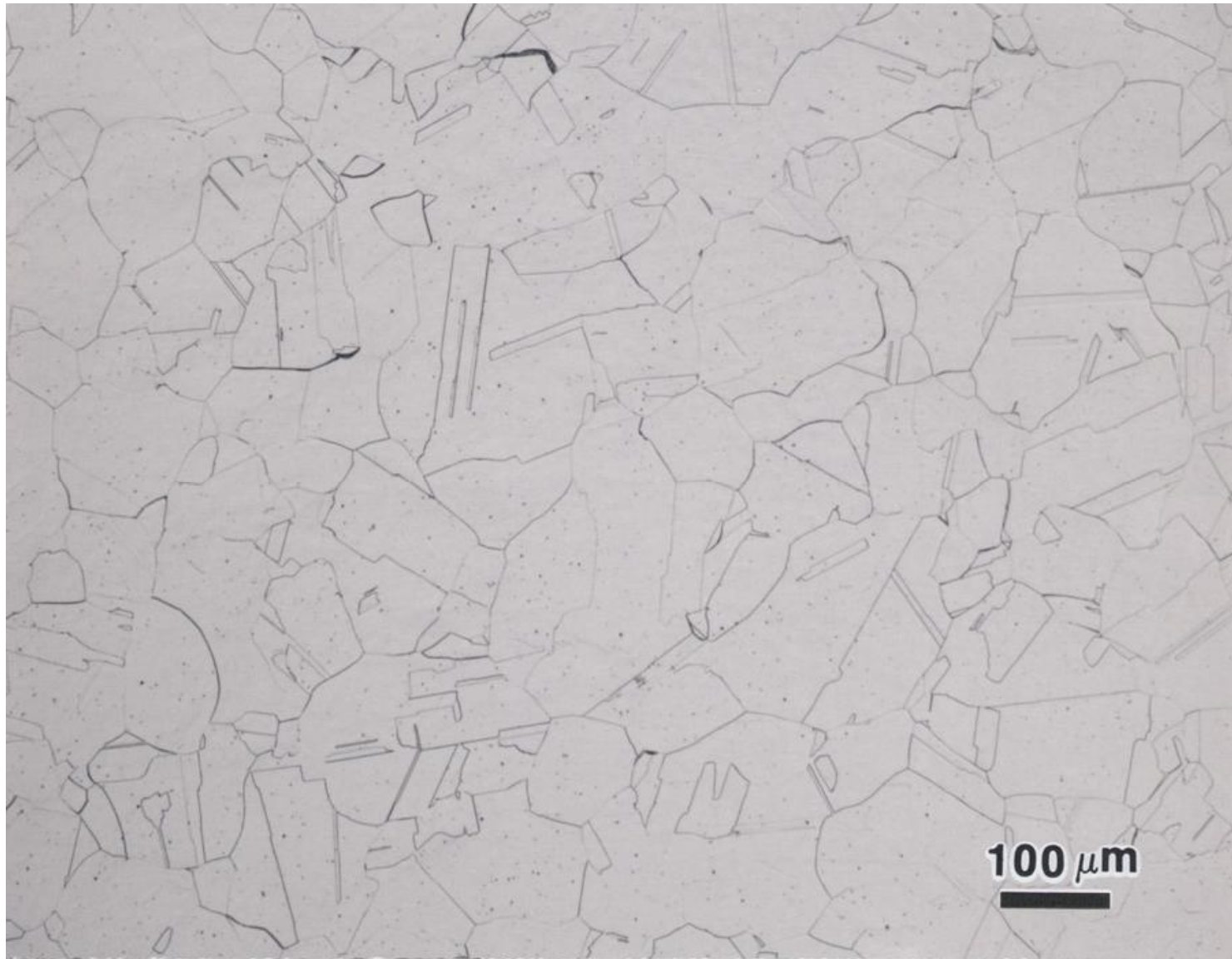
**The face-centered cubic form of pure iron, stable
from 912 to 1394 °C (1674 to 2541 °F)**

AUSTENITE

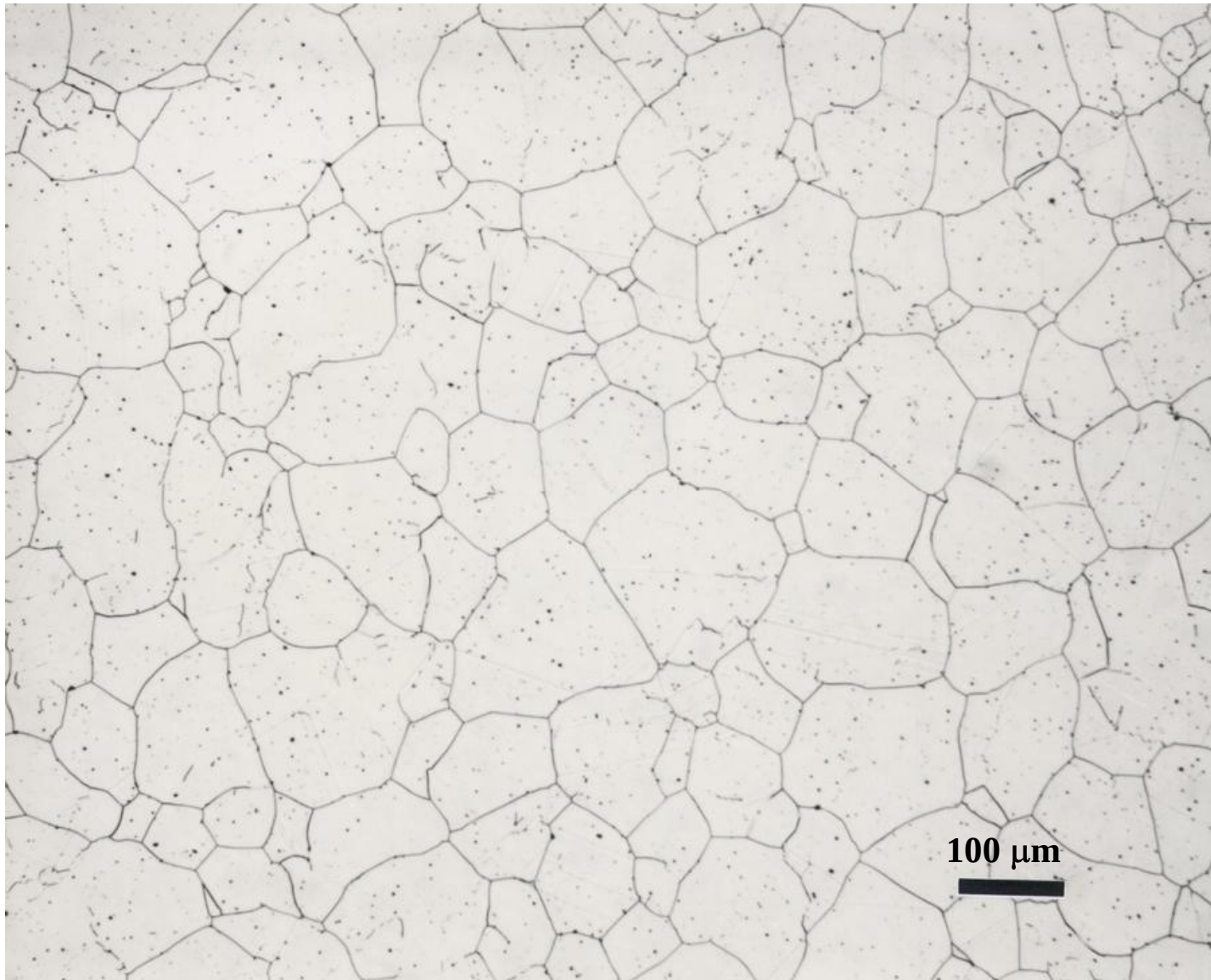
**A solid solution of one or more elements in
face-centered cubic iron.**



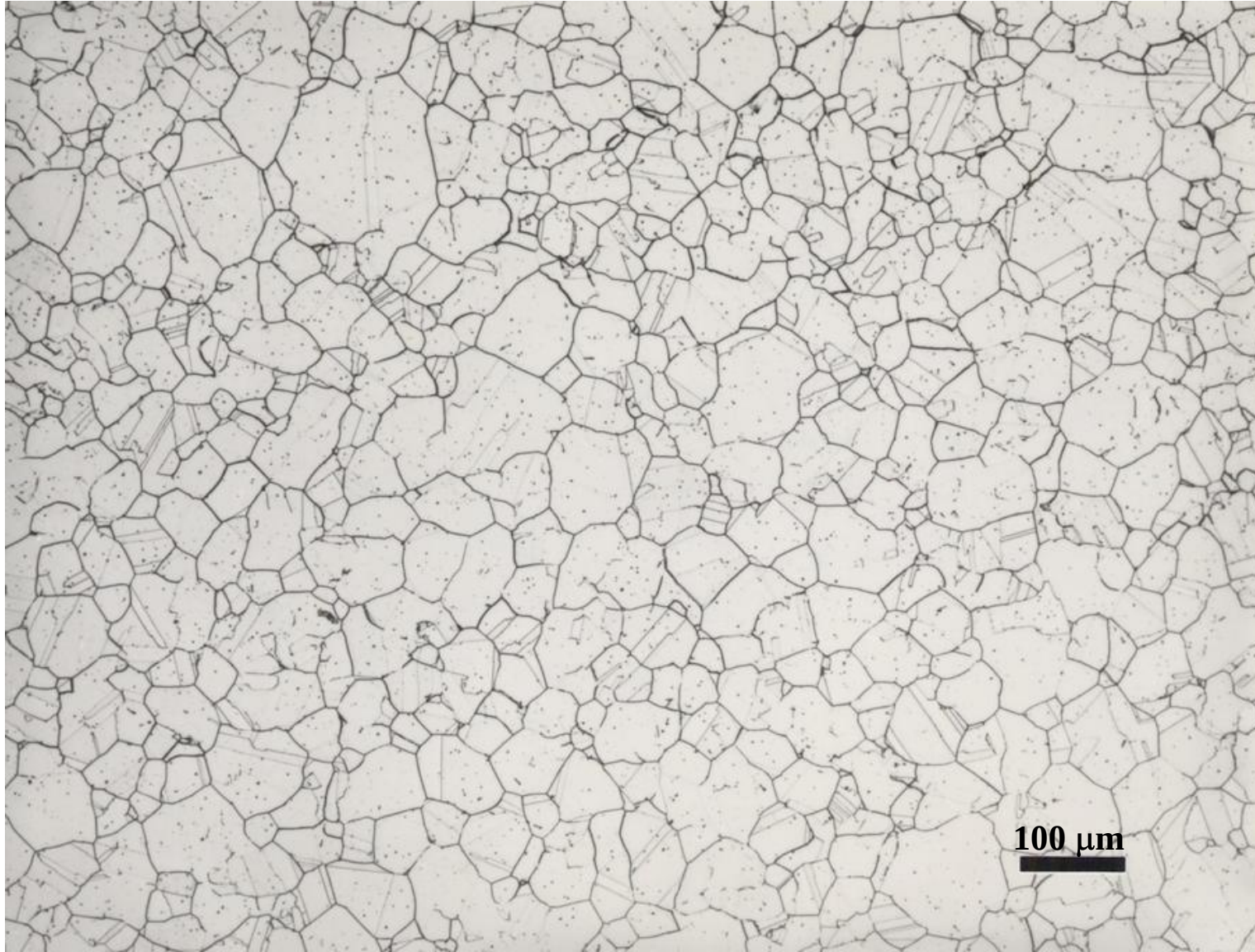
Austenite grains, with annealing twins, in type 316 stainless steel (Fe - <0.08% C - <2% Mn - <1% Si – 17% Cr – 12% Ni – 2.5% Mo) color etched with Beraha's reagent. Original at 100X.



**Austenite grain boundaries and annealing twins in type 316 stainless steel
(Fe - <0.08% C - <2% Mn - <1% Si – 17% Cr – 12% Ni – 2.5% Mo)
etched with glycereregia.**



Only the austenite grain boundaries are revealed in this specimen of type 316 austenitic stainless steel (Fe - $<0.08\%$ C – 17% Cr – 12% Ni – 2.5% Mo) that was solution annealed (1093 °C – 1 h, water quench) and etched with aqueous 60% HNO_3 at 1.5 V dc, 2 min.



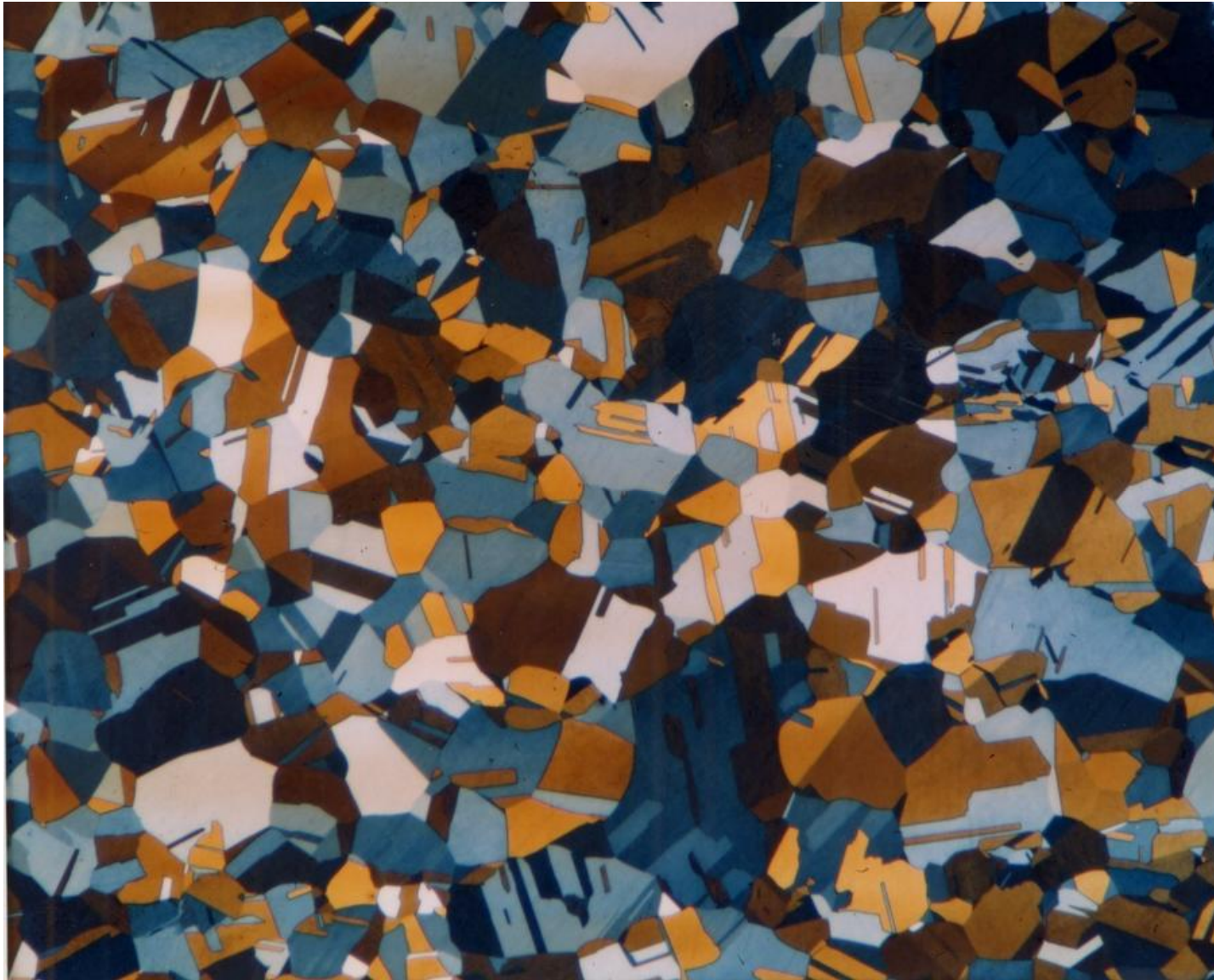
Carbides were precipitated on the austenite grain boundaries of this specimen of type 304 austenitic stainless steel that was solution annealed (1038 °C – 1 h, water quench) and aged at 650 °C for 2 h. Precipitation on twin boundaries is in a much lesser amount. The carbide was darkened by etching with aqueous 10% ammonium persulfate, 6 V dc, 10 s.



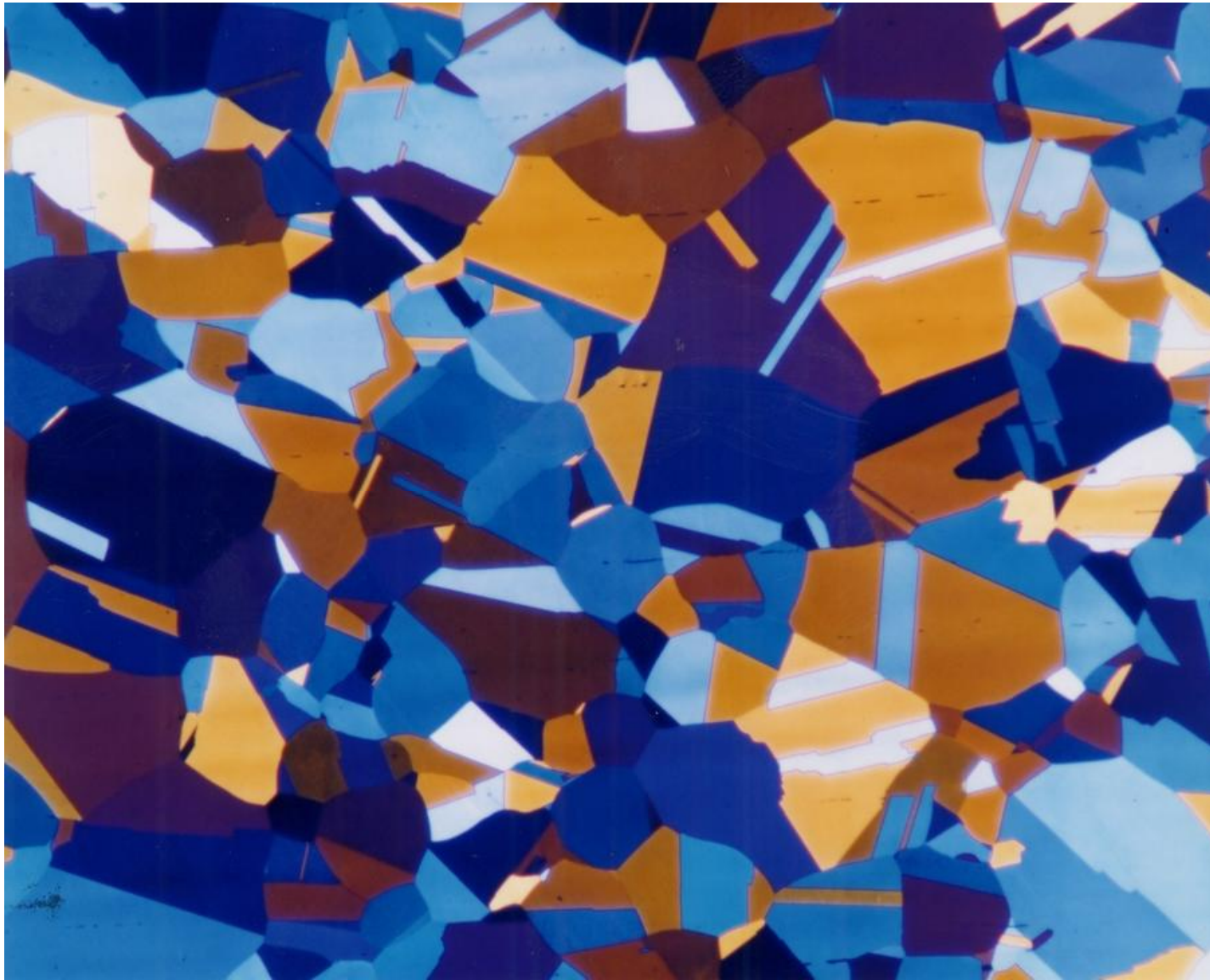
Duplex grain size distribution in solution annealed 22-13-5 austenitic stainless steel (Fe - <0.06% C – 5% Mn – 22% Cr – 12.5% Ni – 2.25% Mo – 0.2% Nb – 0.2% V – 0.3% N) etched with “waterless” Kalling’s reagent.



Necklace-type duplex grain size distribution in SCF-19 austenitic stainless steel (Fe – 0.03% C – 5% Mn – 20% Cr – 18% Ni – 5% Mo – 0.35% N) that was cold reduced 10% in thickness and solution annealed at a low temperature (900 °C – 1 h, water quench). Etched with aqueous 60% HNO₃, 1 V dc, 60 s. Complete recrystallization, and grain growth, would occur at a higher temperature, 1180 °C is the recommended solution annealing temperature.

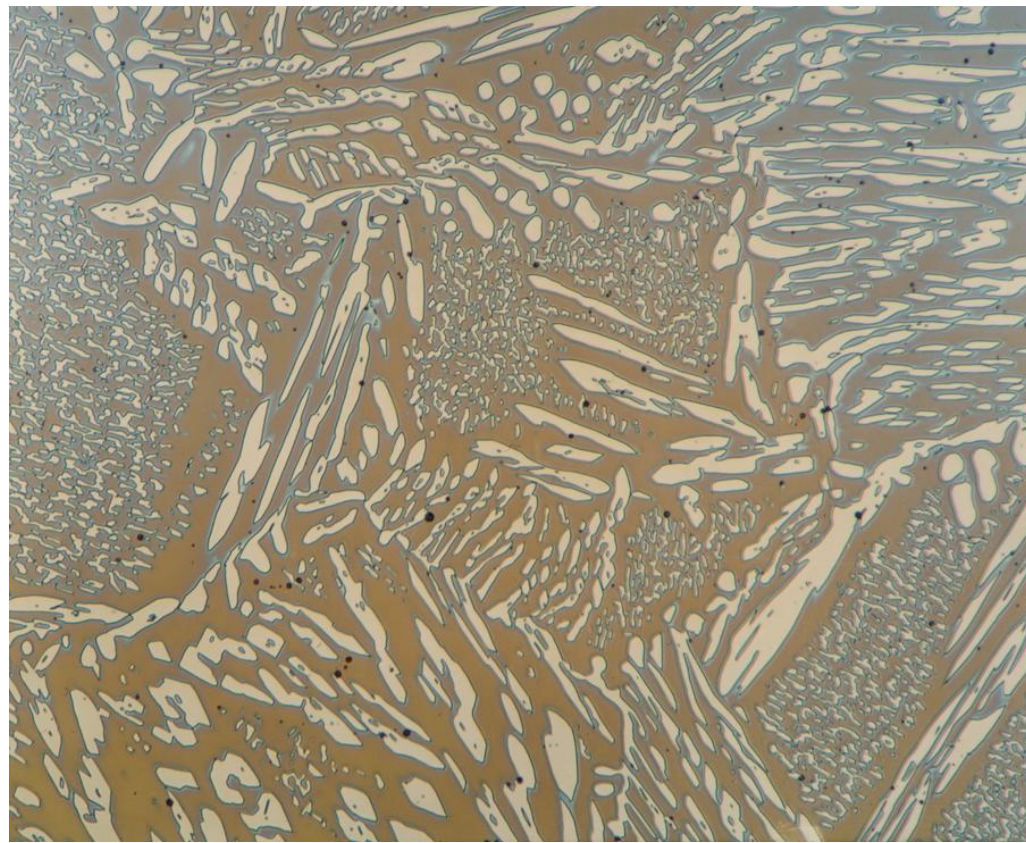


Twinned austenitic grain structure of wrought, annealed Fe – 39% Ni color etched with Beraha's sulfamic acid solution (100 mL water, 3 g potassium metabisulfite, 2 g sulfamic acid) and viewed with polarized light plus sensitive tint. Original at 100X.

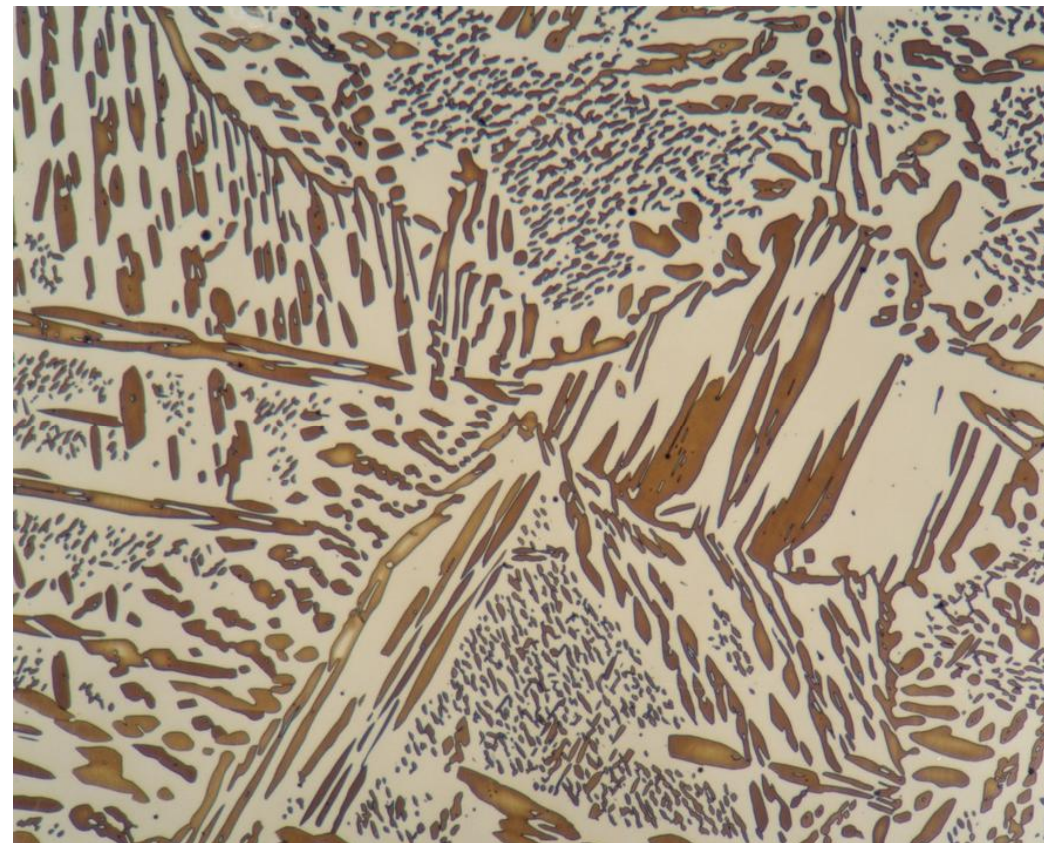


Twinned austenitic grain structure of solution annealed, wrought Hadfield manganese steel (Fe – 1.12% C – 12.7% Mn – 0.31% Si) tint etched with Beraha's sulfamic acid reagent (100 mL water, 3 g potassium metabisulfite and 2 g sulfamic acid) and viewed with polarized light plus sensitive tint. Original at 100X.

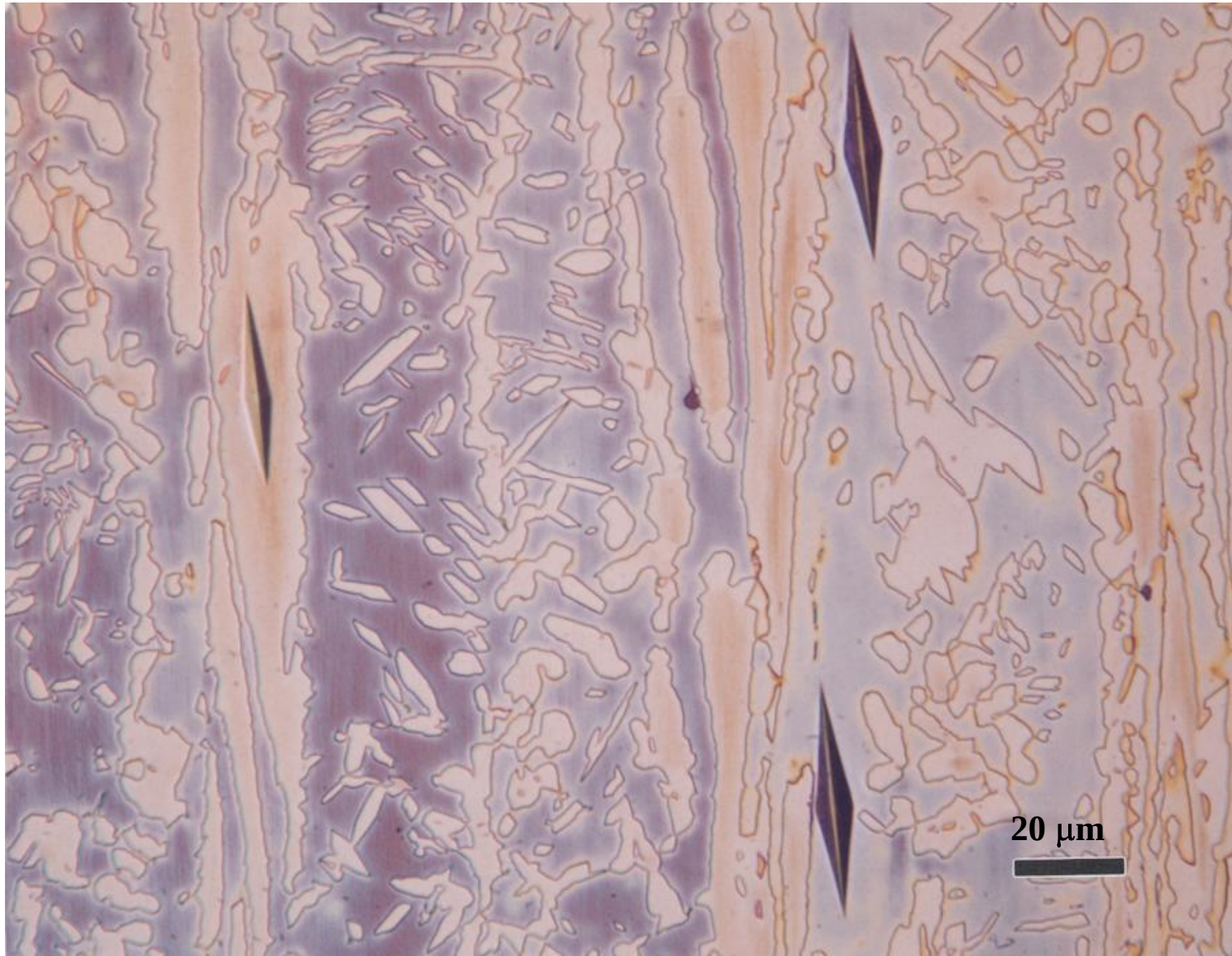
Color Ferrite



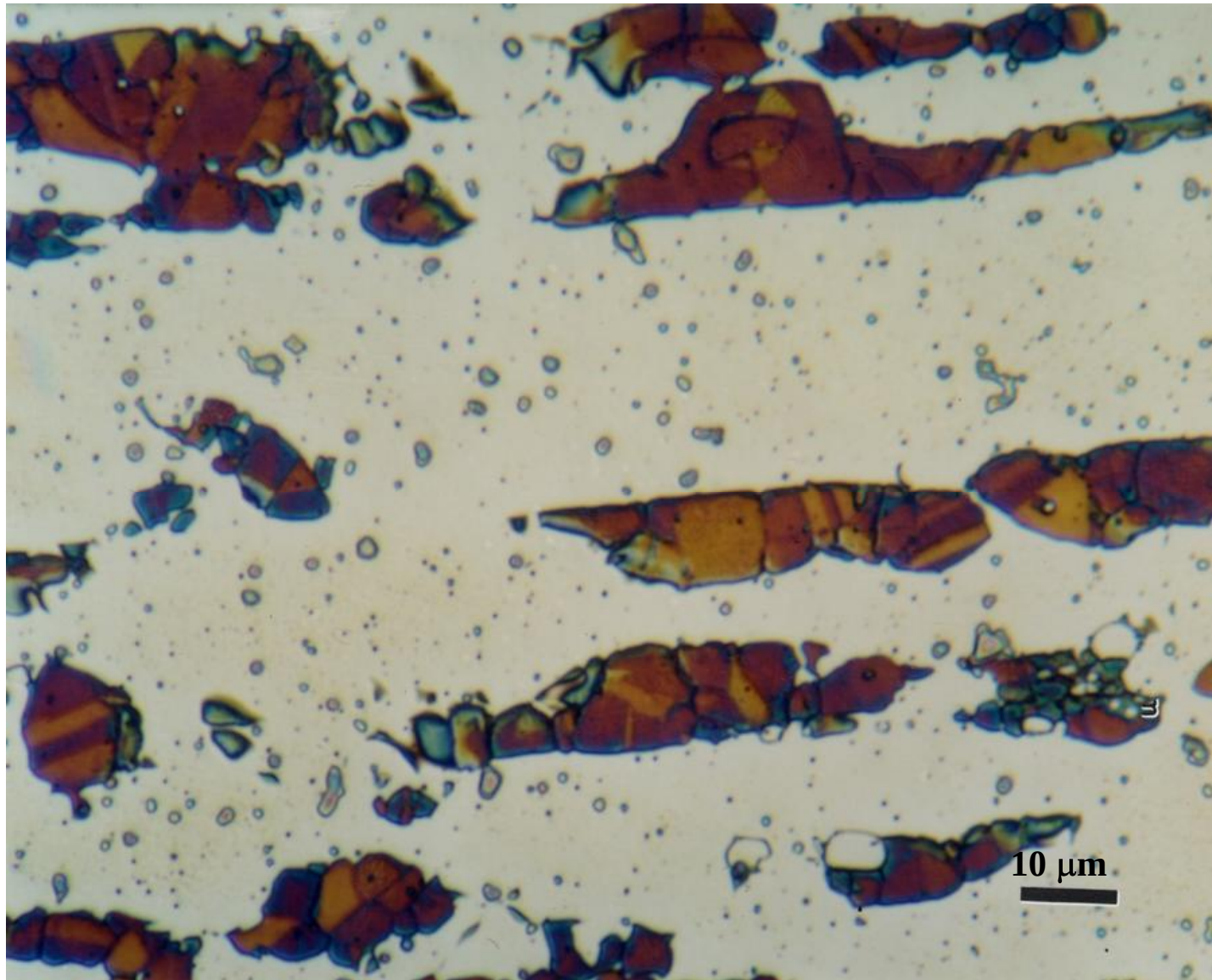
Color Austenite



As-cast ASTM A890, Grade 5A (UNS J93404) (Fe – 0.03% C – 1.5% Mn – 25% Cr – 7% Ni – 4.5% Mo – 0.2% N) duplex stainless steel after etching with aqueous 20% NaOH (3 V dc, 10 s) to color the ferrite (left) or with 100 mL water – 20 g NH_4FHF – 0.5 g $\text{K}_2\text{S}_2\text{O}_5$ to color the austenite (right) . Originals at 100X magnification. The composition is adjusted so that ferrite and austenite are stable at room temperature.



Microstructure of wrought 7-Mo Plus duplex stainless steel (Fe - <0.03% C – 27.5% Cr – 4.2% Ni – 1.75% Mo – 0.25% N) electrolytically etched with aqueous 20% NaOH (3 V dc, 5 s) to color the ferrite blue. There is some light yellow-tan coloring of the austenite. The average Knoop hardness of the austenite was 361.8 HK and that of the ferrite was 263.5 HK. 7-Mo Plus is a registered trademark of Carpenter Technology Corp., Reading, Pennsylvania.



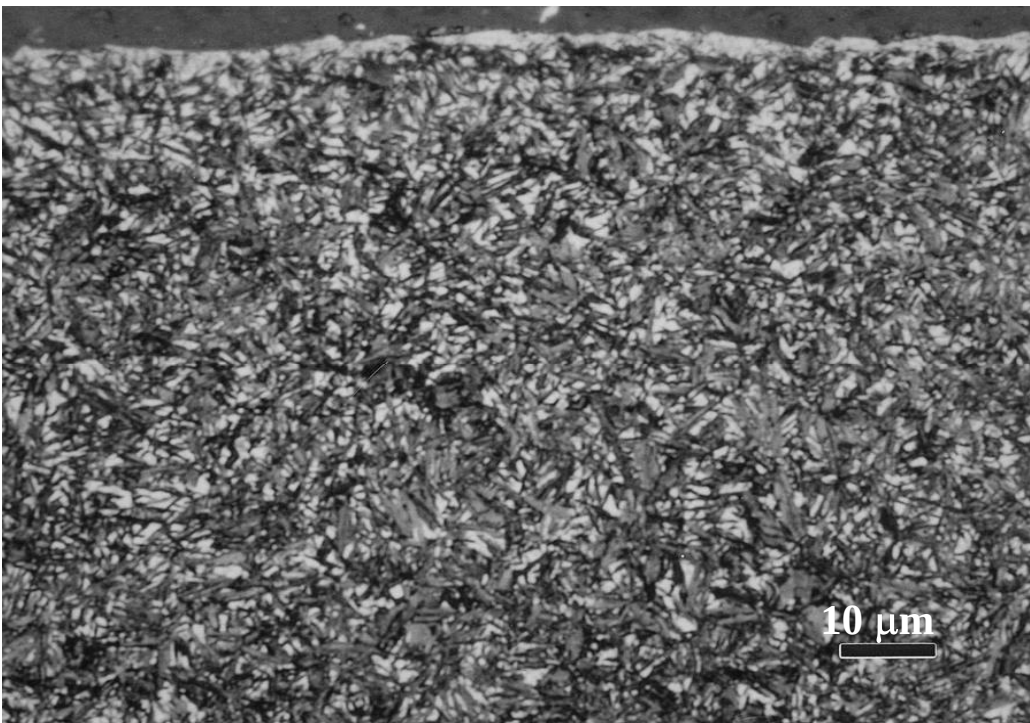
Austenite grains colored by the LB1 reagent in a duplex stainless steel, 7 Mo (Fe - <0.1% C – 27.5% Cr – 4.5% Ni – 1.5% Mo) in the hot rolled and annealed condition. The ferrite grains are not colored. The other particles appear to be carbides. The etch contains 100 mL water, 20 g NH_4FHF and 0.5 g $\text{K}_2\text{S}_2\text{O}_5$. 7-Mo is a registered trademark of Carpenter Technology Corp.

RETAINED AUSTENITE

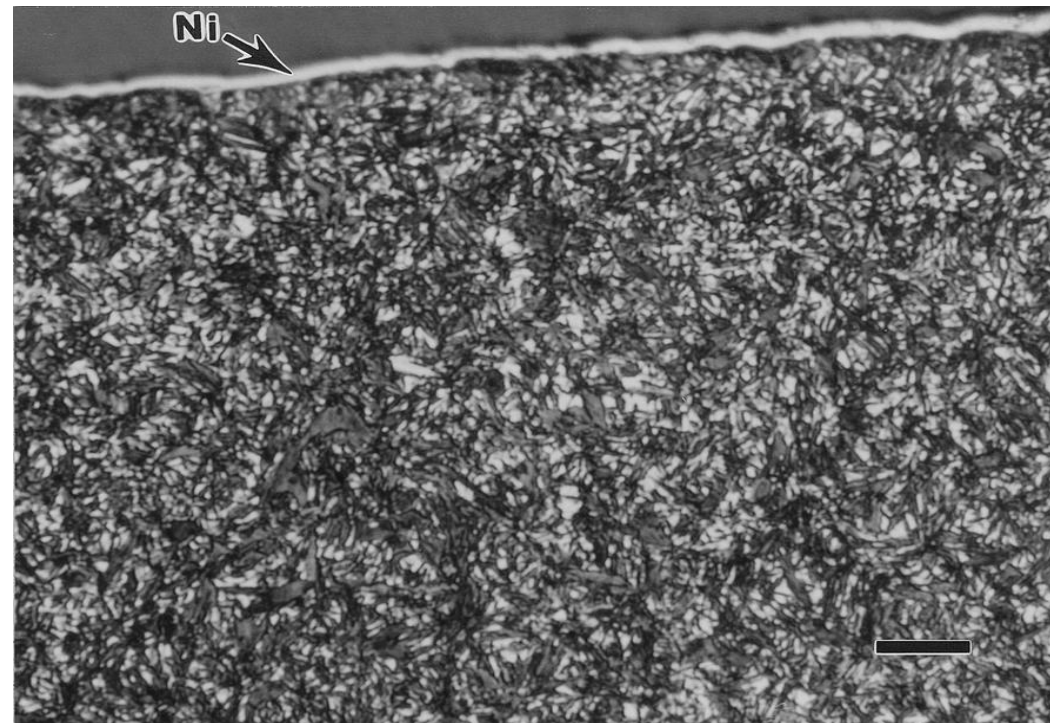
Austenite not converted to martensite during cooling (quenching); the cooling did not reach the martensite finish, M_f , temperature.

Carburized 8720 Alloy Steel

(Fe – 0.2% C – 0.8% Mn – 0.5% Cr – 0.55% Ni – 0.25% Mo)

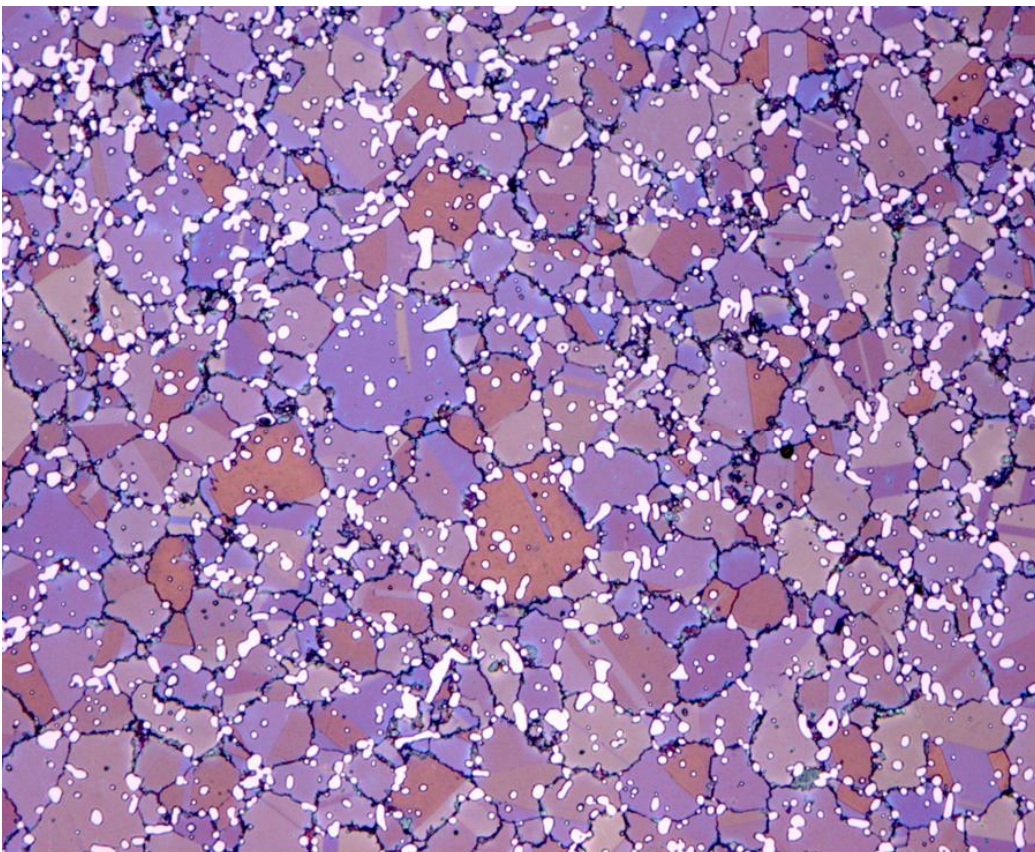


25.4% Retained Austenite

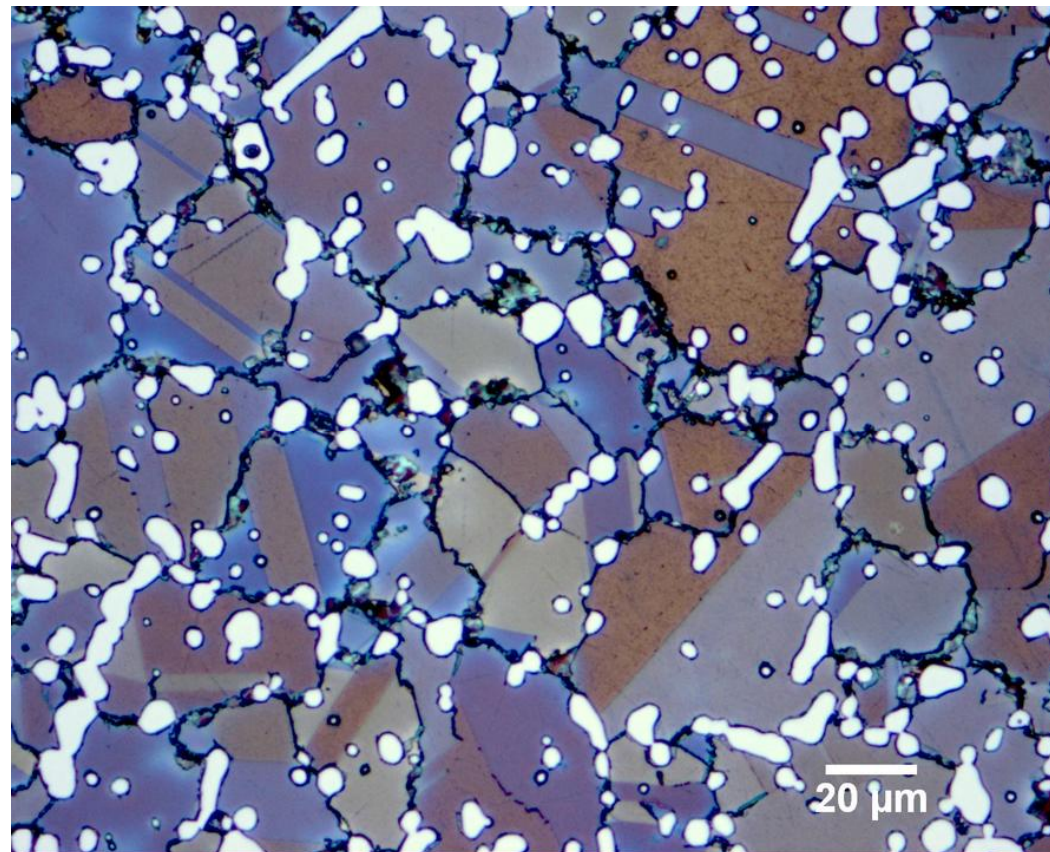


19.7% Retained Austenite

X-ray diffraction revealed 25.4% (left) and 19.7% (right) retained austenite in these specimens. The x-ray beam senses the surface to a depth of at least half a mm. Image analysis performed on the specimens gave much lower values. Specimens etched with 10% $\text{Na}_2\text{S}_2\text{O}_5$.

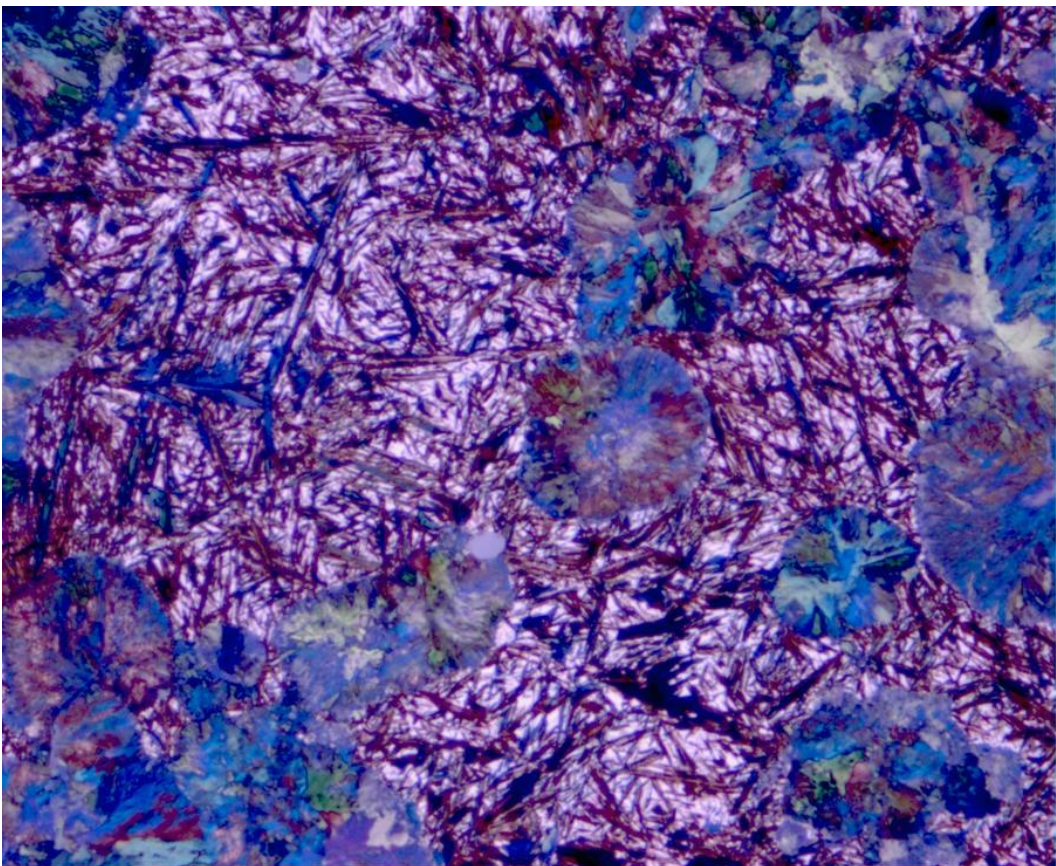


200X

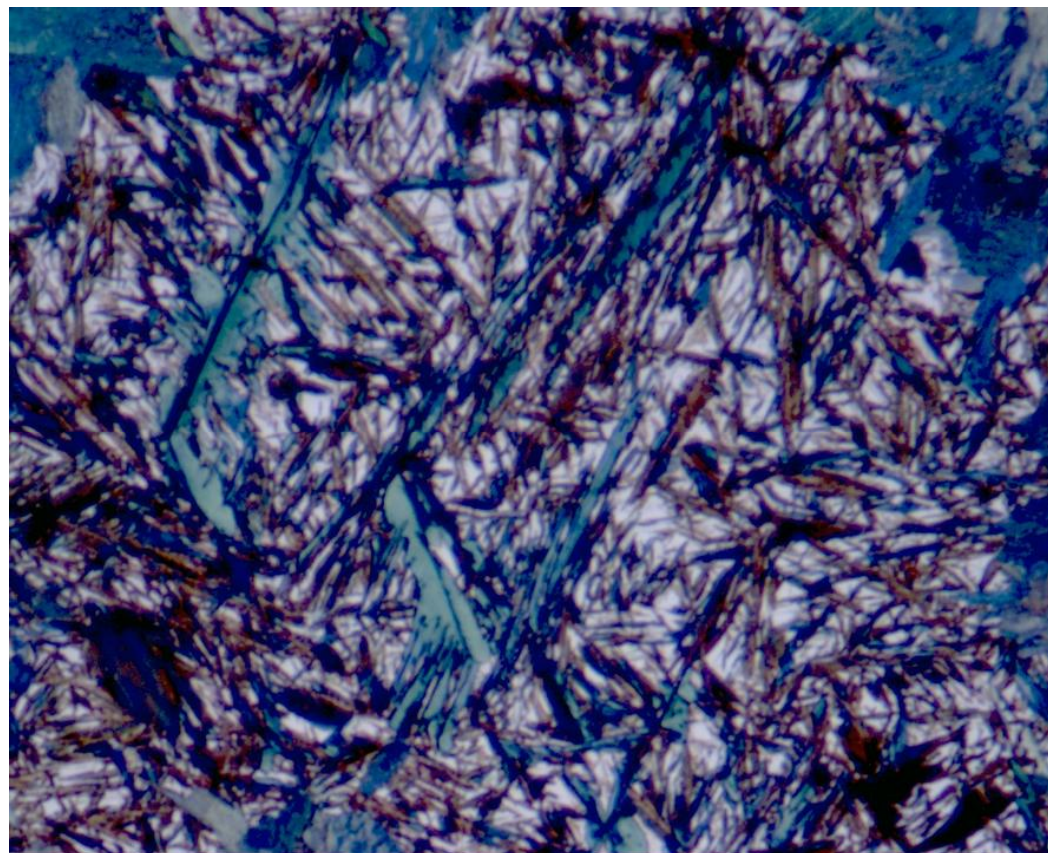


500X

Retained austenite composes the majority of the matrix (plus some undissolved Cr_7C_3 carbide) in this specimen of AISI D3 tool steel that was austenitized at 2200 °F, much higher than recommended. The specimen was etched with Beraha's sulfamic acid reagent number 4.

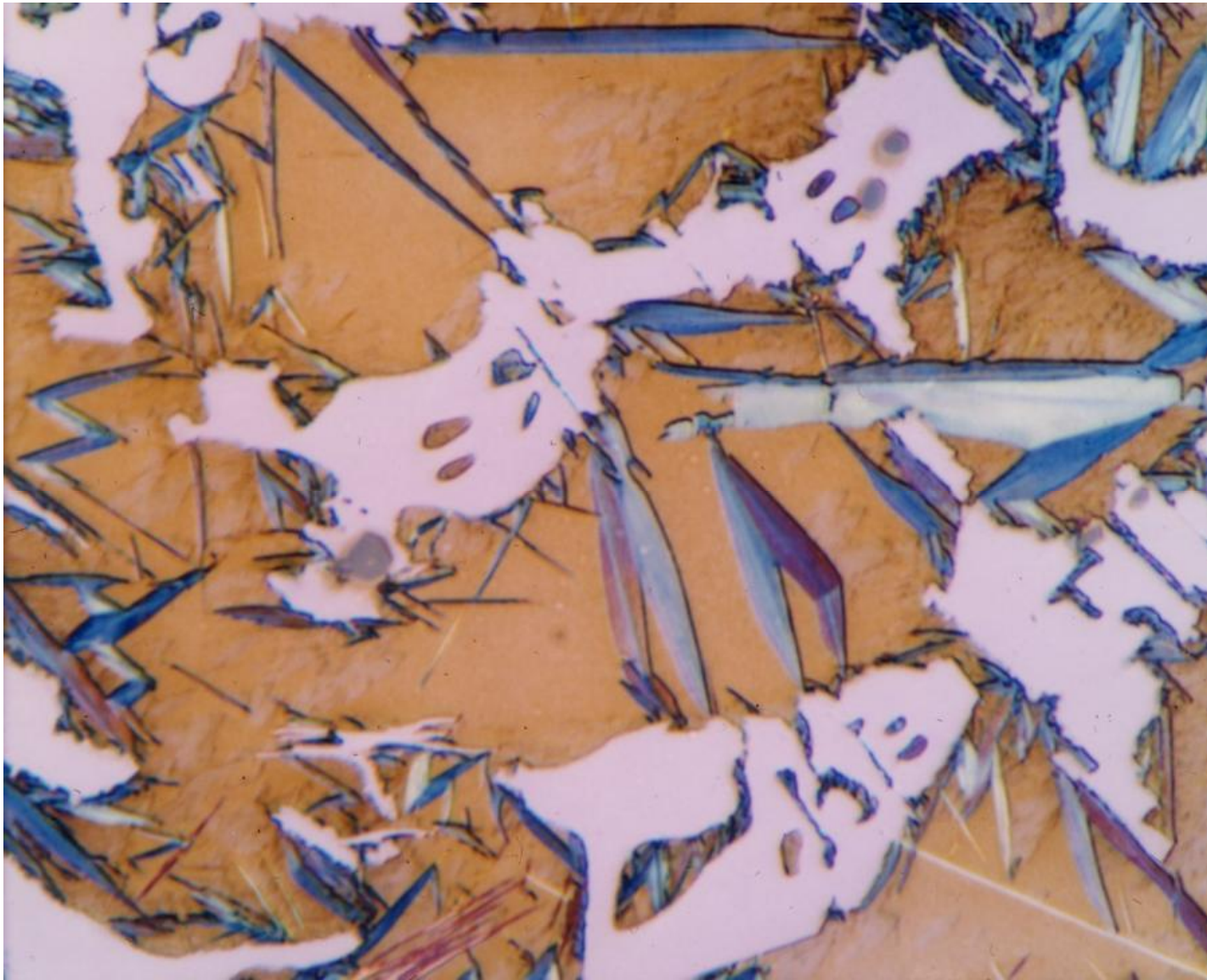


500X



1000X

Patches of retained austenite are observed within a matrix of fine pearlite in the cast of this W1 water-hardening tool steel; etched with Klemm's I.



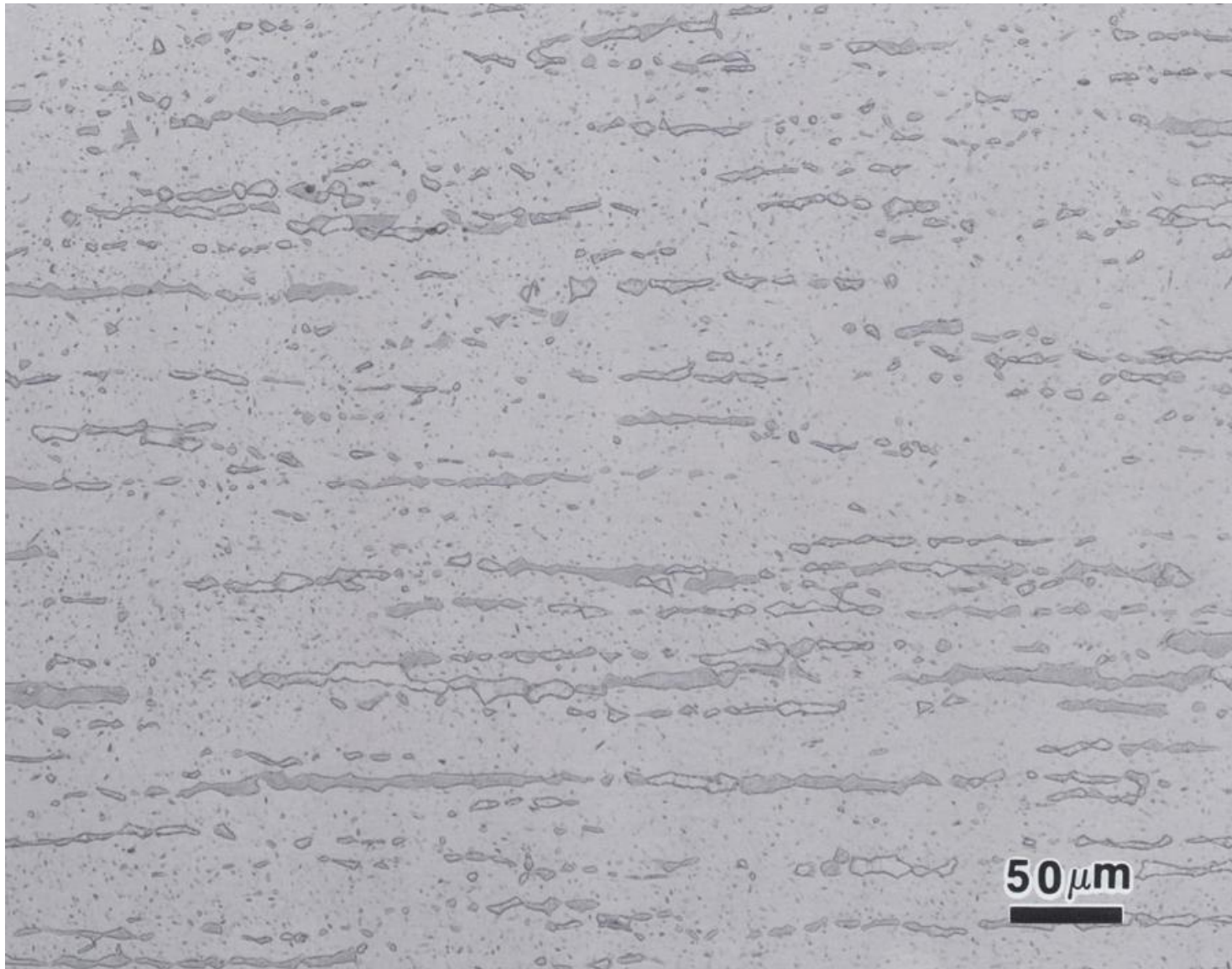
As-cast NiHard cast iron (Fe – 2.98% C – 0.64% Mn – 0.85% Si – 4.4% Ni – 2.34% Cr) containing cementite (white), retained austenite (light brown), manganese sulfides (gray particles) and plate martensite “needles” (light blue and medium blue) after tint etching with Beraha’s CdS reagent and viewing with polarized light plus sensitive tint. Original at 1000X.

DELTA IRON

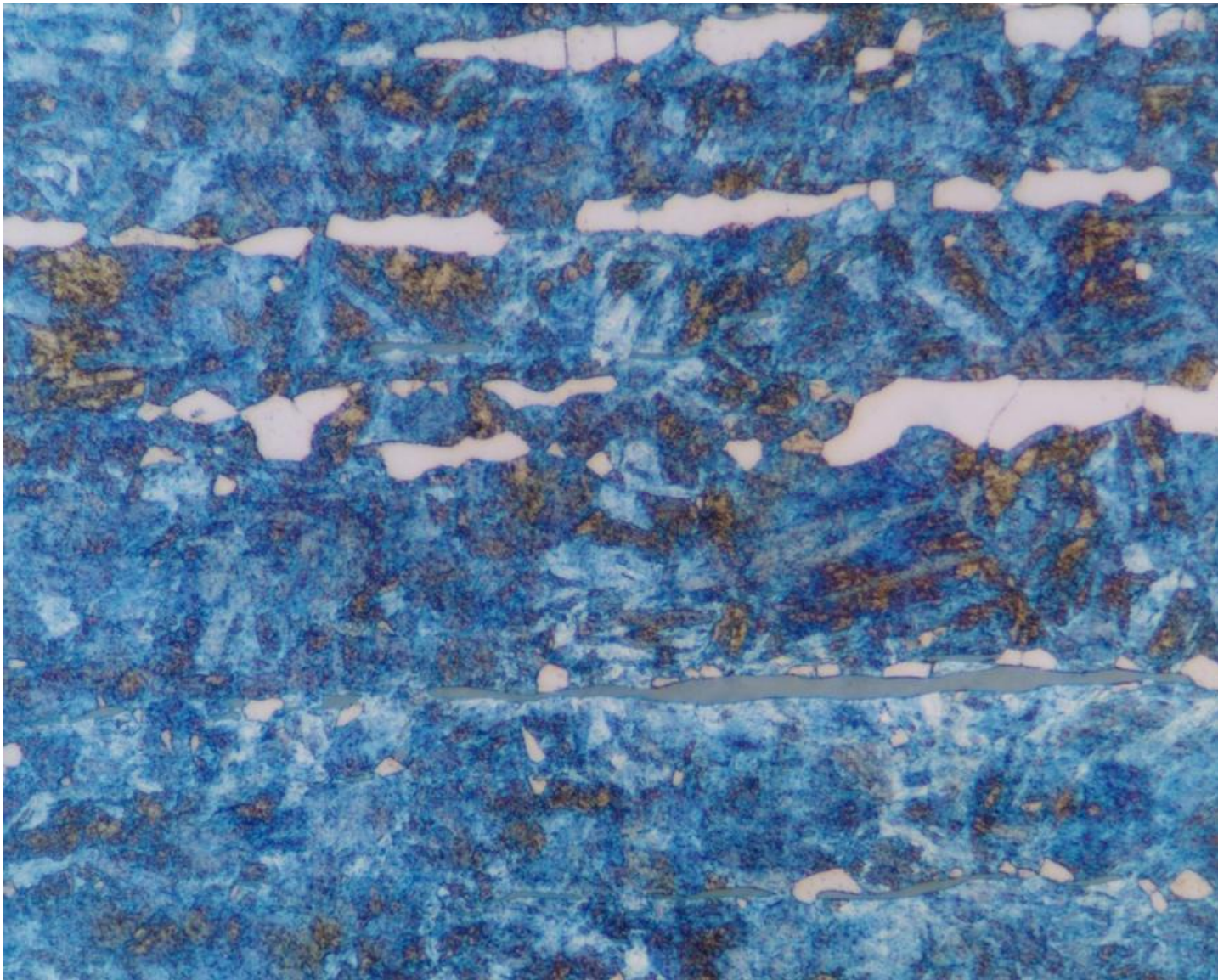
The body-centered cubic form of iron stable above 1394 °C (2541 °F) to the melting point, 1538 °C (2800 °F).

DELTA FERRITE

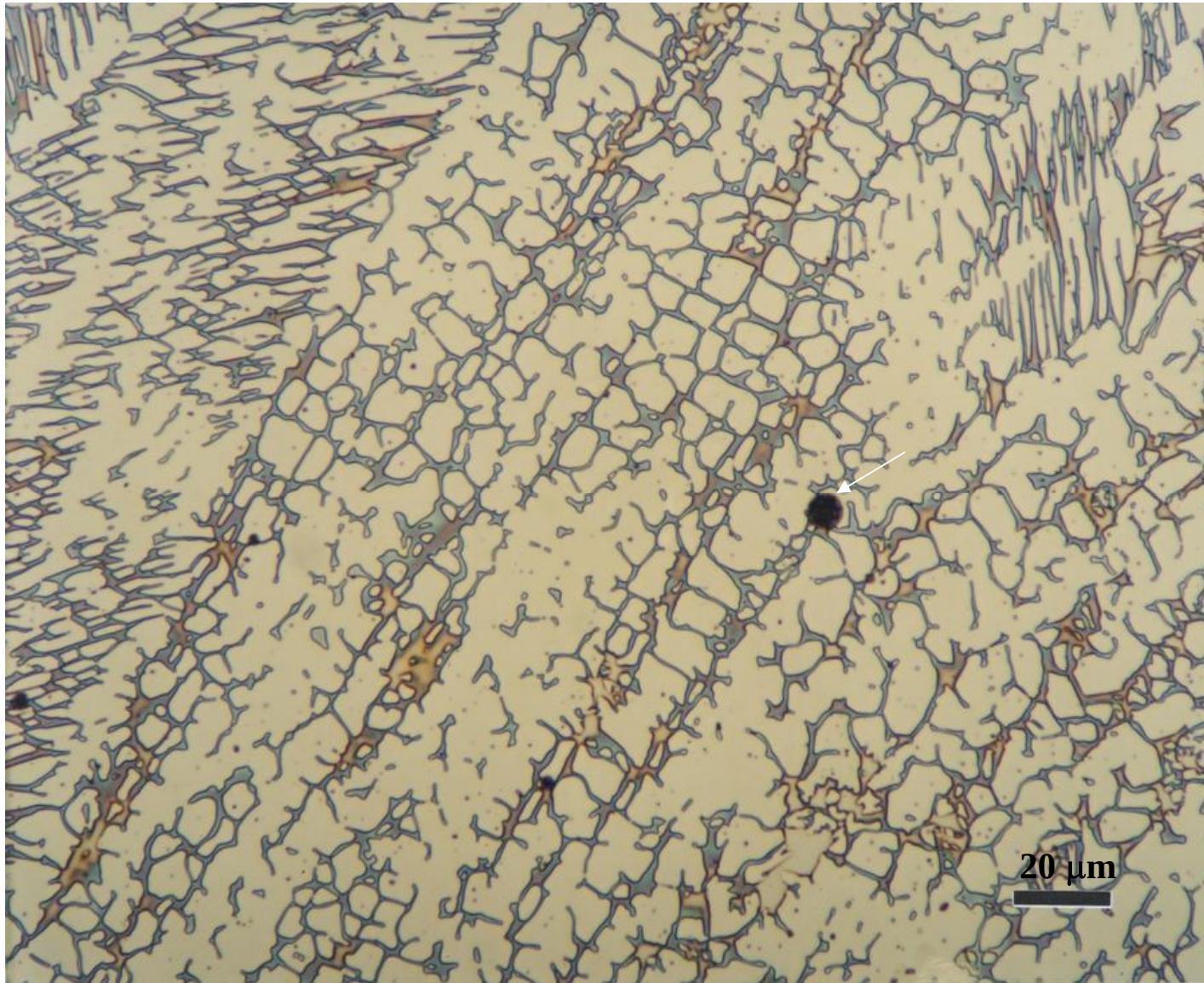
The high temperature stable form of alloyed body-centered cubic iron that may be present at room temperature in tool steels and martensitic stainless steels



Delta ferrite stringers in AM 350 precipitation hardenable stainless steel (Fe – 0.09% C – 0.8% Mn – 16.5% Cr – 4.5% Ni – 2.85% Mo – 0.1% N) revealed by electrolytic etching with aqueous 20% NaOH (3 V dc, 10 sec.). Longitudinal direction is horizontal. Martensite matrix is unetched.



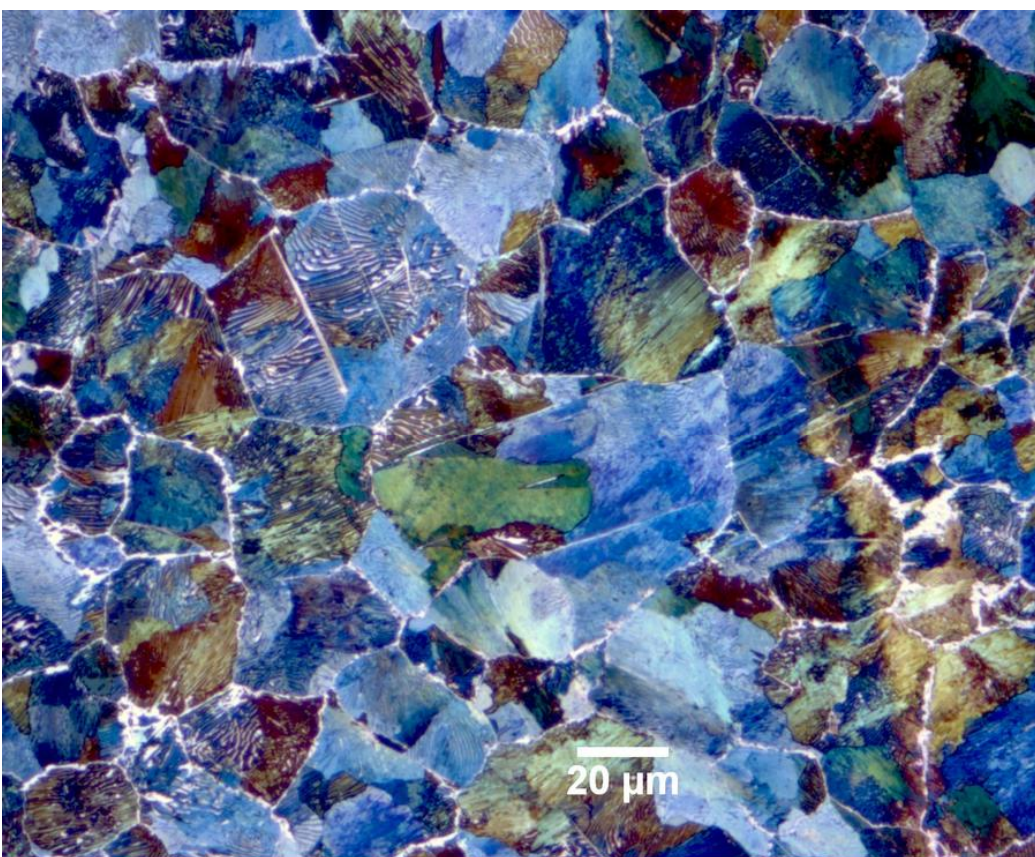
Microstructure of wrought, Project 70 type 416 martensitic stainless steel (Fe - $<0.15\%$ C, 0.35% S – 13% Cr) consisting of manganese sulfides (gray), delta ferrite (white) and tempered martensite (blue and brown) after tint etching with Beraha's CdS reagent and viewing with polarized light plus sensitive tint. Original at 500X.



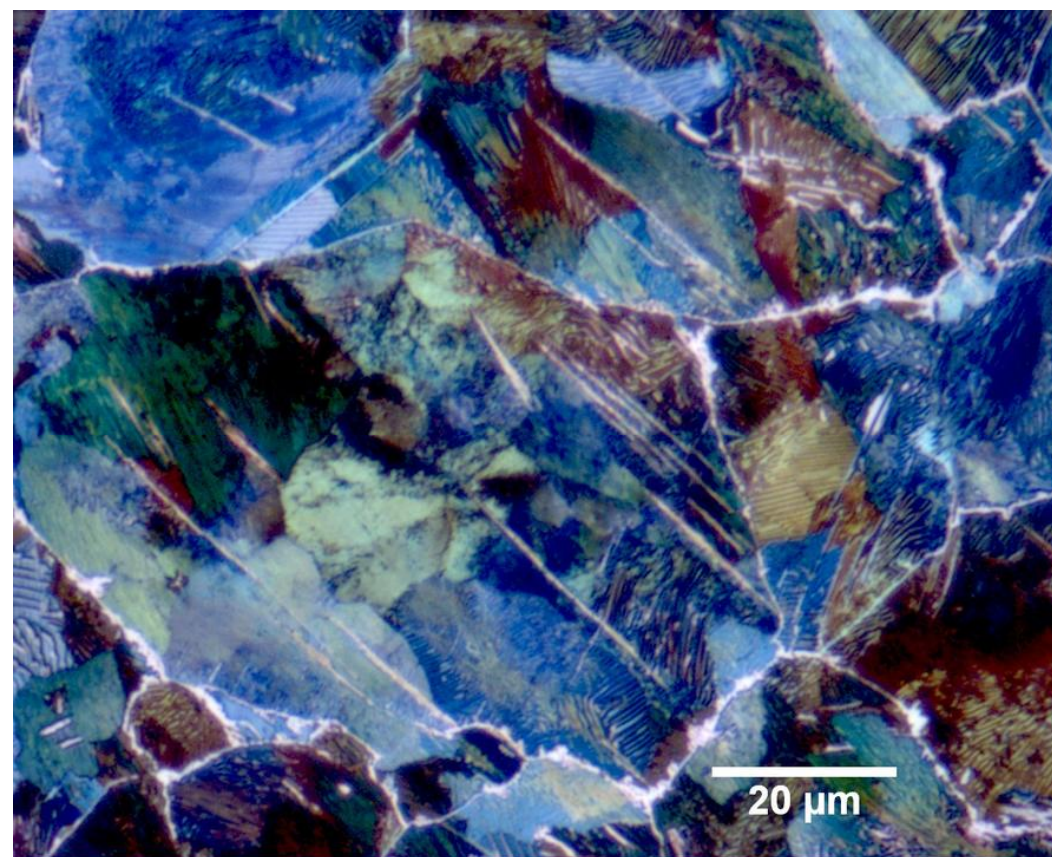
Delta ferrite colored blue and brown in an austenitic matrix in type 312 stainless steel weld metal (as-welded) using modified Murakami's reagent (30 g sodium hydroxide, 30 g potassium ferricyanide, 100 mL water, 100 °C – 10 s). The arrow points to a slag inclusion in the weld nugget.

CEMENTITE

A compound of iron and carbon, also called iron carbide, with the approximate formula Fe_3C and an orthorhombic crystal structure. Other elements, such as Mn and Cr, will substitute for Fe.



500X

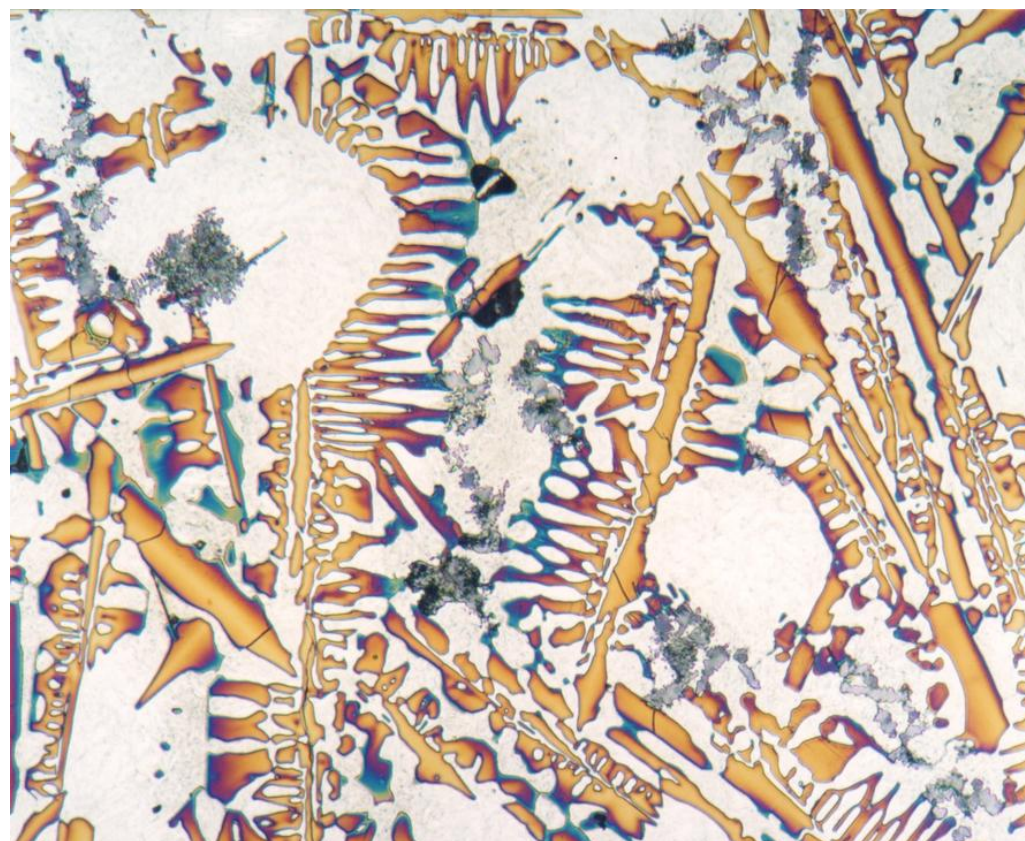
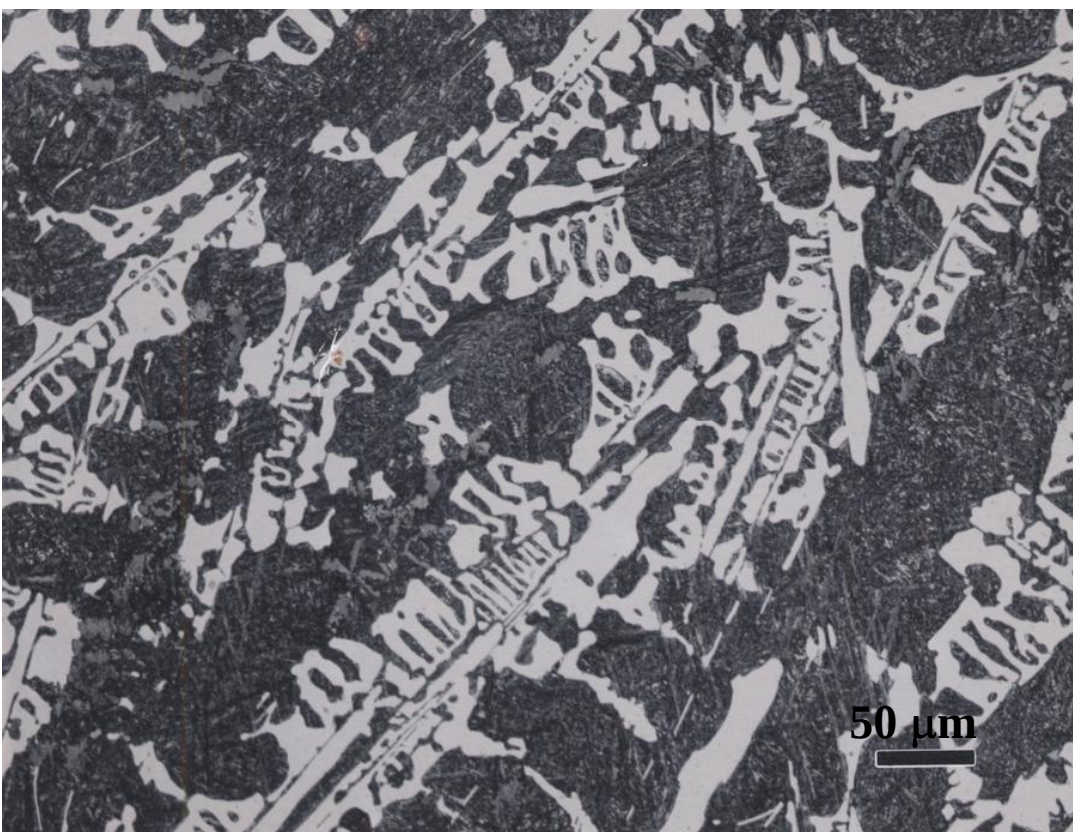


1000X

Intergranular and intragranular cementite (as well as cementite in the pearlite) are present in this hot-rolled specimen of W1 water-hardening carbon tool steel; etched with Klemm's I.



Cementite in white cast iron revealed by etching with 2% nital. The matrix is lamellar pearlite. Original at 1000X.



Proeutectic cementite in heat treated NiHard cast iron (Fe – 3.4% C – 0.9% Mn – 0.9% Si - 1.7% Cr – 4.5% Ni – 0.4% Mo). Etching with Vilella's reagent (left) reveals the cementite (white, unetched) and tempered martensite. Graphite is present but hard to see. Etching with boiling alkaline sodium picrate (right) colors the cementite and the gray graphite is easily seen against the unetched martensite.

EUTECTIC

An isothermal reversible reaction in which a liquid solution is converted into two, or more, intimately mixed solids upon cooling. The number of solid phases is the same as the number of components in the system.

An alloy of the composition of the eutectic point on an equilibrium phase diagram.

A structure formed by a eutectic reaction.

EUTECTOID

An isothermal reversible reaction in which a solid solution is converted into two, or more, intimately mixed solids upon cooling. The number of solid phases is the same as the number of components in the system.

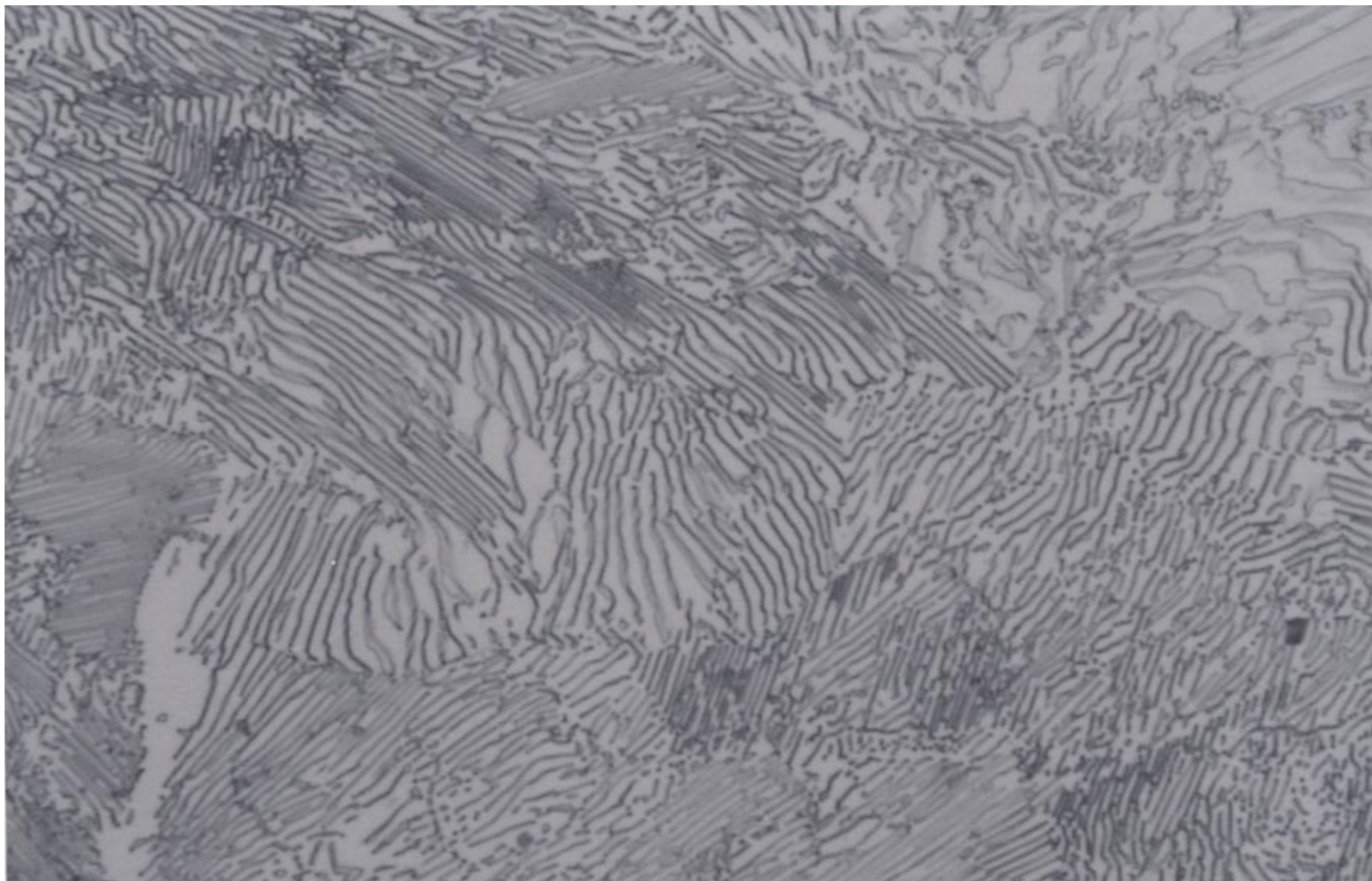
An alloy of the composition of the eutectoid point on an equilibrium phase diagram.

A structure formed by a eutectoid reaction.

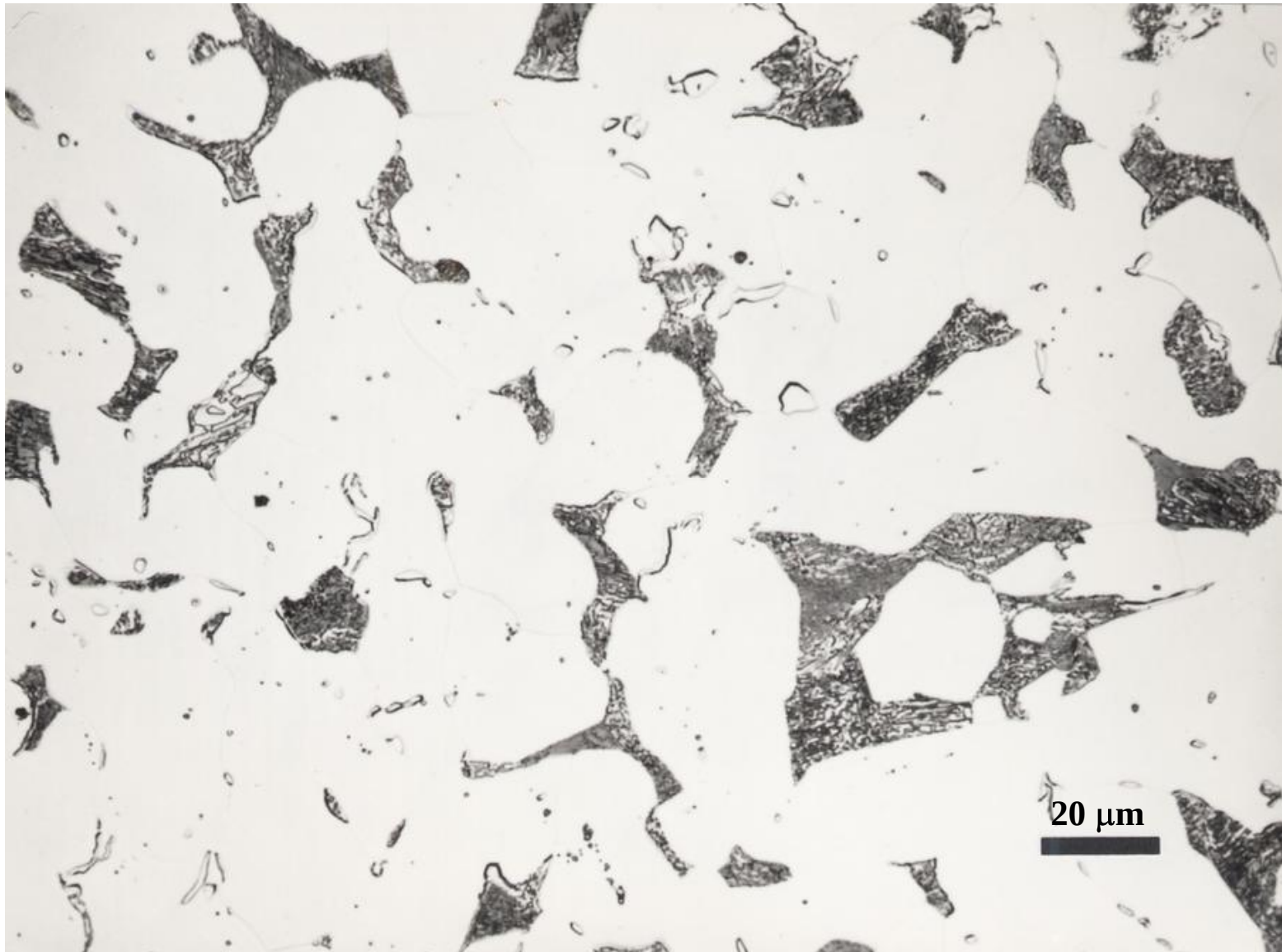
PEARLITE

A metastable lamellar aggregate of ferrite and cementite due to transformation of austenite above the bainite region.

The eutectoidal transformation product in steels.



Coarse pearlitic structure in isothermally annealed (780 °C, 1436 °F – 1 h, isothermally transformed) 1080 steel (Fe – 0.8% C – 0.75% Mn) etched with 4% picral. All of the lamellae are resolvable. Original at 1000X.



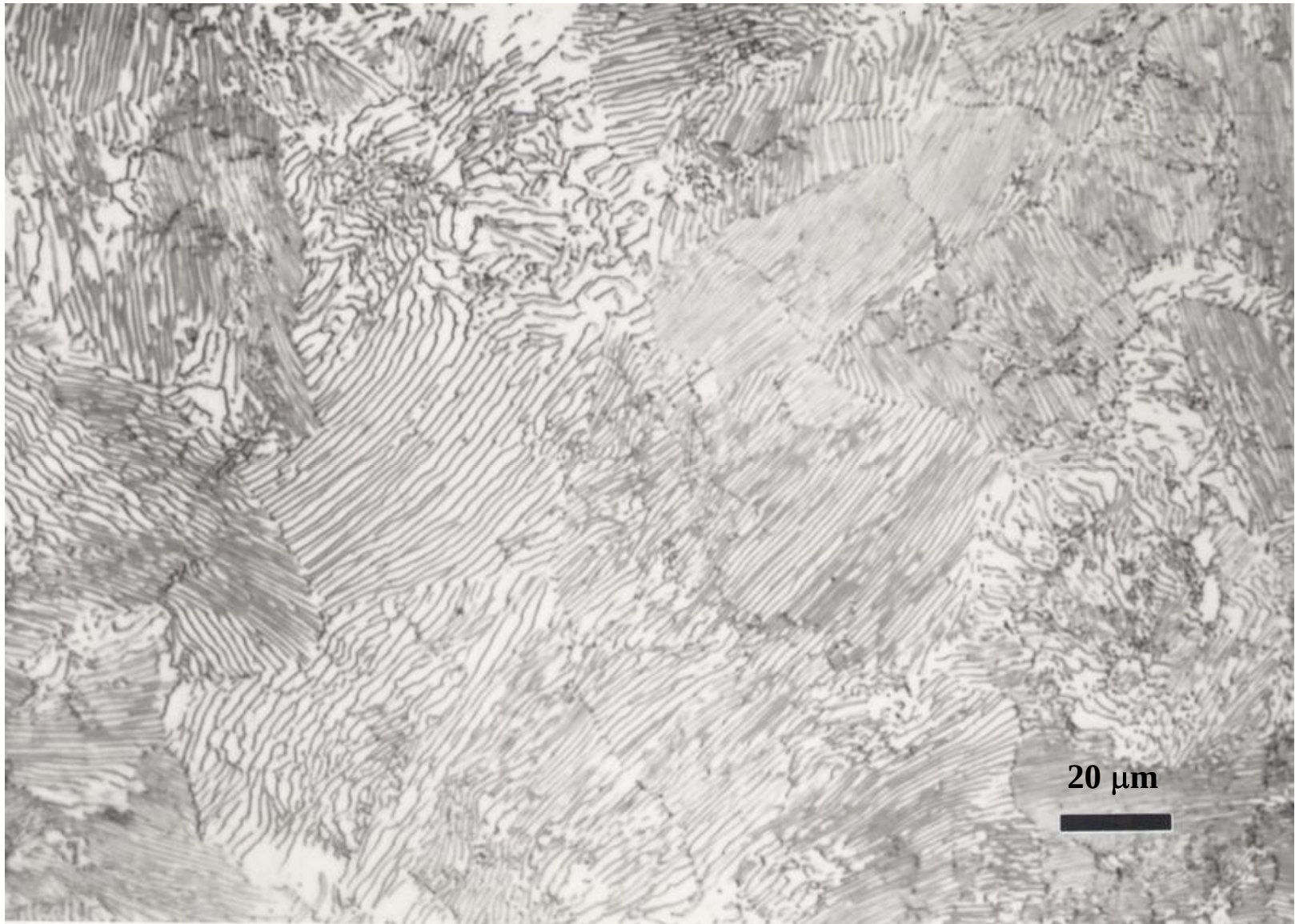
Ferrite (white) and pearlite in a hot-rolled Fe – 0.2% C binary alloy. Picral etch.



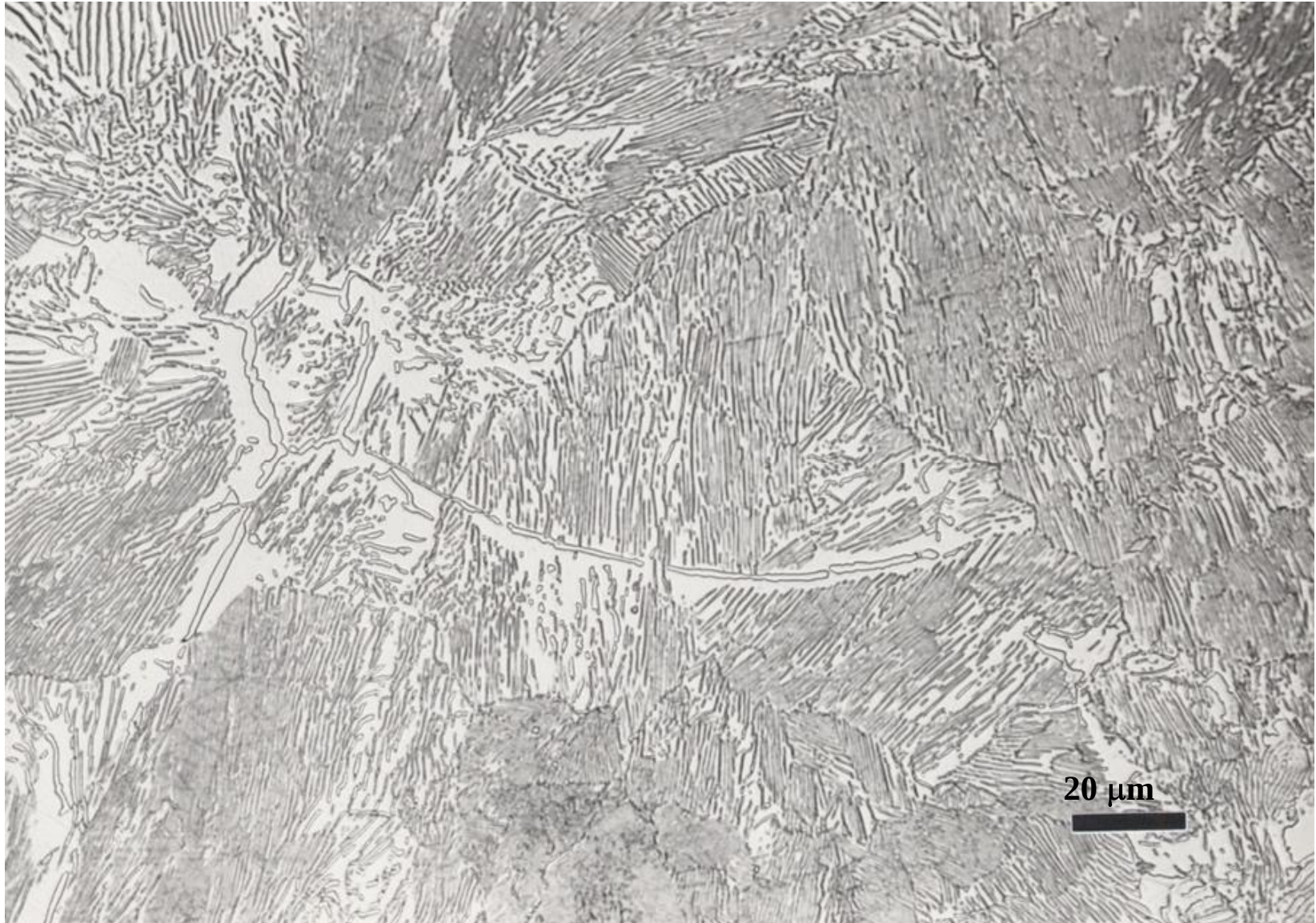
Ferrite (white) and pearlite in a hot-rolled Fe – 0.4% C binary alloy. Picral etch.



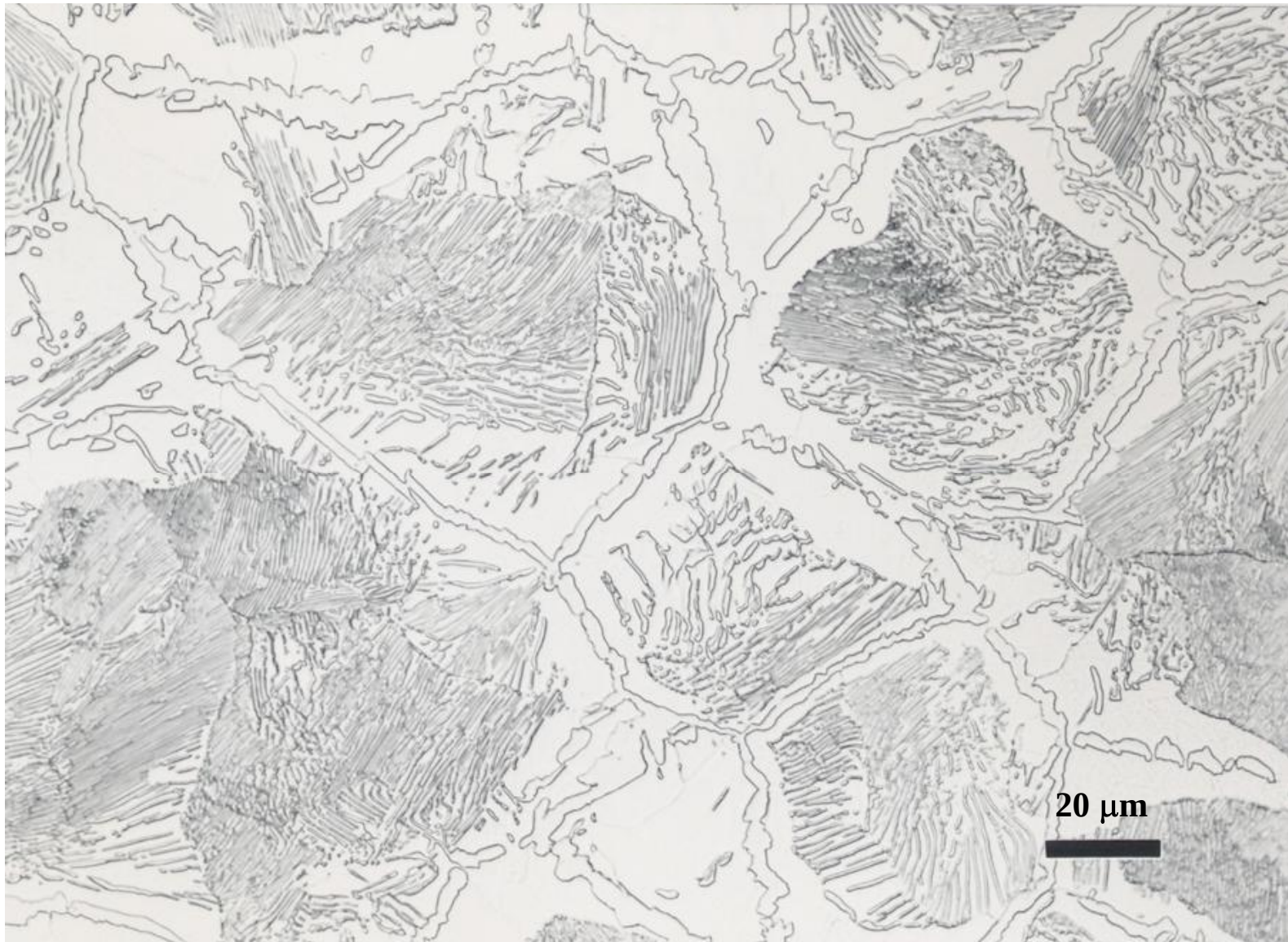
Ferrite (white) and pearlite in a hot-rolled Fe – 0.6% C binary alloy. Picral etch.



Coarse lamellar pearlite in a hot-rolled Fe – 0.8% C binary alloy. Picral etch.



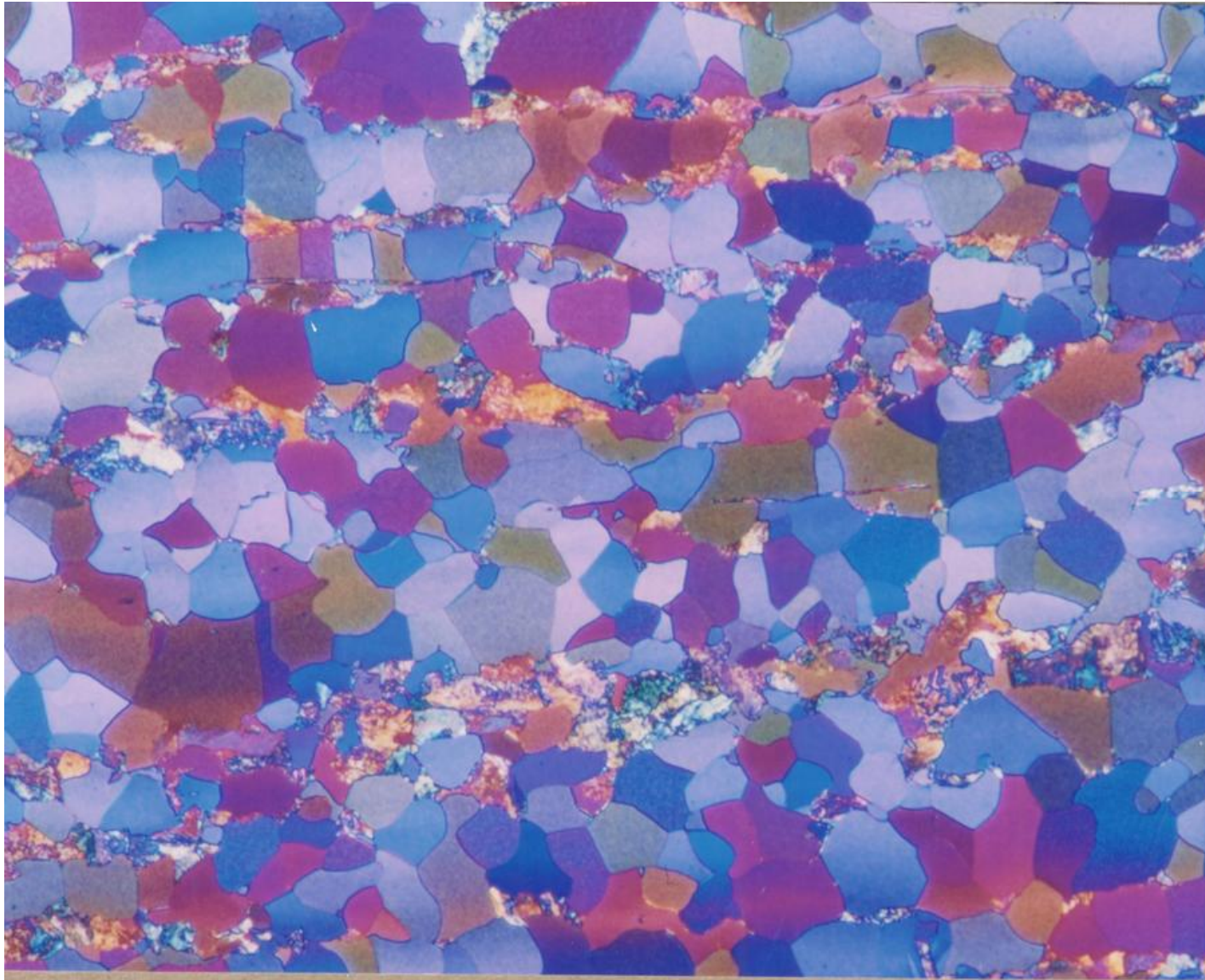
**Intergranular proeutectoid cementite and pearlite in a hot-rolled Fe – 1.0% C binary alloy.
Picral etch.**



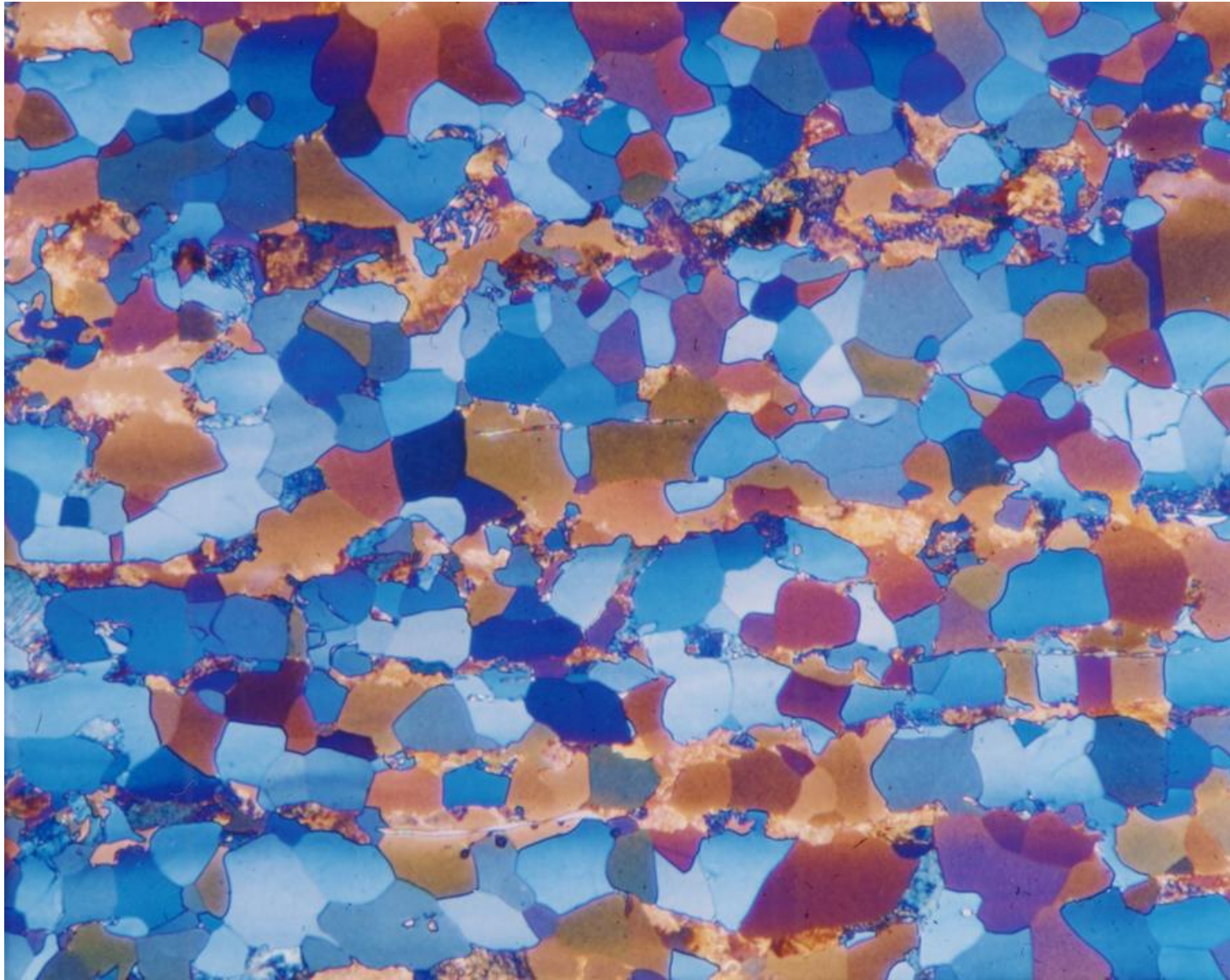
Coarse intergranular proeutectoid cementite (white, outlined) and pearlite in a hot-rolled Fe – 1.2% C binary alloy. Picral etch.



Coarse intergranular and intragranular proeutectoid cementite (dark) and pearlite in a hot-rolled Fe – 1.4% C binary alloy. Alkaline sodium picrate (boiling) was used to color the cementite. The cementite in the pearlite (within the grains) is not revealed as well.

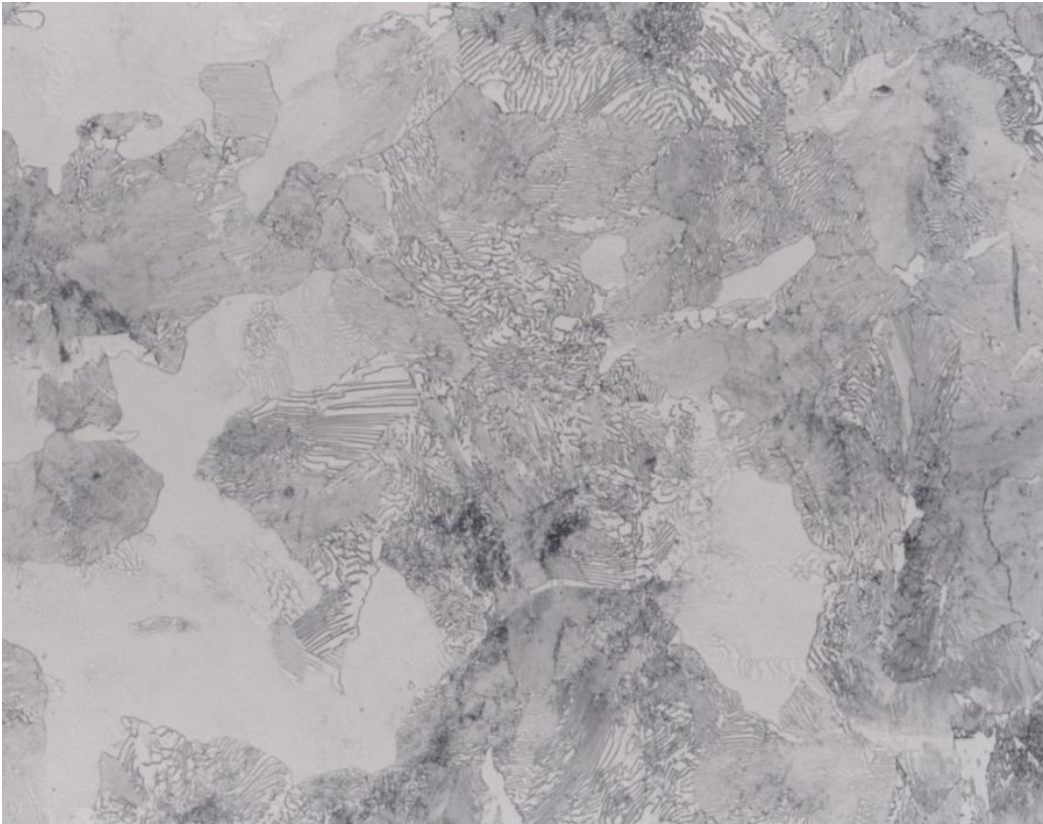


Ferrite-pearlite microstructure of a wrought carbon steel (Fe – 0.11% C – 0.85% Mn – 0.21% Si) etched with Klemm's I reagent and viewed with polarized light and sensitive tint. Klemm's colors ferrite. Original at 500X.

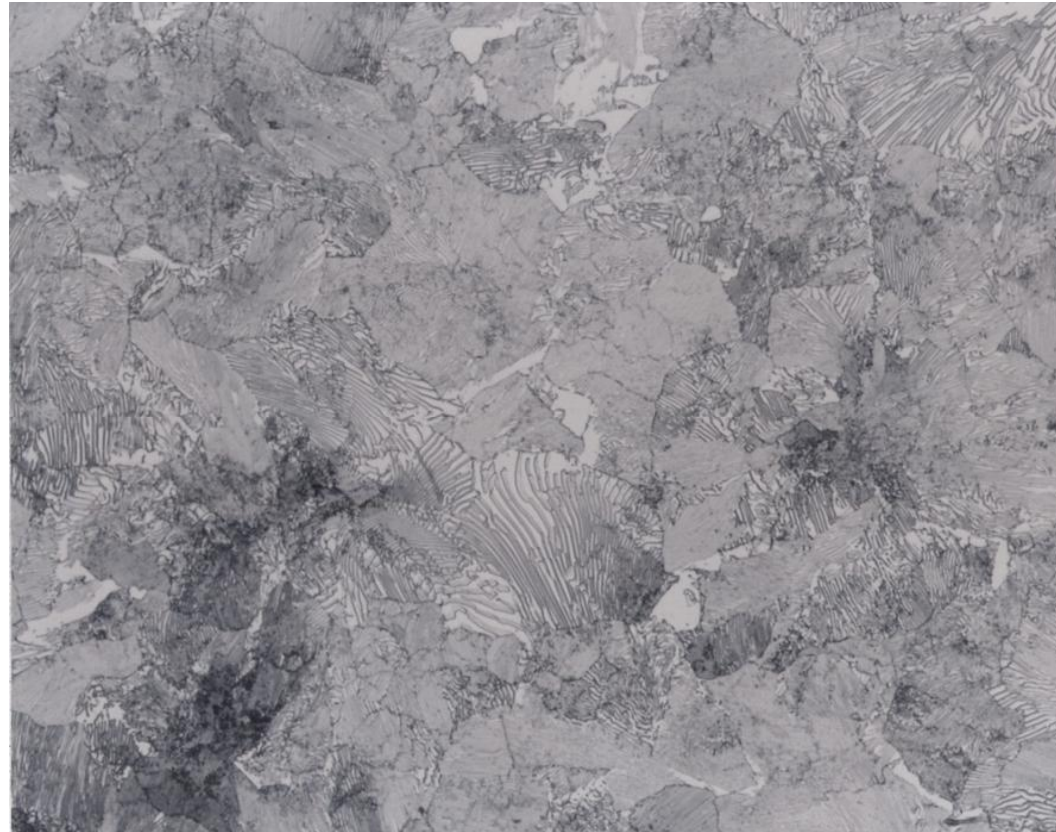


Ferrite-pearlite microstructure of a wrought carbon steel (Fe – 0.23% C – 0.85% Mn – 0.22% Si) etched with Klemm's I reagent and viewed with polarized light and sensitive tint. Original at 500X.

2% Nital



4% Picral



Microstructure of eutectoid Fe – 0.80% C – 0.21% Mn – 0.22% Si carbon steel in the as-rolled condition after etching with nital and picral. Note that nital etches pearlite non-uniformly as there are several lightly etched patches of pearlite that could be misidentified as as-quenched martensite.

Picral etches pearlite uniformly. Original images at 500X magnification.

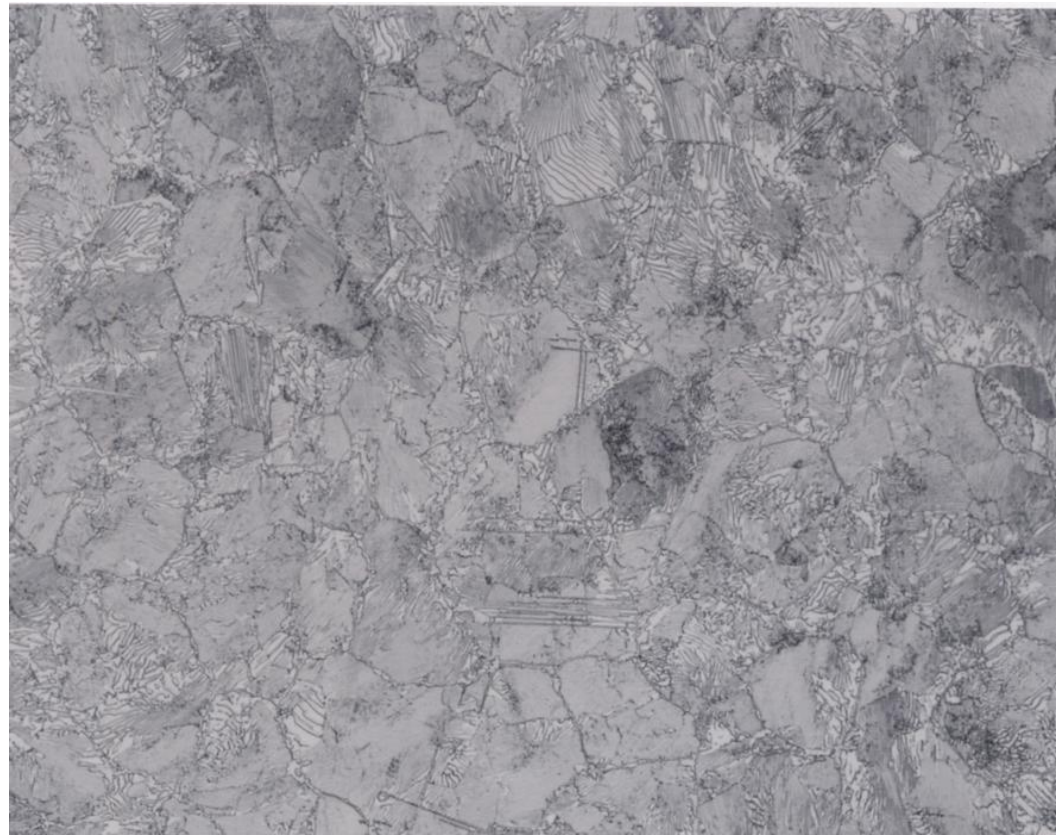


Fully pearlitic microstructure of eutectoid carbon steel (Fe – 0.80% C – 0.21% Mn – 0.22% Si) revealed using Klemm's I (left) and Beraha's Sulfamic acid (100 mL water, 3 g potassium metabisulfite, 1 g sulfamic acid) etch (right) with viewing under polarized light and sensitive tint. Originals at 500X.

2% Nital



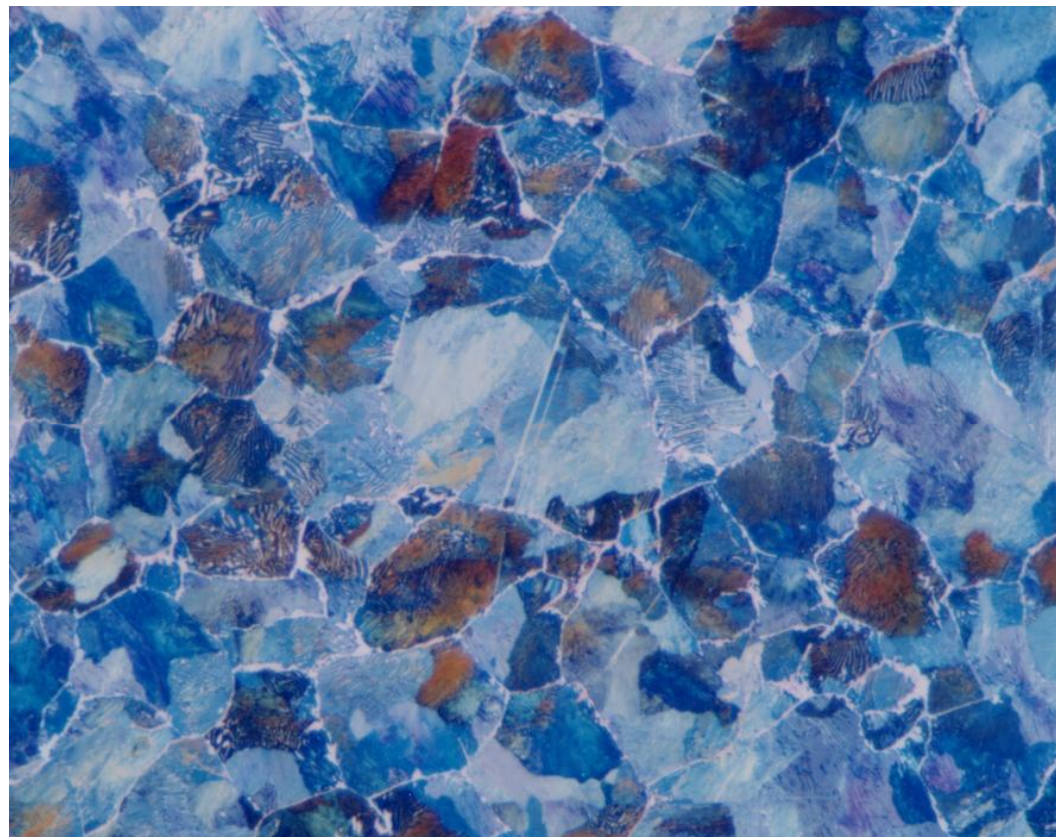
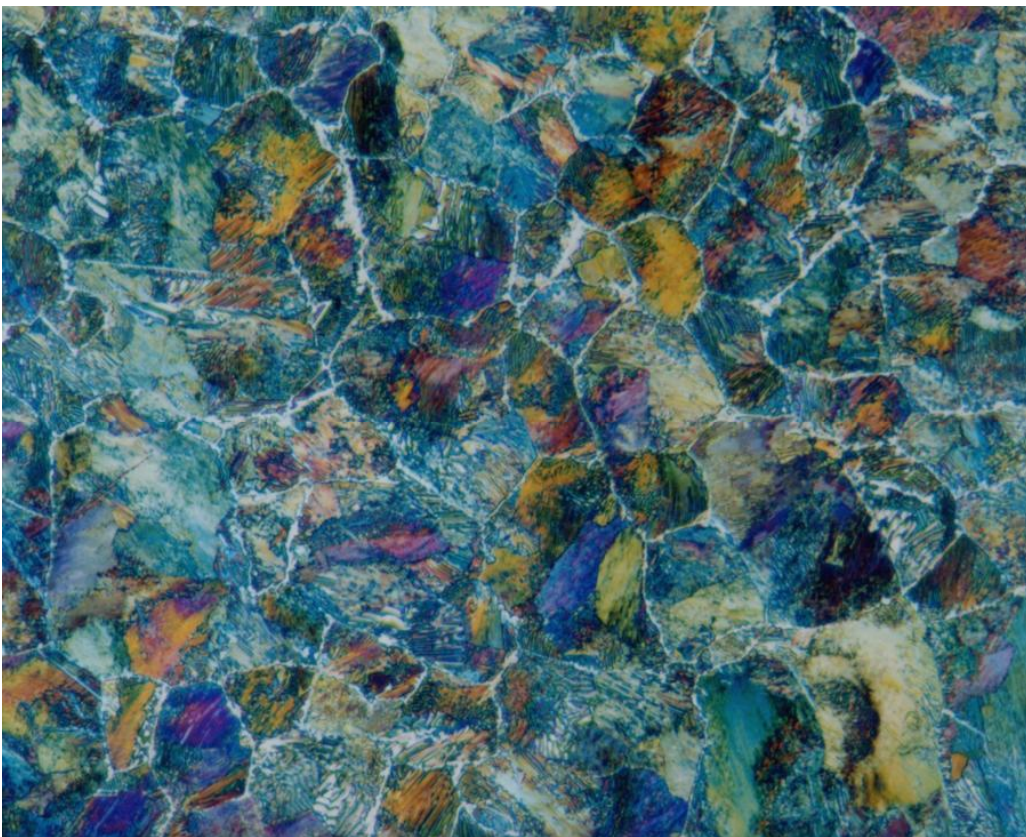
4% Picral



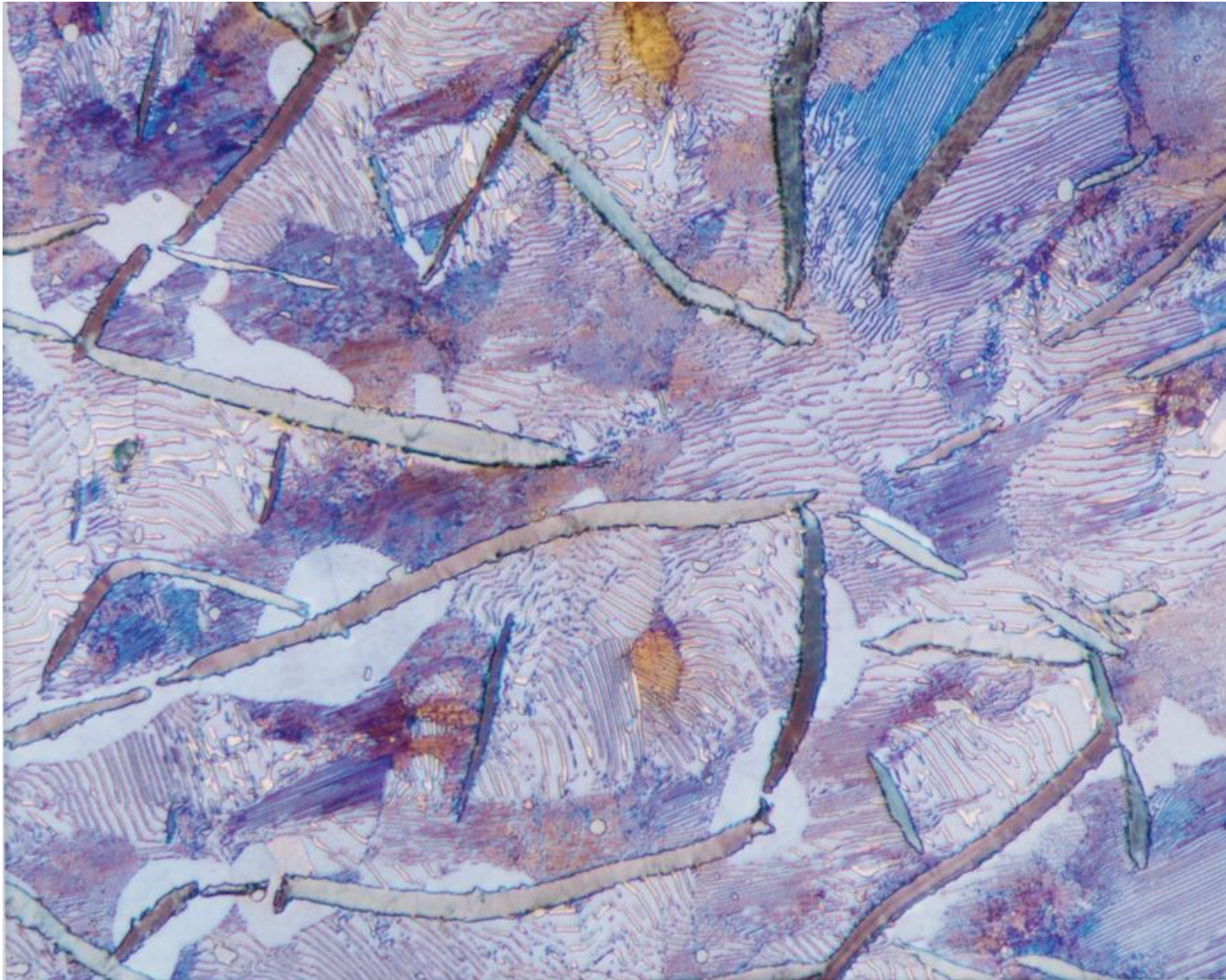
Microstructure of as-rolled Fe – 1.31% C – 0.35% Mn – 0.25% Si hypereutectoid carbon steel.
Although there is a well-developed intergranular cementite grain boundary network in this specimen, neither nital, nor picral, reveal it clearly. The matrix is fine pearlite. Note that nital etches pearlite non-uniformly in that some patches display little detail and could be mistaken for as-quenched martensite. Originals at 500X magnification. Compare with the next two slides.



Microstructure of the as-rolled Fe – 1.31% C – 0.35% Mn – 0.25% Si specimen with the intergranular carbide network clearly visible after etching with alkaline sodium picrate, 90 °C – 60 s. Original at 500X magnification. Note also some intragranular Widmanstätten cementite.



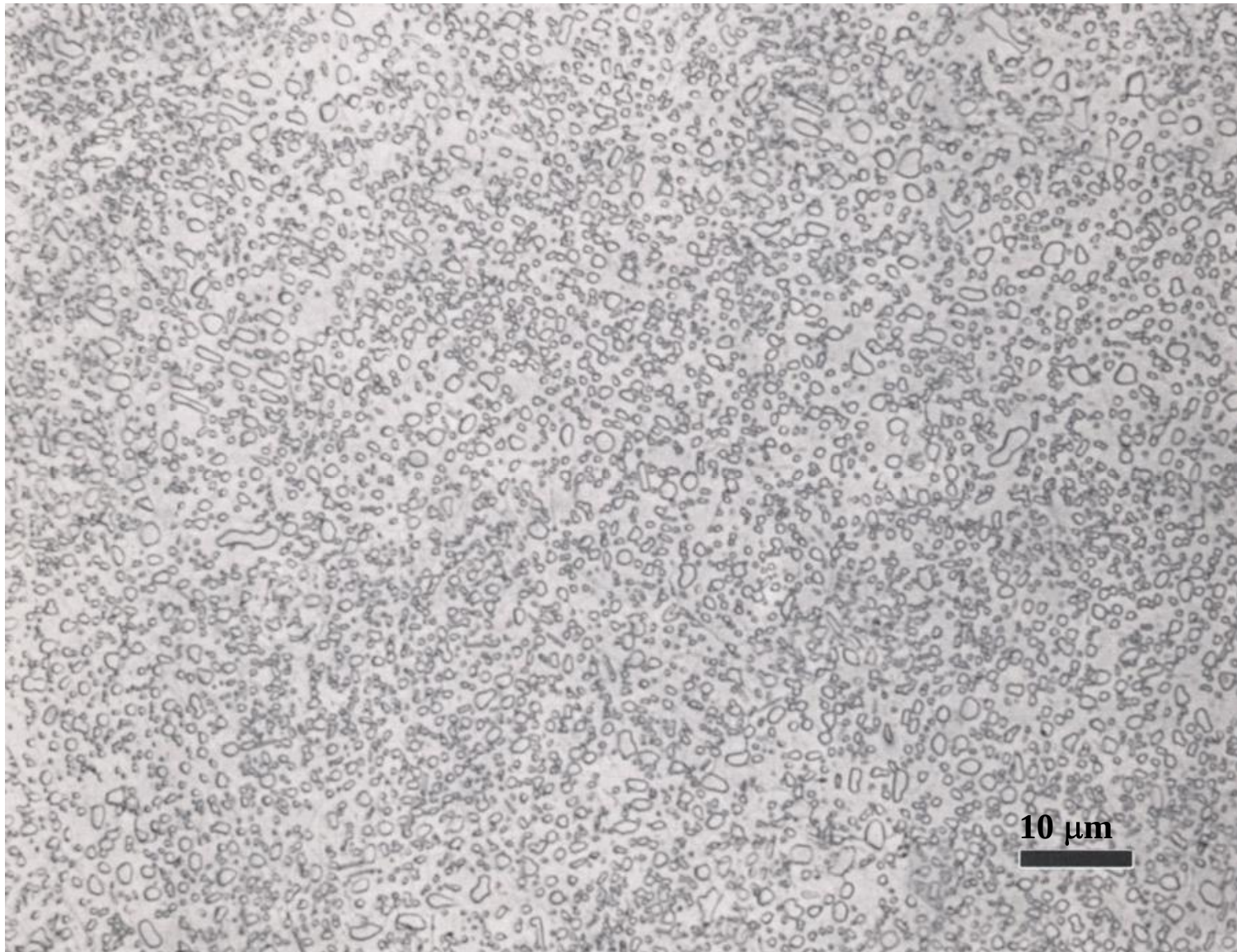
Color etching of the as-rolled hypereutectoid Fe-1.31% C – 0.35% Mn – 0.25% Si specimen clearly revealed the intergranular cementite films. Beraha's sulfamic acid etch (100 mL water, 3 g $K_2S_2O_5$ and 2 g NH_2SO_3H) (left) and Klemm's I reagent (right) were used. Original magnifications were 500X. Taken with polarized light and sensitive tint.



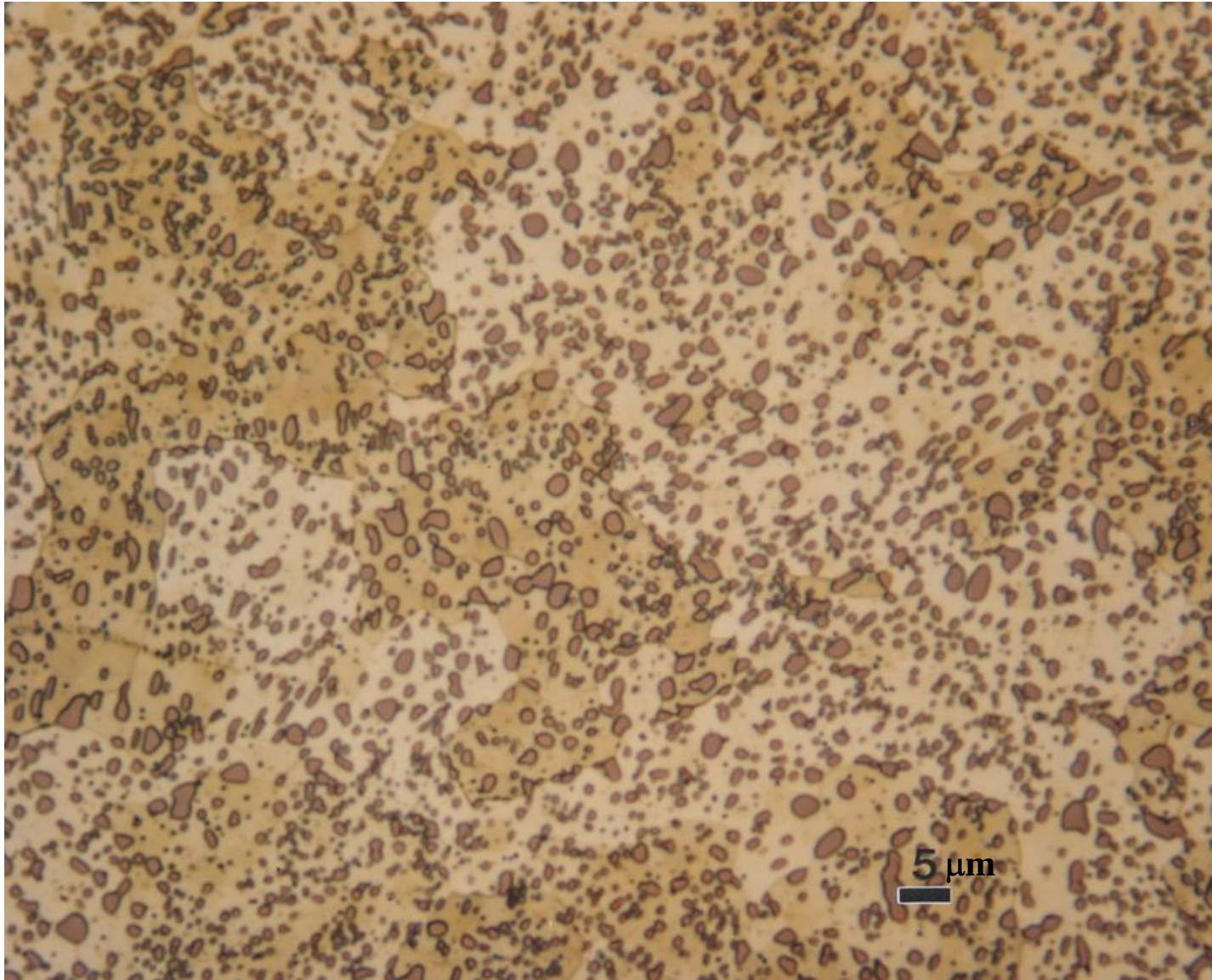
**Microstructure of pearlitic gray cast iron tint etched with Beraha's CdS reagent.
Original at 500X.**

SPHEROIDITE

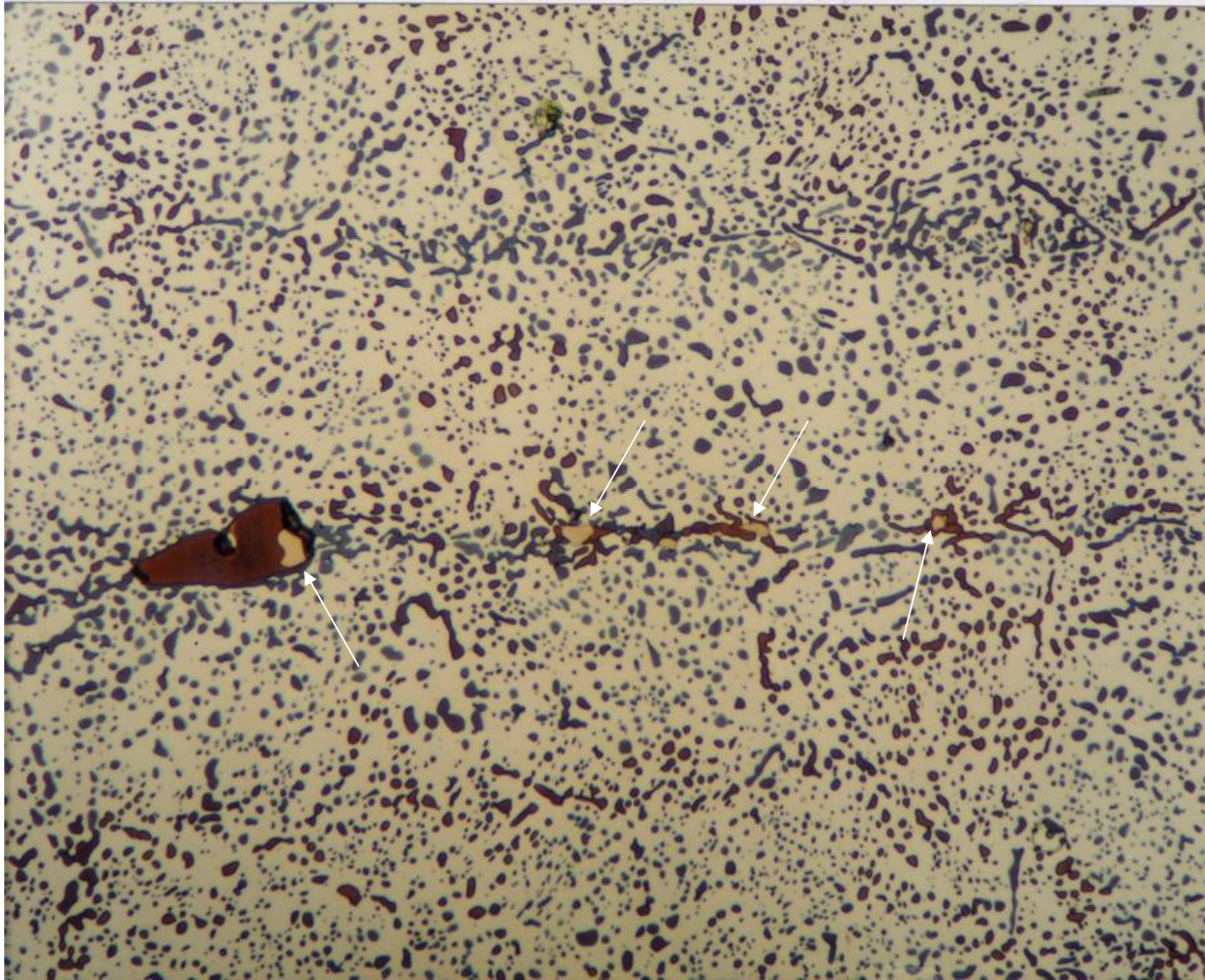
An aggregate of cementite or alloy carbides of essentially spherical shape dispersed in a ferritic matrix.



Spheroidized cementite in a ferritic matrix in annealed type 52100 (Fe – 1% C – 1.5% Cr) bearing steel. Etched with 4% picral.



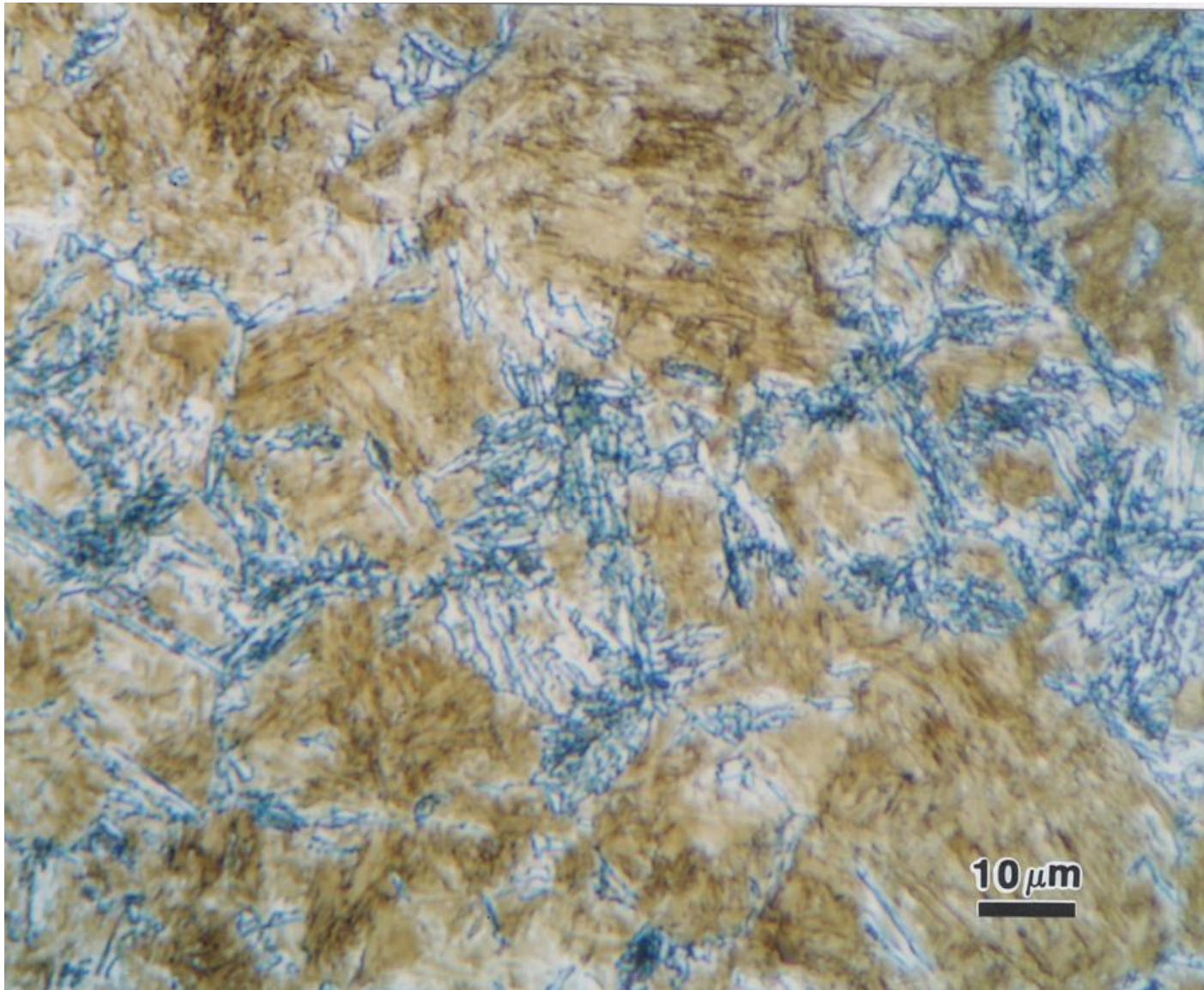
Spheroidize annealed microstructure of type W1 carbon tool steel (Fe - 1.05% C – 0.25% Mn – 0.2% Si) etched with Beraha's sodium molybdate reagent which colored both the cementite particles (brownish red) and the ferrite matrix.



**Spheroidize annealed microstructure of wrought type O6 graphitic tool steel (Fe – 1.45% C – 0.85% Mn – 1.0% Si – 0.25% Mo) after coloring the cementite with alkaline sodium picrate (80 °C). There is some variation in carbide composition, and perhaps type, in the segregate stringer (arrows).
Longitudinal plane shown; original at 500X.**

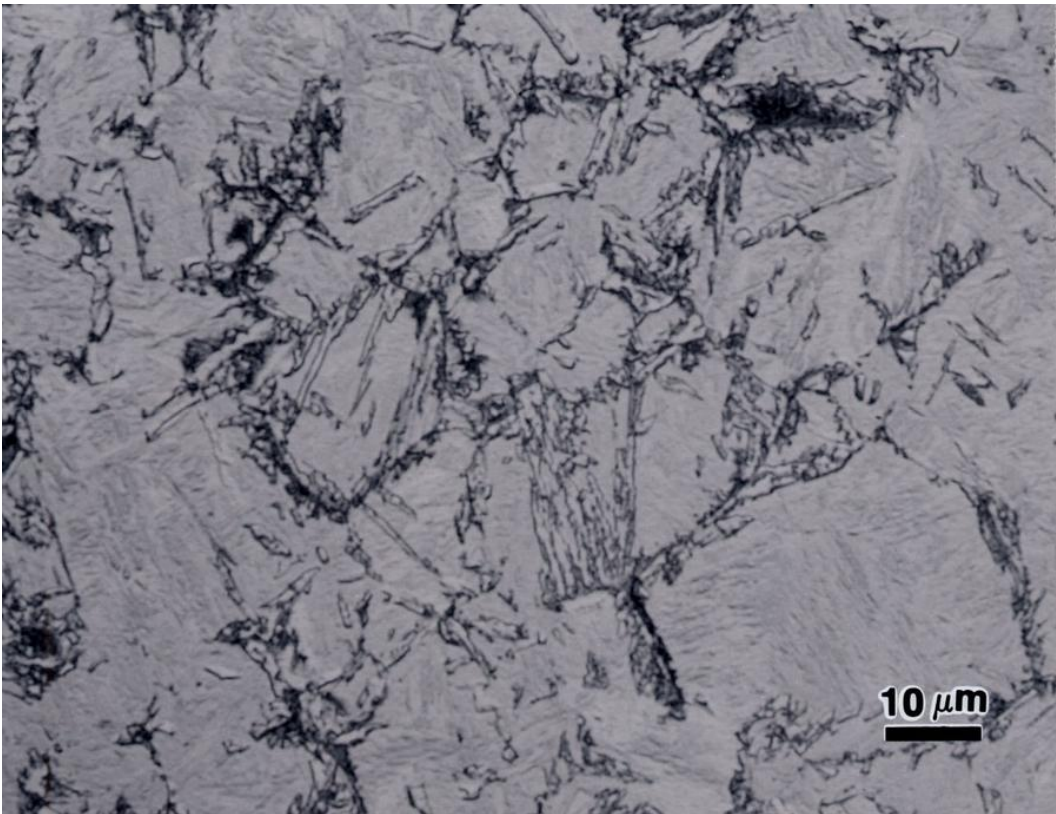
BAINITE

A metastable aggregate of ferrite and cementite from austenite transformation at temperatures below the pearlite range and above the martensite start, M_s , temperature. For high-carbon steels, bainite appears feathery in the upper range and acicular in the lower range.

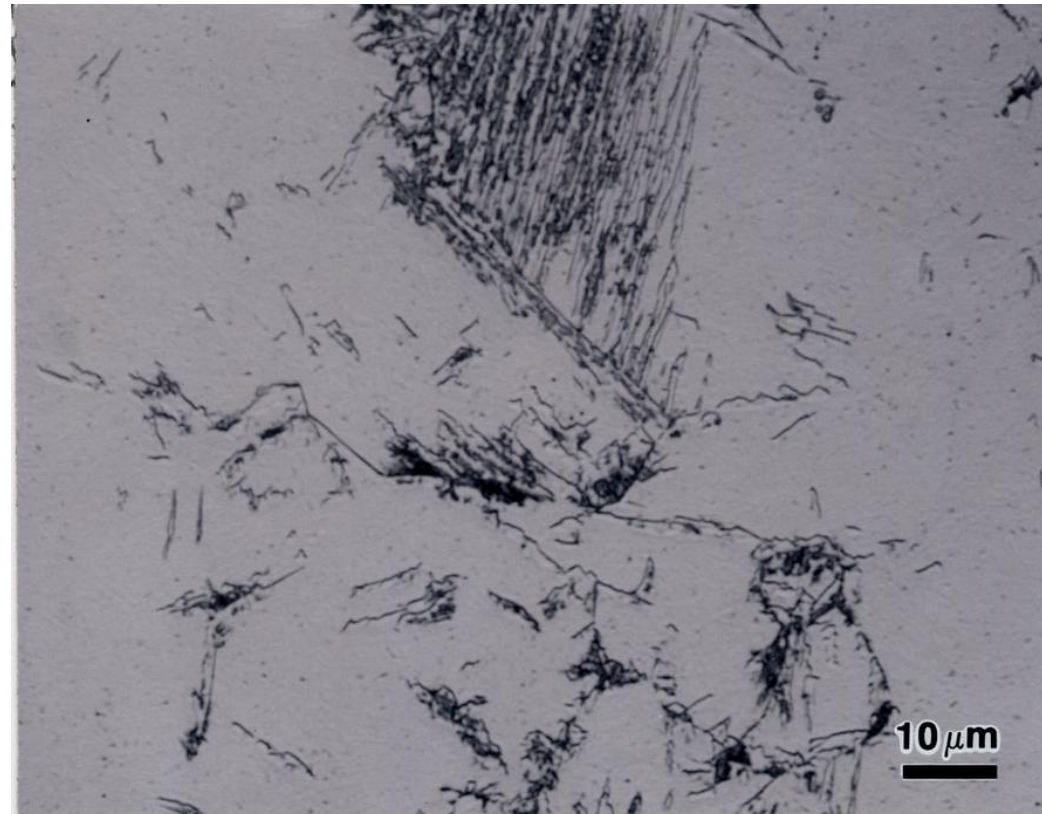


Upper bainite (blue and white) and as-quenched martensite (brown) in 5160 alloy steel (Fe – 0.6% C – 0.85% Mn – 0.25% Si – 0.8% Cr) that was austenitized at 830 °C (1525 °F) for 30 min., isothermally held at 538 °C (1000 F°) for 30 sec to partially transform the austenite, and then water quenched (untransformed austenite forms martensite). Etched with aqueous 10% Na₂S₂O₅.

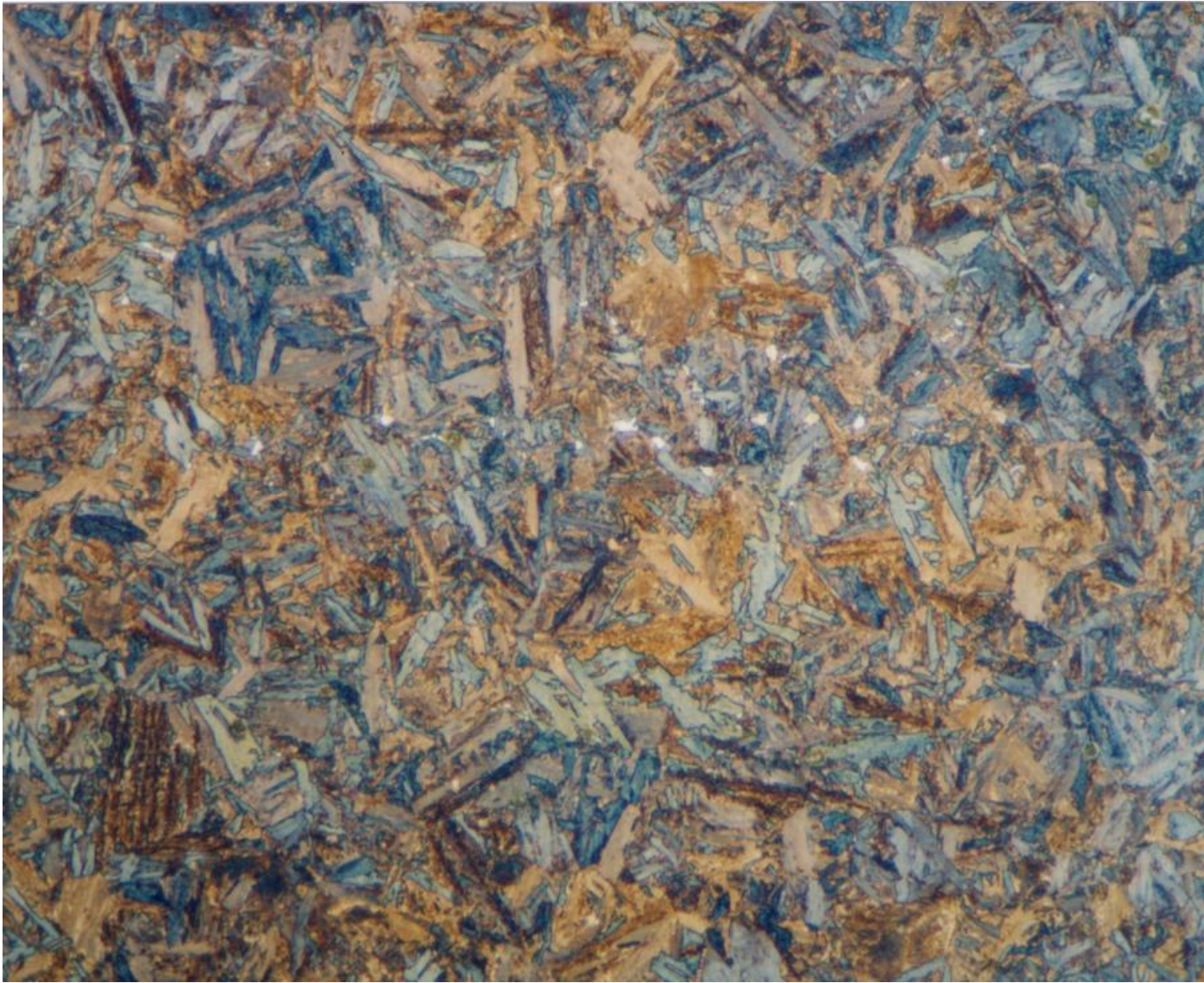
2% Nital



4% Picral

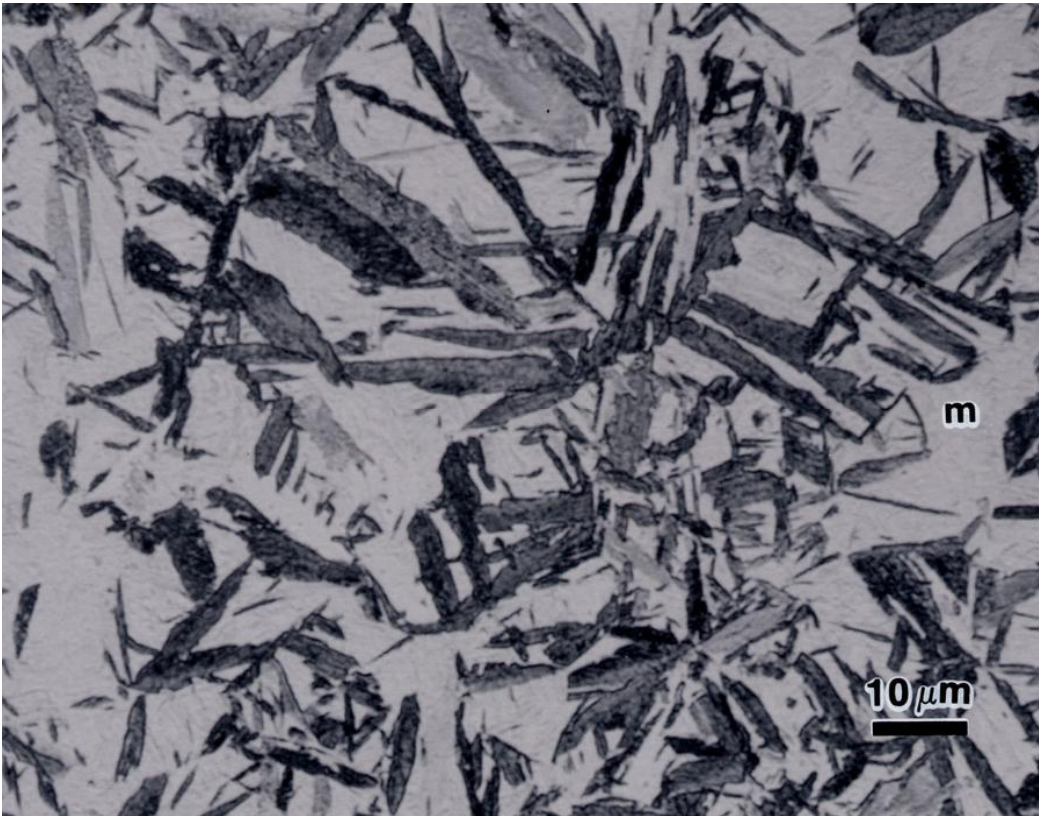


Upper bainite (dark or outlined) and as-quenched martensite (gray or white) in 5160 alloy steel (Fe – 0.6% C - 0.85% Mn – 0.25% Si – 0.8% Cr) that was austenitized at 830 °C (1525 °F) for 30 min., isothermally held at 538 °C (1000 F°) for 30 sec to partially transform the austenite, and then water quenched (untransformed austenite forms martensite).

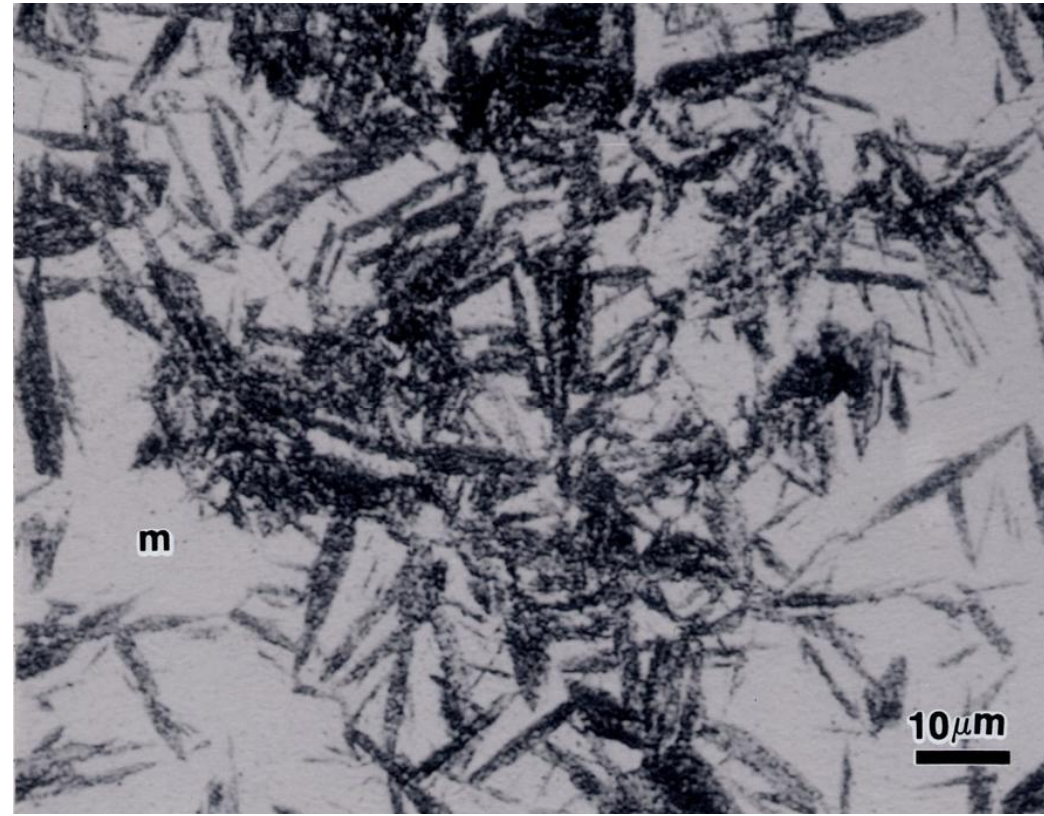


Lower bainite (blue and gray), undissolved cementite (white) and as-quenched martensite (brown) in 5160 alloy steel (Fe – 0.6% C - 0.85% Mn – 0.25% Si – 0.8% Cr) that was austenitized at 830 °C (1525 °F) for 30 min., isothermally held at 343 °C (650 F°) for 20 minutes to partially transform the austenite, and then water quenched (untransformed austenite forms martensite). Etched with aqueous 10% Na₂S₂O₃ and 3% K₂S₂O₅, 500x.

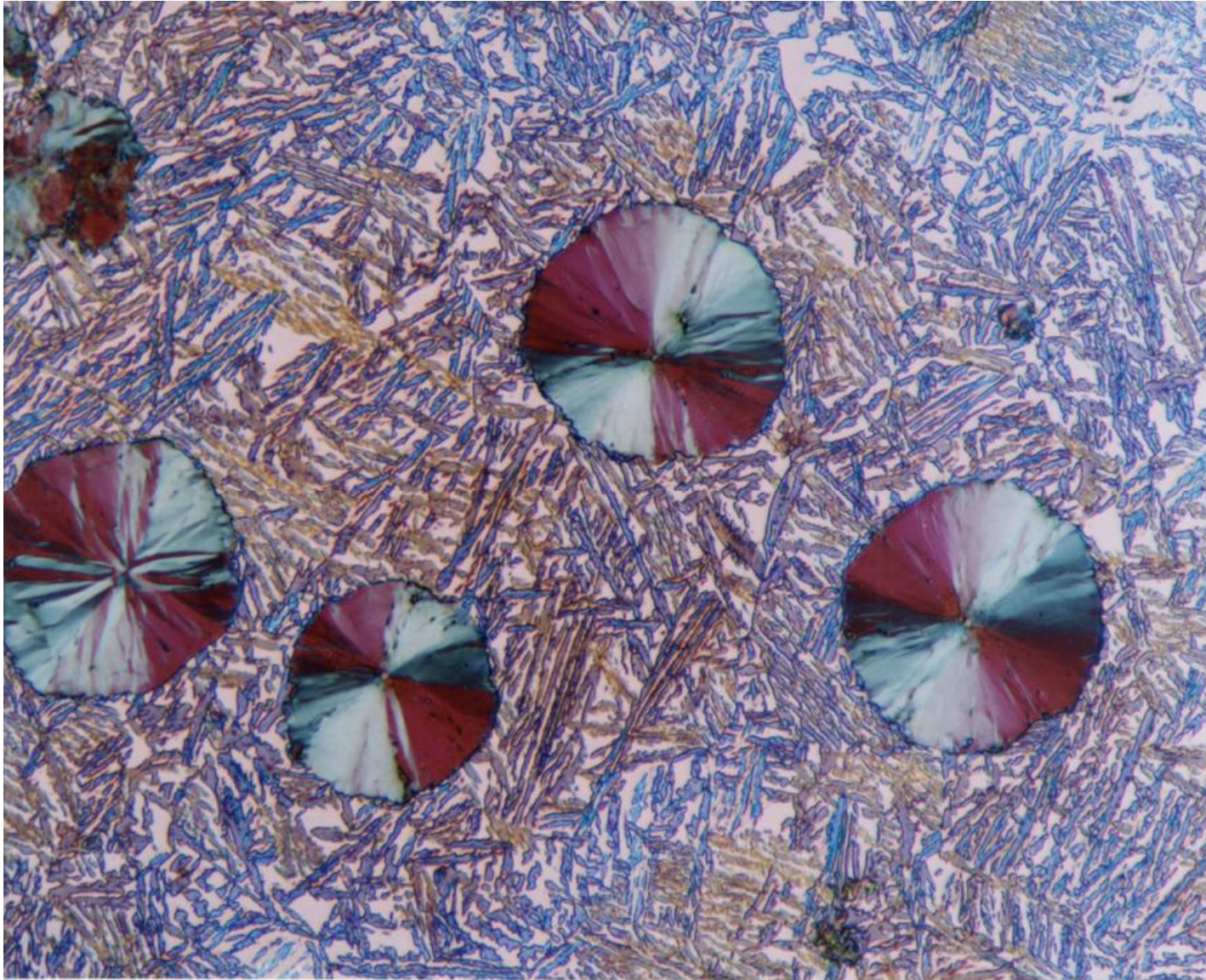
2% Nital



4% Picral



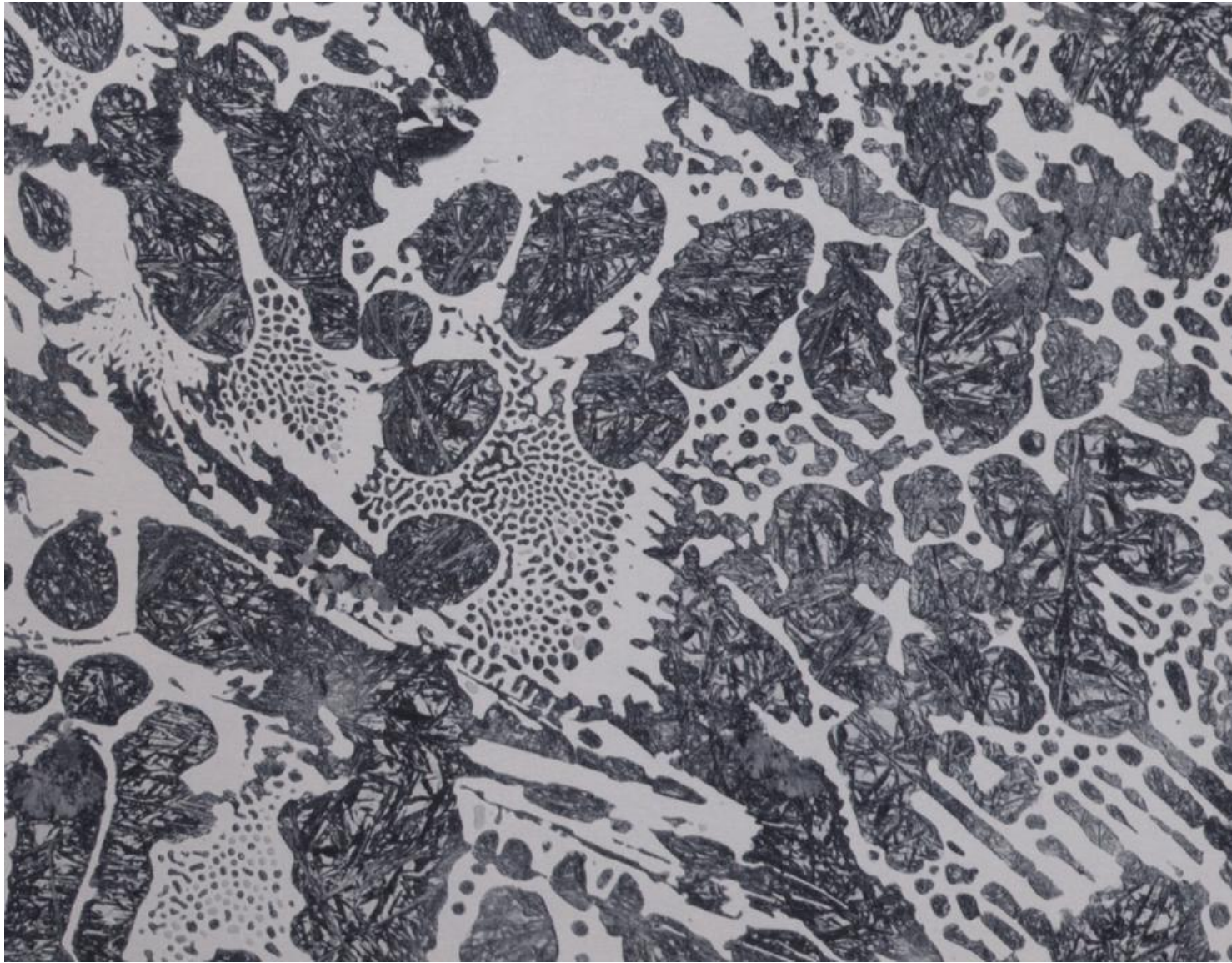
Lower bainite (dark) and as-quenched martensite (white/gray) in 5160 alloy steel (Fe – 0.6% C - 0.85% Mn – 0.25% Si – 0.8% Cr) that was austenitized at 830 °C (1525 °F) for 30 min., isothermally held at 343 °C (650 F°) for 20 minutes to partially transform the austenite, and then water quenched (untransformed austenite forms martensite).



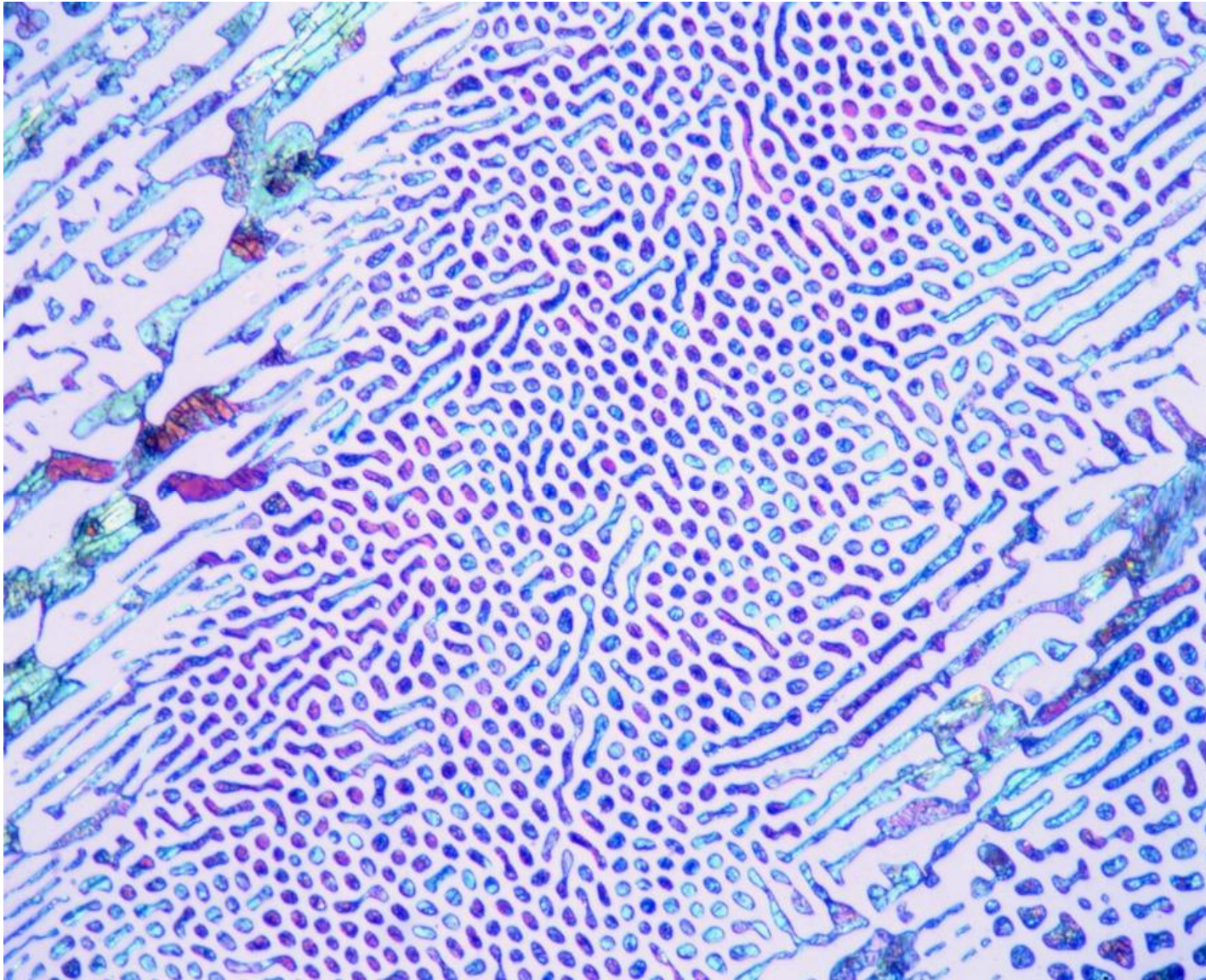
Microstructure of austempered ductile iron tint etched with Beraha's CdS reagent containing large graphite nodules (arrow), ausferrite (blue and brown) and retained austenite (white) when viewed with polarized light plus sensitive tint. Original at 500X.

LEDEBURITE

The eutectic of the iron-carbon system, austenite and cementite. Upon further cooling, the austenite decomposes into the eutectoid, ferrite and cementite.



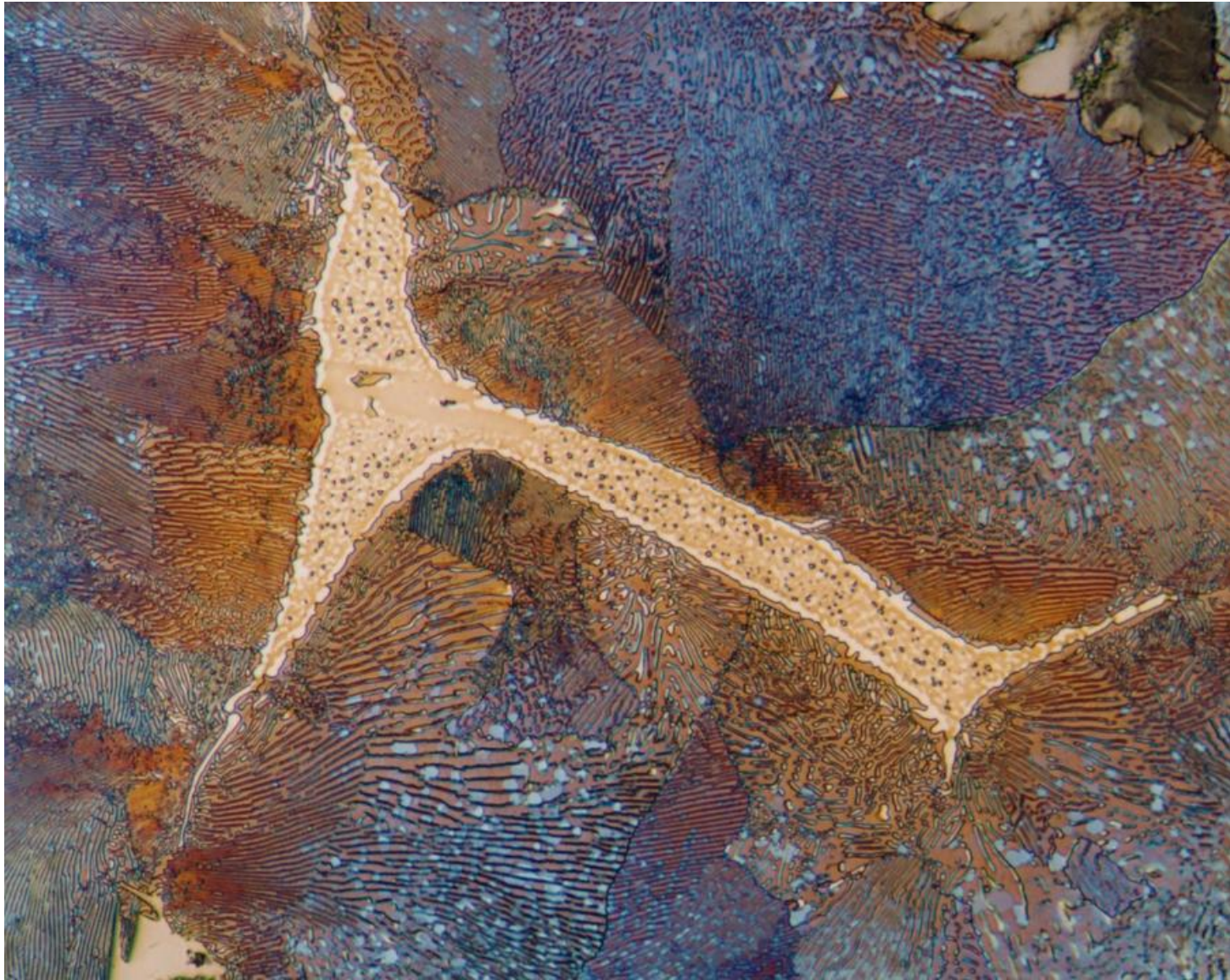
Ledeburite in as-cast NiHard cast iron (Fe – 3.3% C – 0.9% Mn – 0.9% Si – 1.8% Cr – 4.4% Ni – 0.4% Mo). Due to its high alloy content, the austenite within the “holes” in the cementite partially transformed to martensite (retained austenite is present between the martensite “needles”). Etched with aqueous 10% sodium metabisulfite. Original at 200X.



Ledeburite in a white cast iron (Fe – 4.0% C – 0.3% Si – 0.16% Mn – 0.91% Cr) etched with Beraha's sulfamic acid reagent (100 mL water, 3 g K₂S₂O₅ and 2 g NH₂SO₃H) . Original at 500X magnification. Taken in polarized light with sensitive tint.

STEADITE

A psuedo-binary or ternary eutectic of ferrite and iron phosphide or ferrite, iron phosphide and cementite, respectively, that can form in mottled iron or gray iron with a phosphorous content $> 0.06\%$. Small amounts of Cr and V promote the ternary eutectic. The presence of cementite depends upon how much of the carbon is present as graphite.



Ternary eutectic, Steadite, in blast furnace pig iron (pearlitic matrix) revealed by tint etching with Beraha's CdS reagent. Original at 500X.

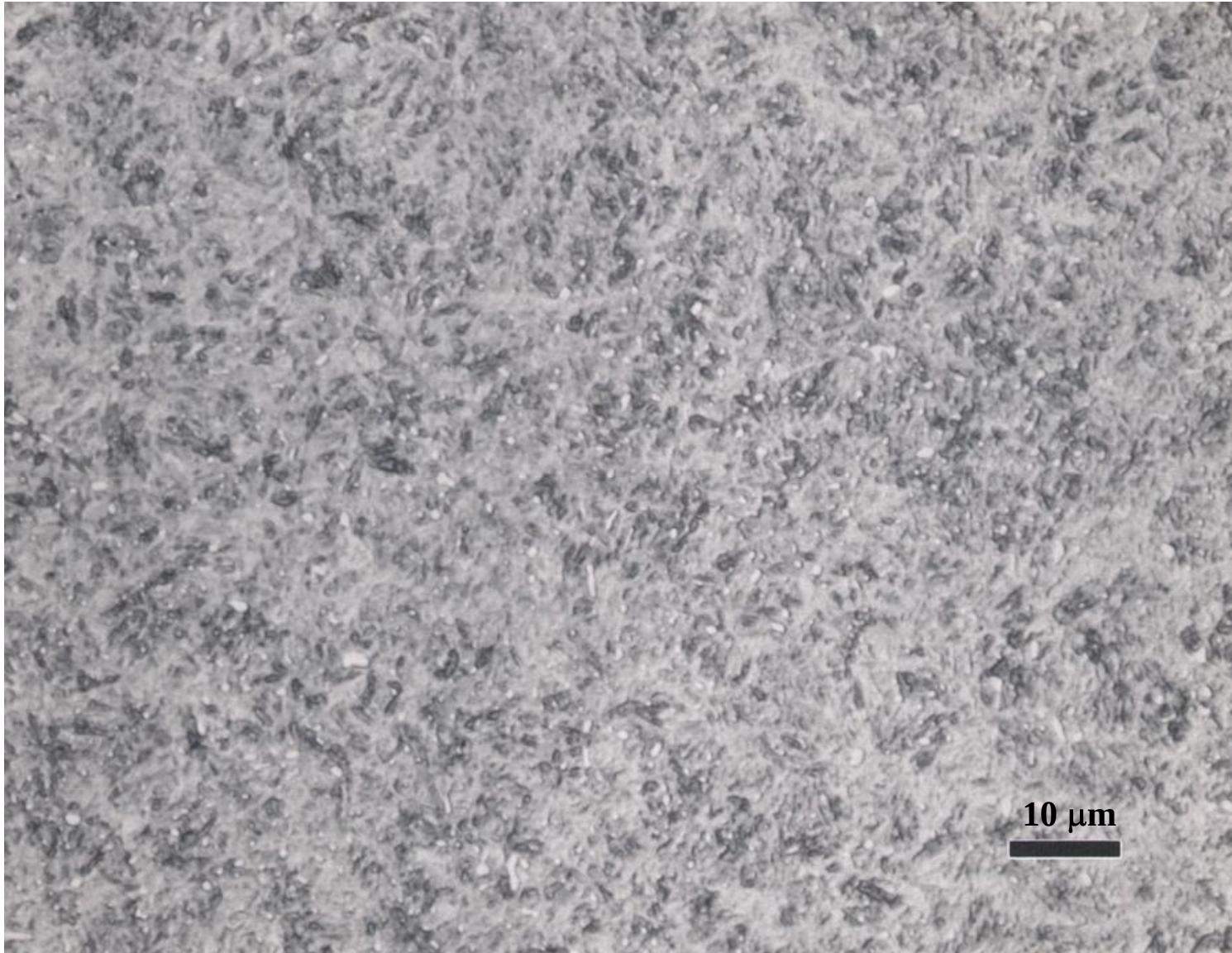
MARTENSITE

Generic term for microstructures that form by diffusionless transformation, where the parent and product phases have a specific crystallographic relationship and the same composition.

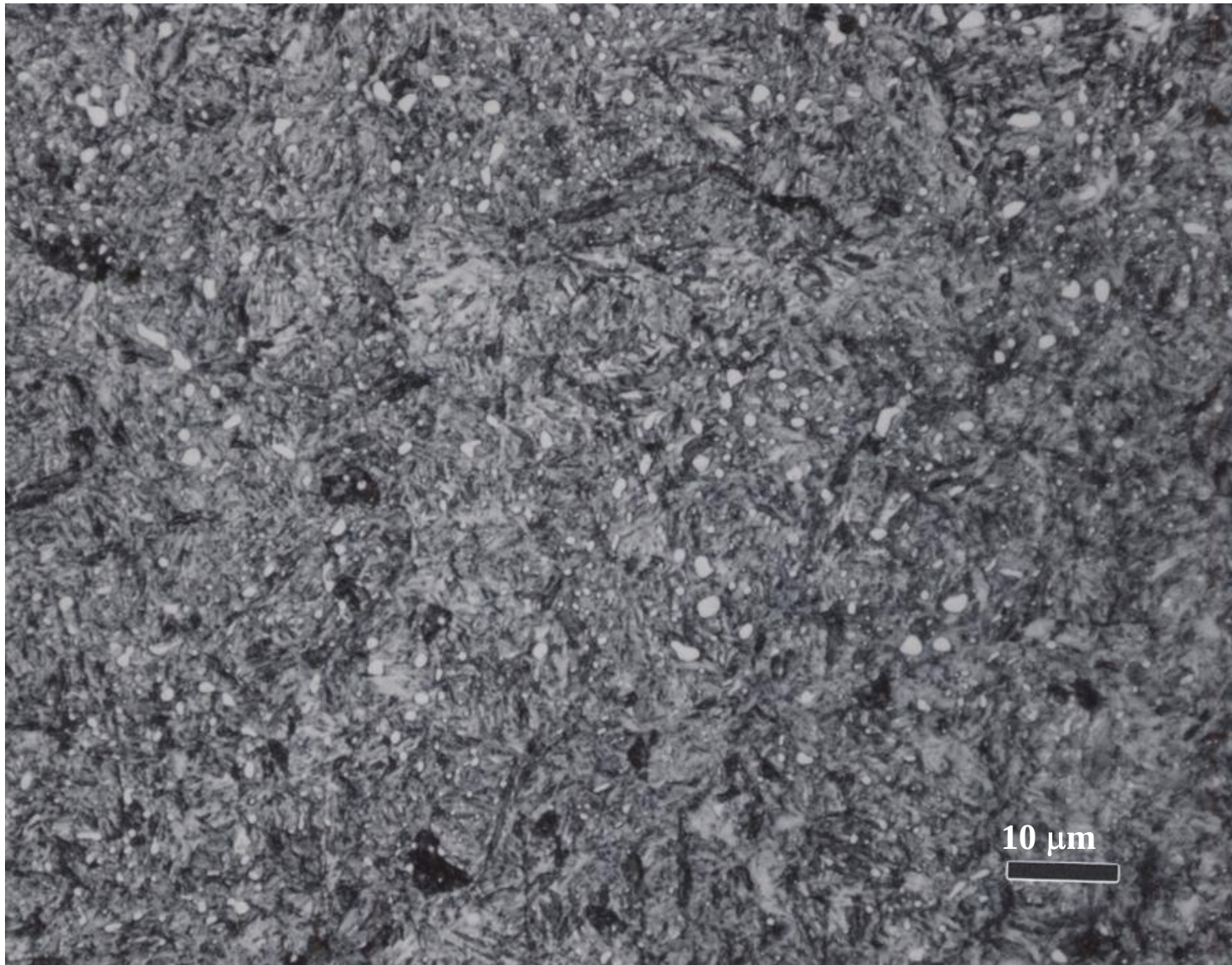
MARTENSITE

Alloys where the solute atoms occupy interstitial sites (C in Fe) – martensite is hard, highly strained and has a high dislocation density.

Alloys where the solute atoms occupy substitutional sites (Ni in Fe) – martensite is soft and ductile.



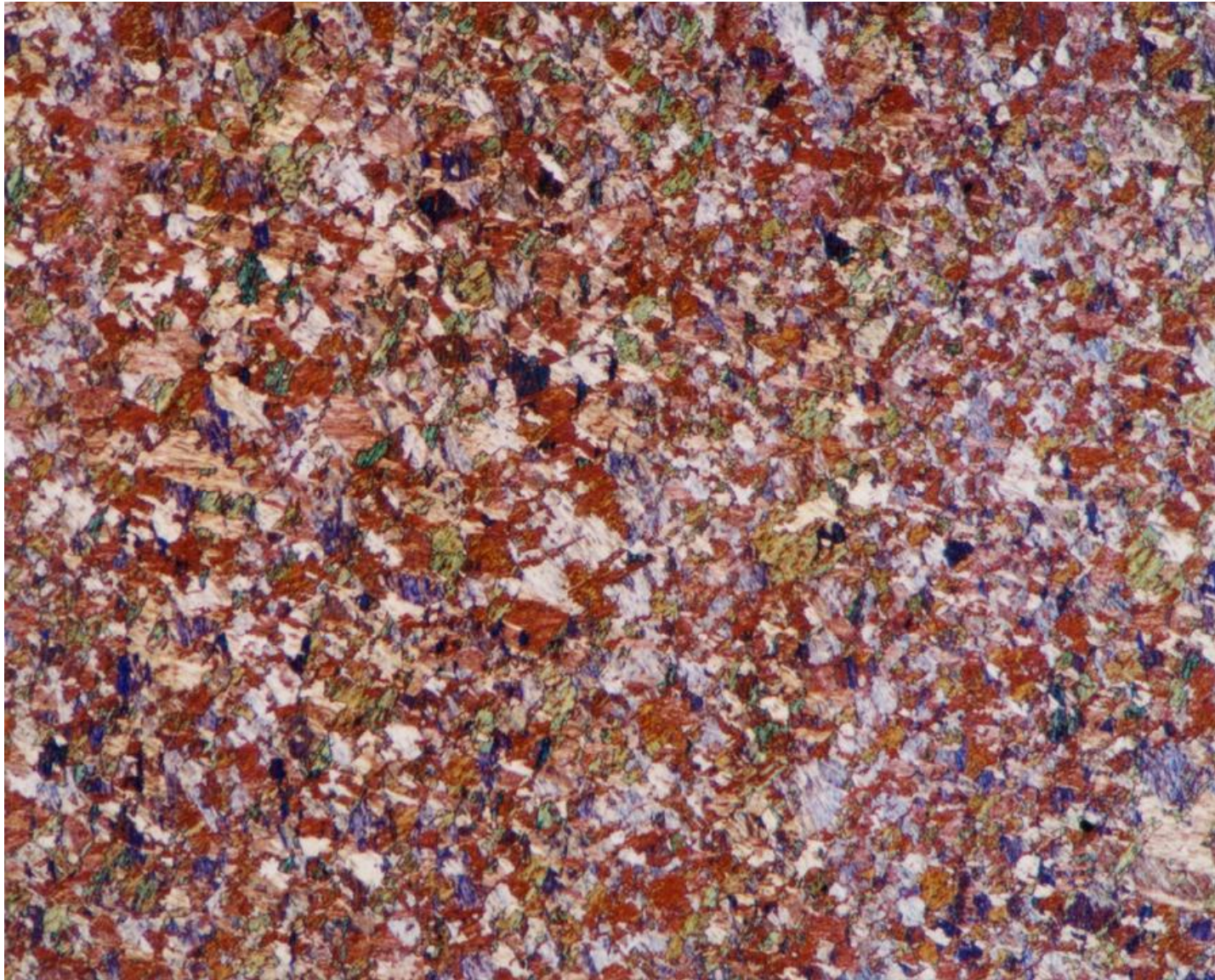
Tempered martensite and residual cementite in quenched and tempered type 52100 (Fe – 1% C – 1.5% Cr) bearing steel with a fine prior-austenite grain size. Etched with 2% nital.



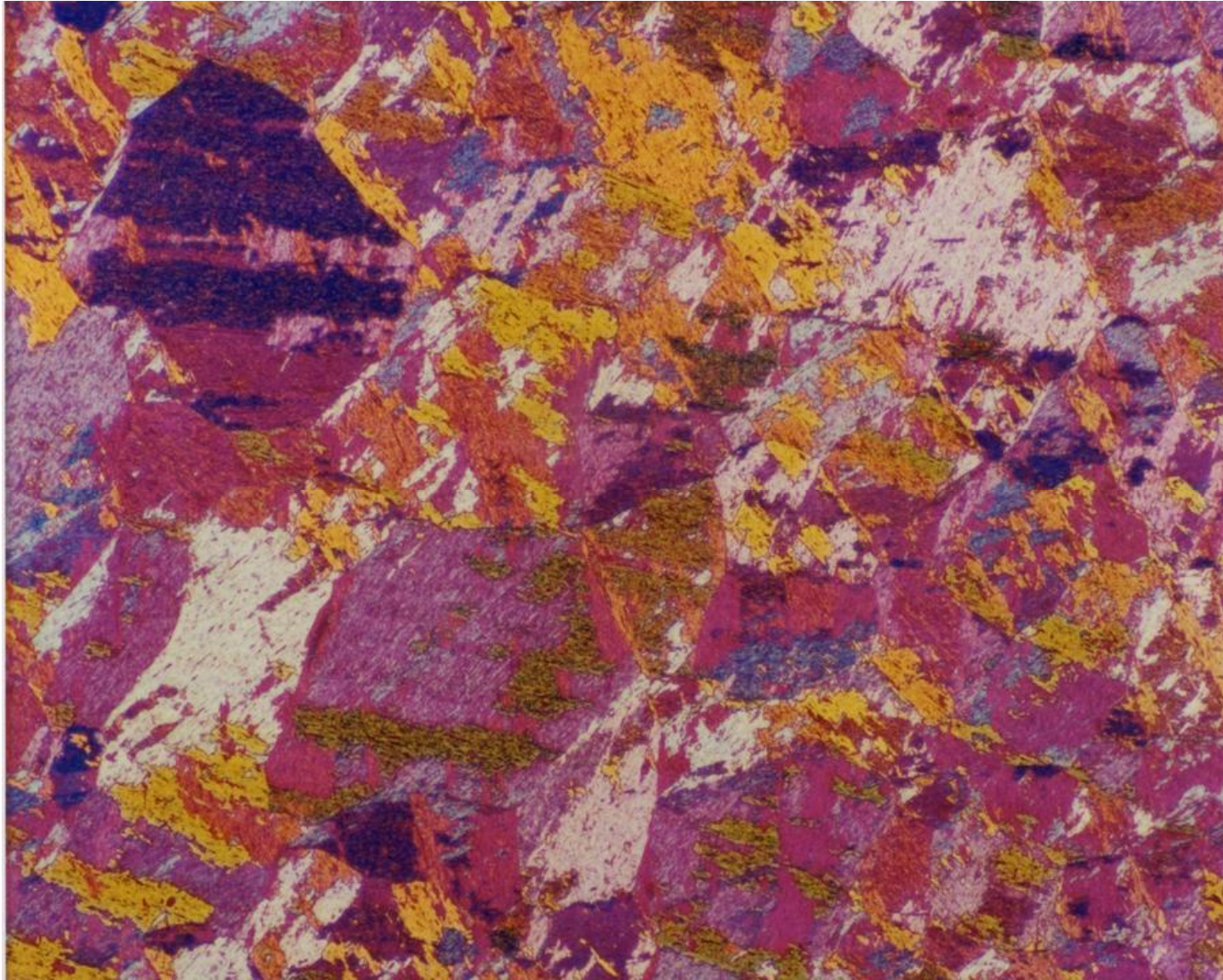
Residual carbide and tempered martensite in properly hardened type W1 tool steel (Fe – 1.1% C – 0.2% Mn – 0.2% Si). Note the similarity to properly hardened 52100 bearing steel, with a similar treatment and hardness. Etched with 2% nital.



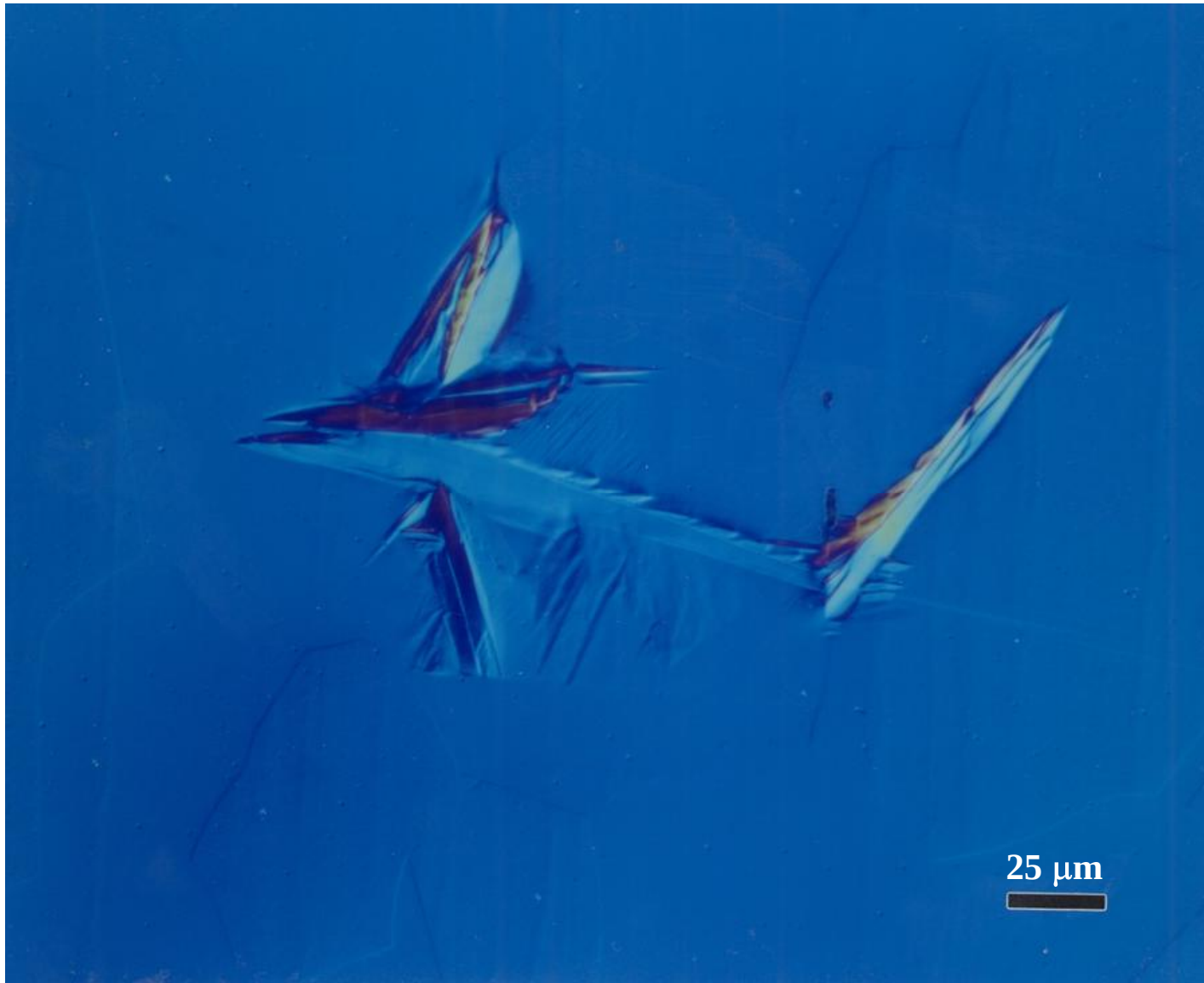
Improperly carburized and hardened 8620 (Fe – 0.2% C – 0.75% Mn – 0.55% Ni – 0.5% Cr – 0.2% Mo) revealing excess cementite (left, arrows) near the surface but not further below in the case (right). The carburized case contains coarse plate martensite (dark “needles”) and retained austenite between the martensite. Etched with 2% nital.



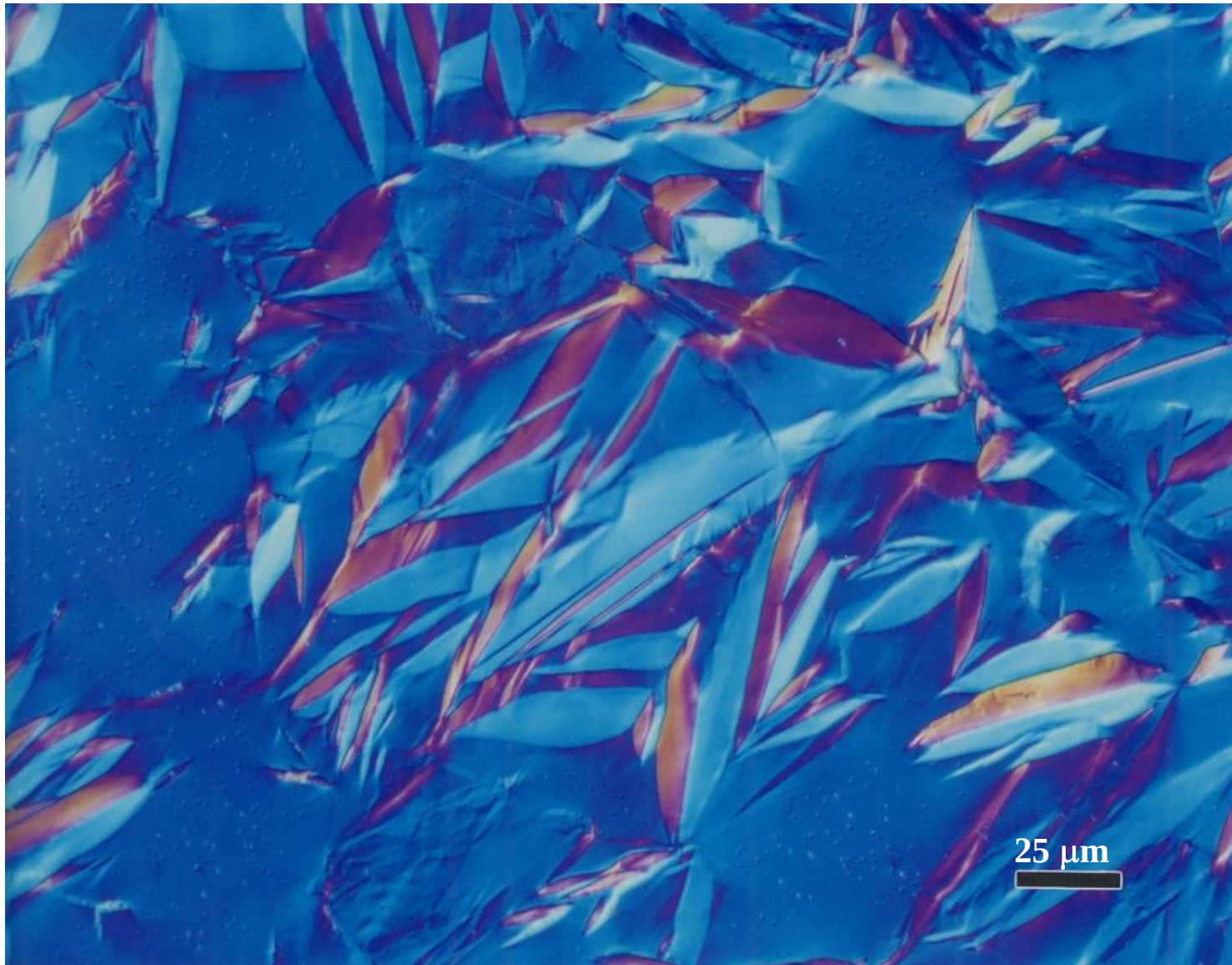
Tempered lath martensitic microstructure of wrought, heat treated AF-1410 ultra-high strength steel (Fe – 0.16% C – 14% Co – 10% Ni – 2% Cr – 1% Mo) tint etched with Beraha's CdS reagent and viewed with polarized light plus sensitive tint. Original at 100X.



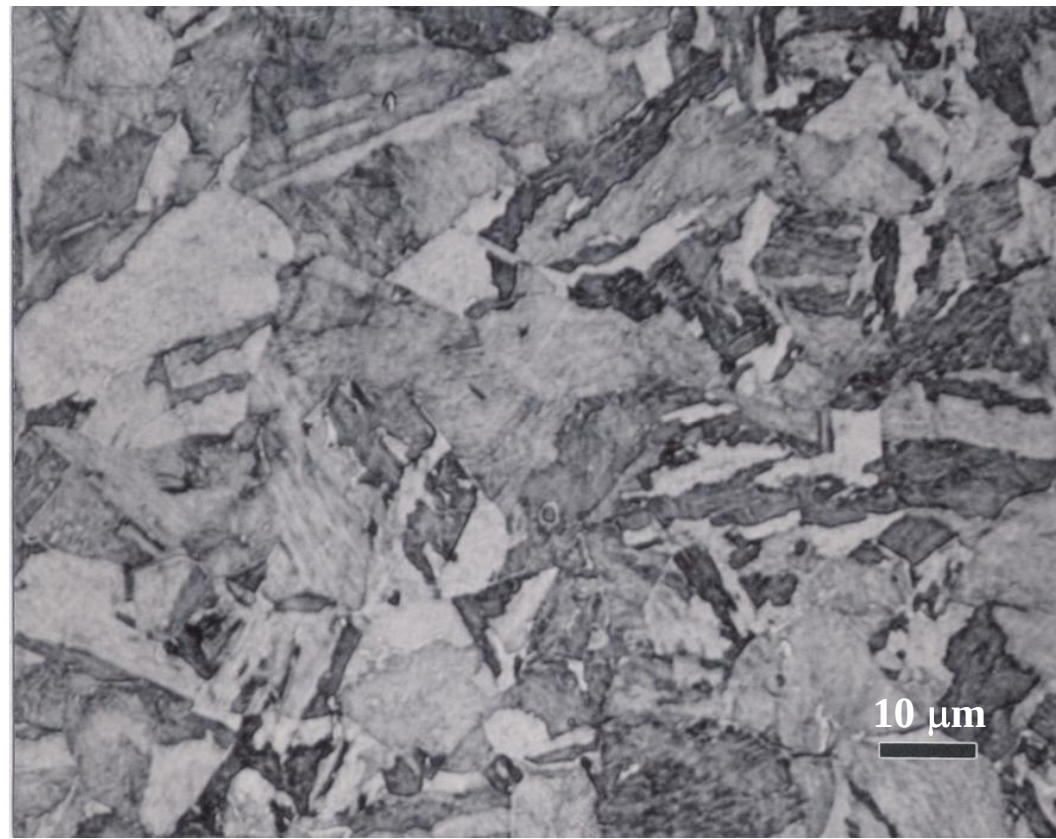
Low-carbon, “lath” martensite in an over-austenitized specimen of AerMet 100 (Fe – 0.23% C – 13.4% Co – 11.1% Ni – 3.1% Cr – 1.2% Mo). The grain size was coarsened by the heat treatment (1093 °C, AC, age at 675 °C for 6 h, AC) making it easier to see the lath structure. Etched with aqueous 10% sodium metabisulfite and viewed with polarized light plus sensitive tint. Original magnification was 100X. AerMet is a trademark of Carpenter Technology Corp., Reading, Pennsylvania.



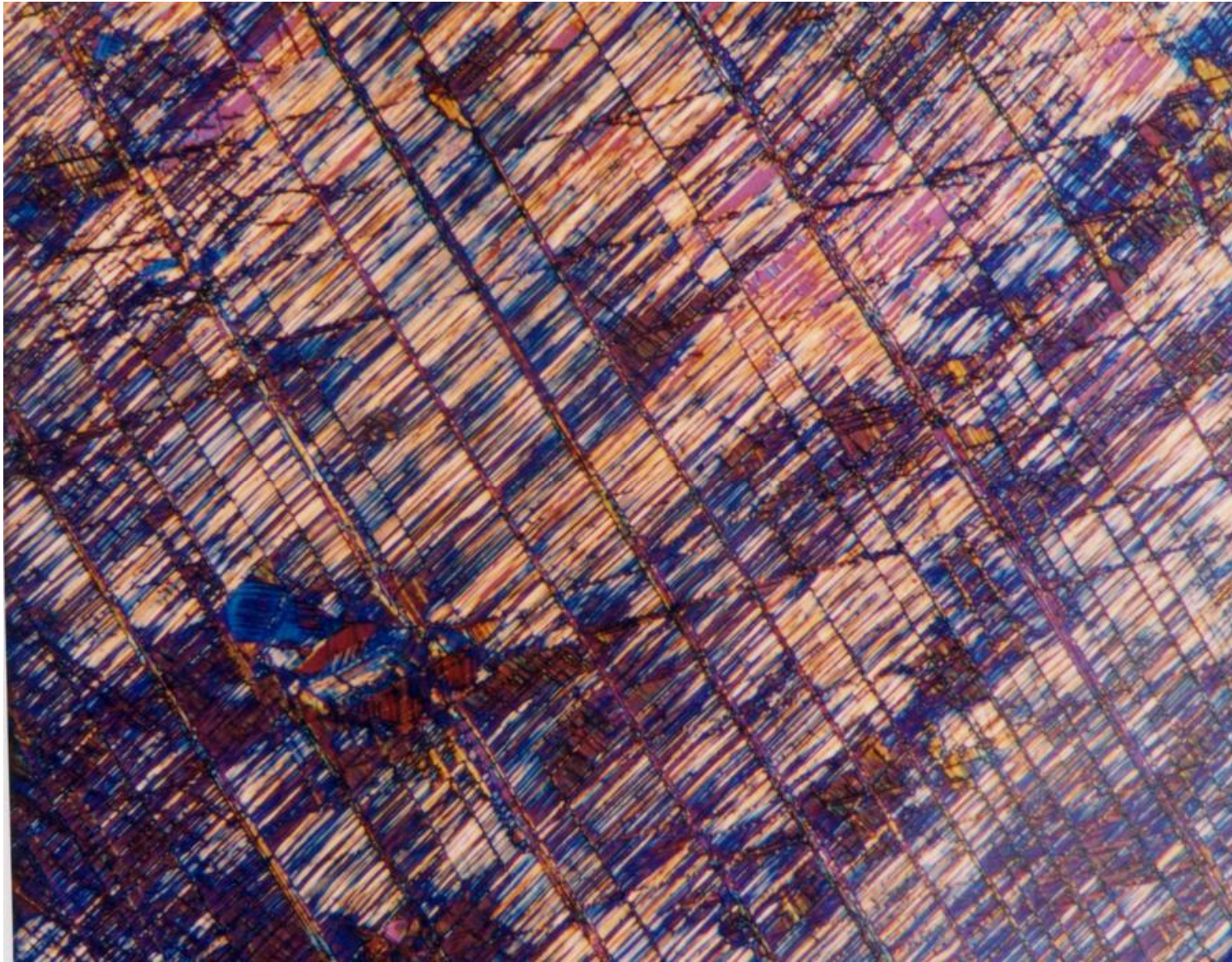
Example of “butterfly” martensite (low carbon martensite) formed on the polished surface of a specimen of High-Expansion 22-3 (Fe – 0.1% C – 0.5% Mn – 0.2% Si – 3.1% Cr – 22% Ni) alloy steel when cooled to $-73\text{ }^{\circ}\text{C}$. Unstable austenite transformed to martensite with its characteristic shear which is visible on a free surface using Nomarski DIC (not etched).



Low-carbon martensite formed on the polished surface of Temperature Compensator “30” alloy, type 2 (Fe – 0.12% C – 0.6% Mn – 0.25% Si – 30% Ni) after refrigeration in liquid nitrogen. The low temperature converted any unstable austenite to martensite with its characteristic shear reaction, visible on a prepared free surface using Nomarski DIC (not etched).



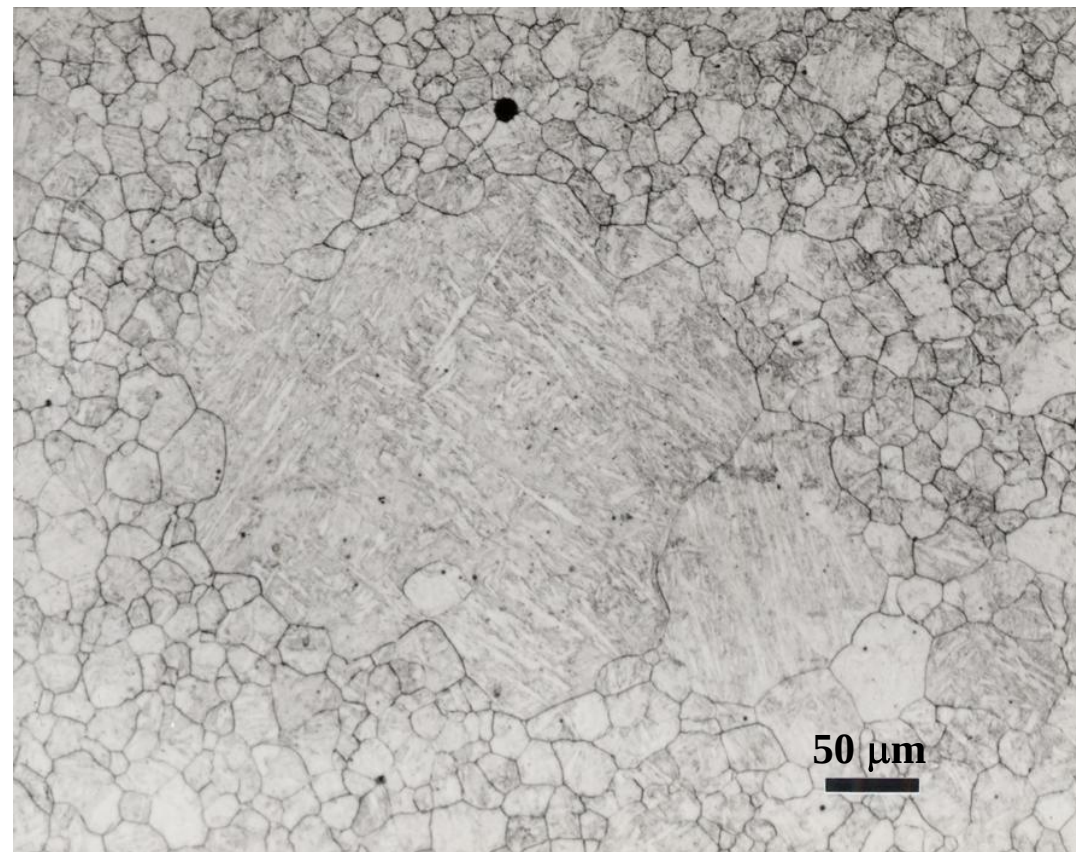
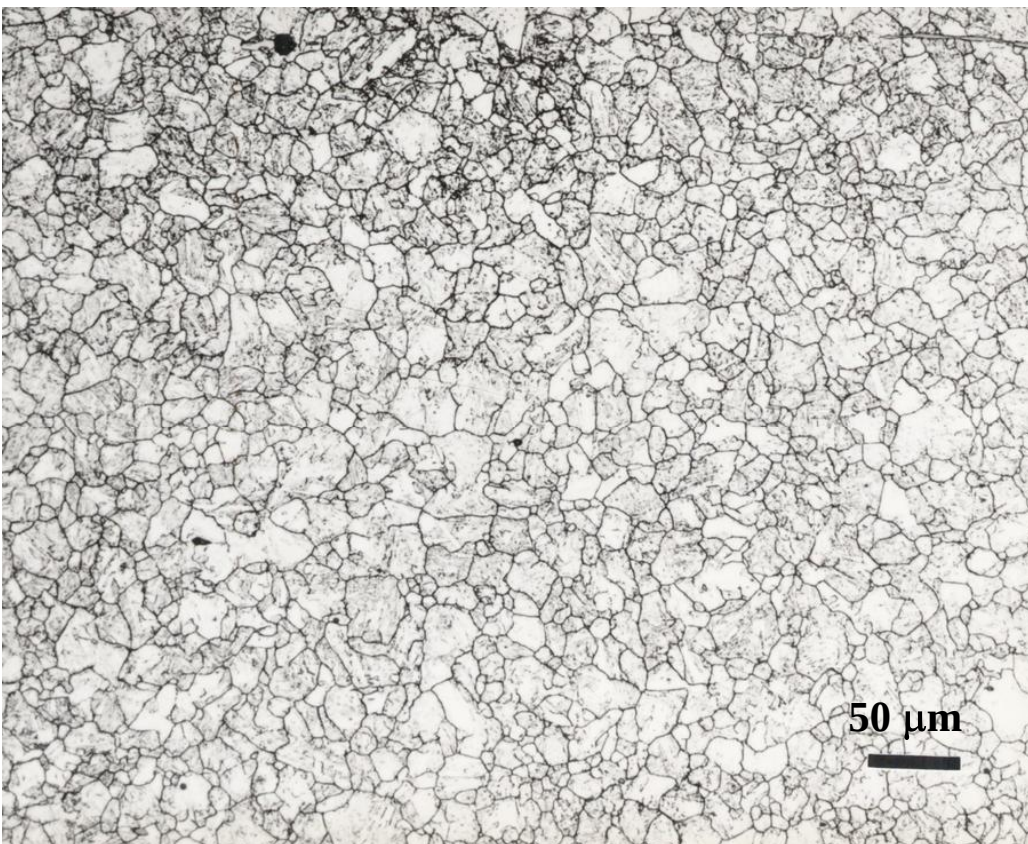
Solution annealed and aged martensite in 18Ni250 (Fe - <0.03% C – 18.5% Ni – 7.5% Co – 4.85% Mo – 0.4% Ti – 0.1% Al) maraging steel. Etched with Fry's reagent.



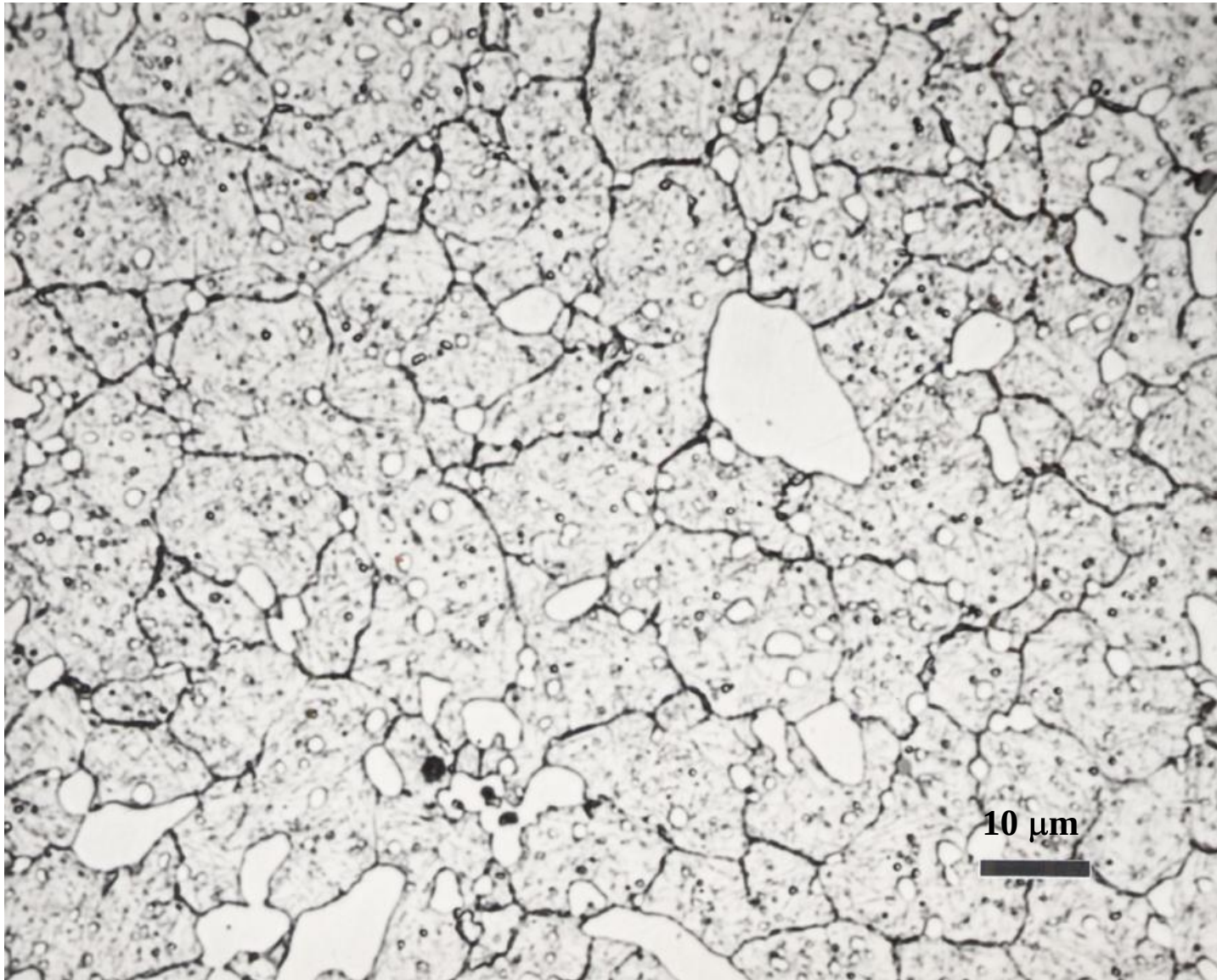
Epsilon martensite formed by explosive-hardening of as-cast Hadfield manganese steel (Fe – 1.3% C – 13% Mn) as revealed by tint etching with Beraha's sulfamic acid reagent (100 mL water, 3 g potassium metabisulfite, 2 g sulfamic acid, 0.8 g ammonium bifluoride) and viewed with polarized light plus sensitive tint. Original at 500X.

Prior-Austenite Grain Boundaries

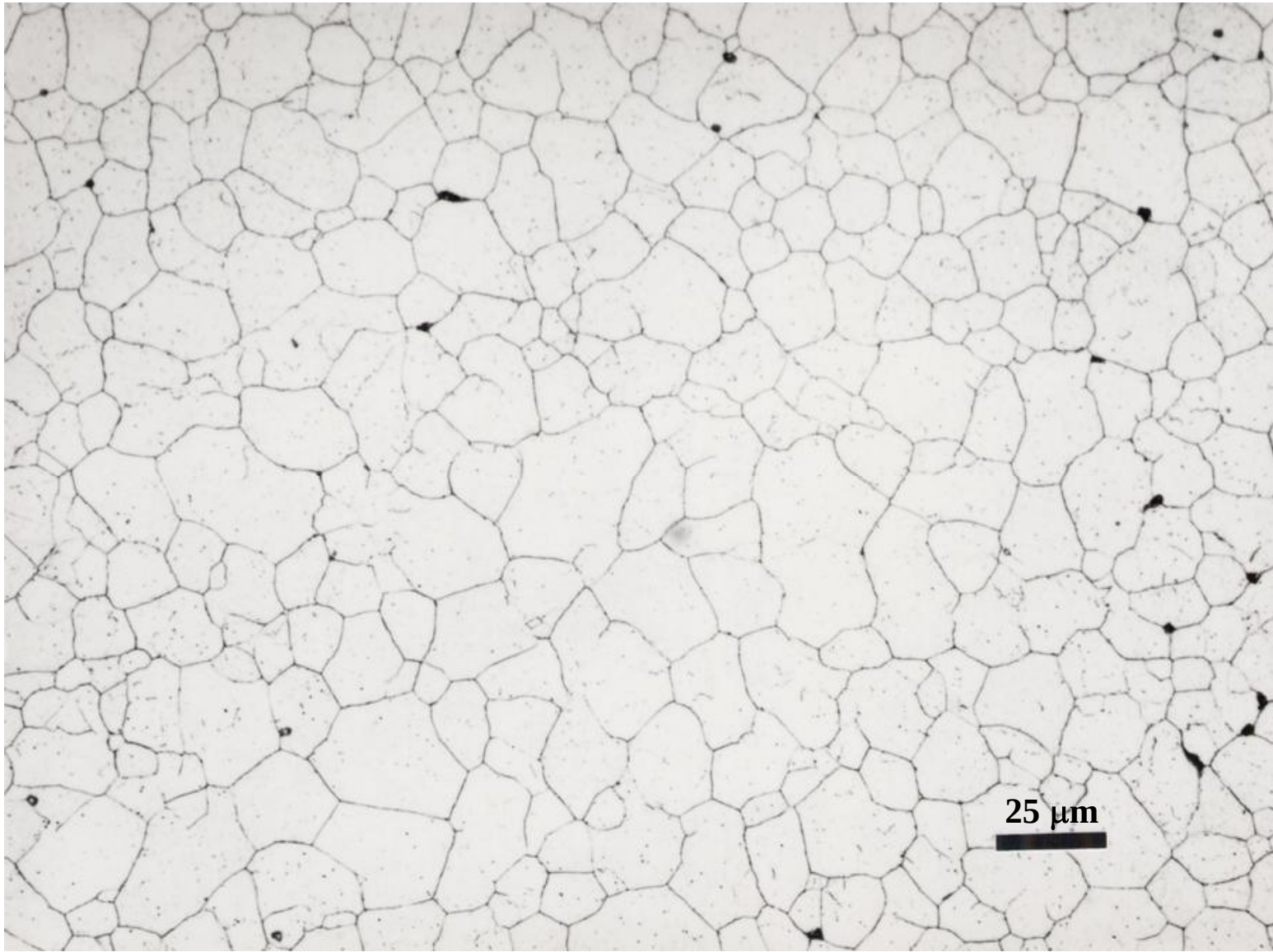
When steels are to be heat treated to form martensite, they are heated into the austenite phase field (if they are hypereutectoid, they are heated into the austenite plus carbide phase field), held for an appropriate time period, and quenched at a rate required to form martensite. The grain size of the parent phase, austenite, can be correlated with many important mechanical properties. Revealing the prior-austenite grain boundaries is difficult and requires special etchants, that do not always work well and do not always reveal a high percentage of the grain boundaries.



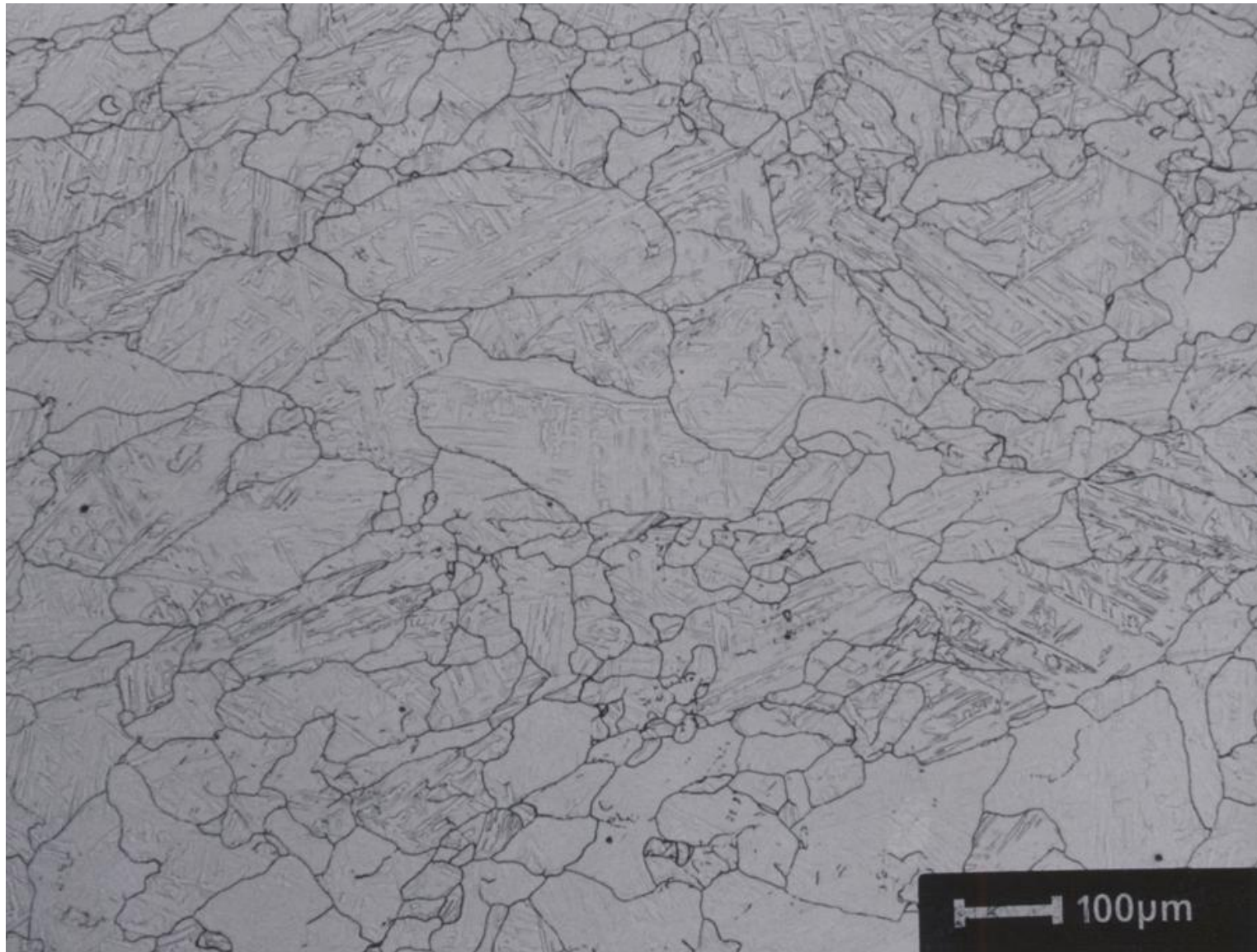
Prior-austenite grain boundaries in 43B17 alloy steel (Fe – 0.17% C – 1.85% Ni – 0.75% Cr – 0.25% Mo – 0.003% B) revealed by etching with aqueous saturated picric acid plus 1% HCl and a wetting agent. The specimen at the left was given a normal heat treatment (843 °C – 30 min, oil quench, 230 °C temper) and has a fine grain size. The specimen on the right was “mock carburized” (925 °C for 8 h, oil quench and tempered at 230 °C) and rapid grain growth has begun. This specimen is not suitable for carburizing.



Prior-austenite grain boundaries can be revealed by etching with nital in many highly alloyed tool steels, as long as they are in the as-quenched or lightly tempered (<250 °C) condition. This specimen of type D2 tool steel (Fe – 1.5% C – 12% Cr – 0.9% V – 0.8% Mo) was austenitized at 1010 °C, air quenched and tempered 2 h at 200 °C. It was etched with 10% nital.



In high-alloy steels it is often possible to precipitate a second phase, such as pearlite, on the prior-austenite grain boundaries so that they can be easily examined. This experimental 5% Cr hot-work die steel specimen was austenitized at 1080 °C, cooled to 700 °C and held for 1 h, then air cooled to precipitate pearlite on the grain boundaries. It was etched with glyceresia.



Prior-austenite grain boundaries can be revealed in precipitation hardenable stainless steels and in maraging steels if the are in the solution annealed condition (before aging). This specimen of 17-4PH stainless steel (Fe - <0.07% C – 17% Cr – 4% Ni – 4% Cu – 0.3% Nb) was solution annealed at 1038 °C. It was etched with aqueous 60% HNO₃, 1.1 V dc, Pt cathode, 30 s.

Other Constituents in Steels

Nonmetallic Inclusions

Nitrides and Carbonitrides

Alloy Carbides

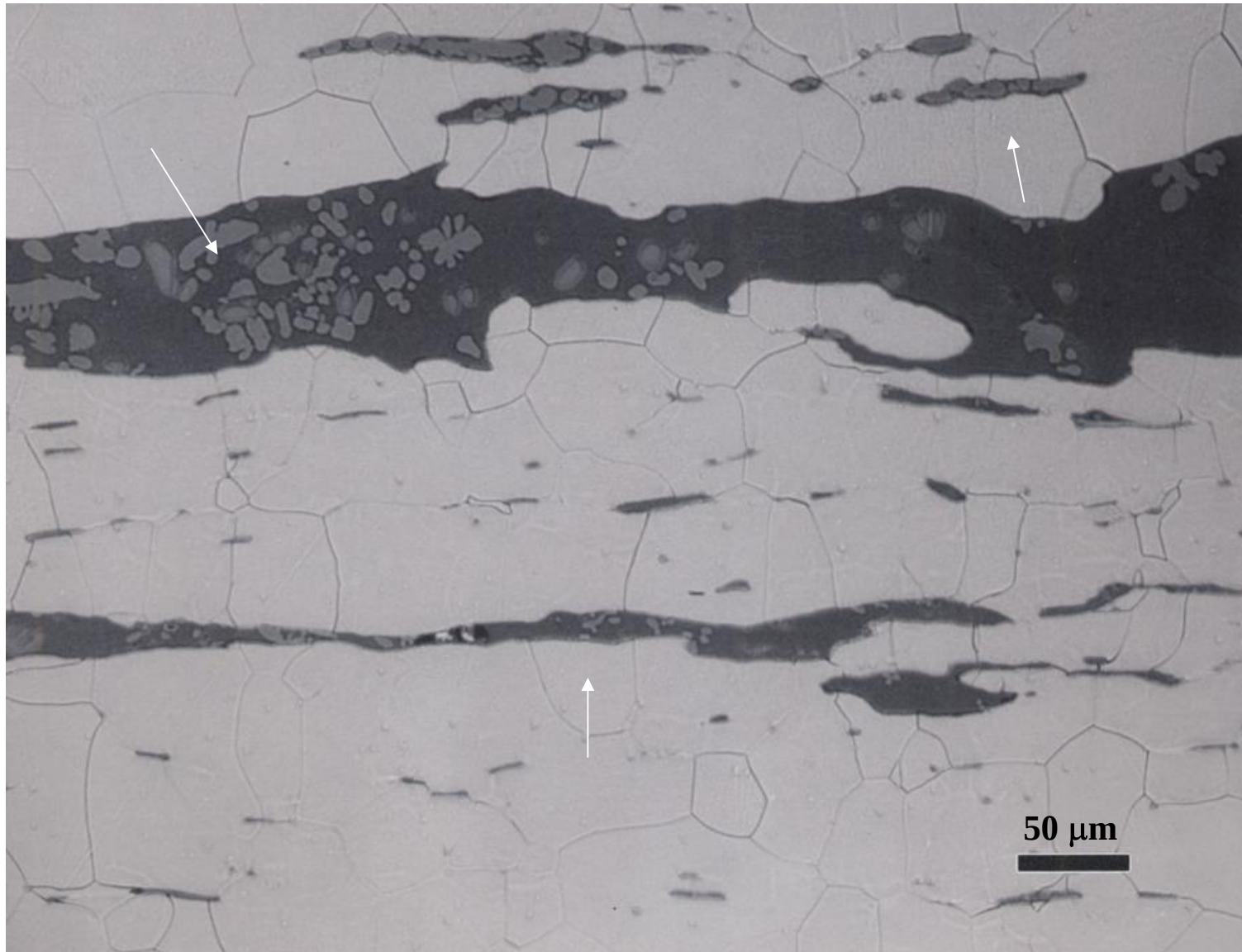
Graphite

Sigma

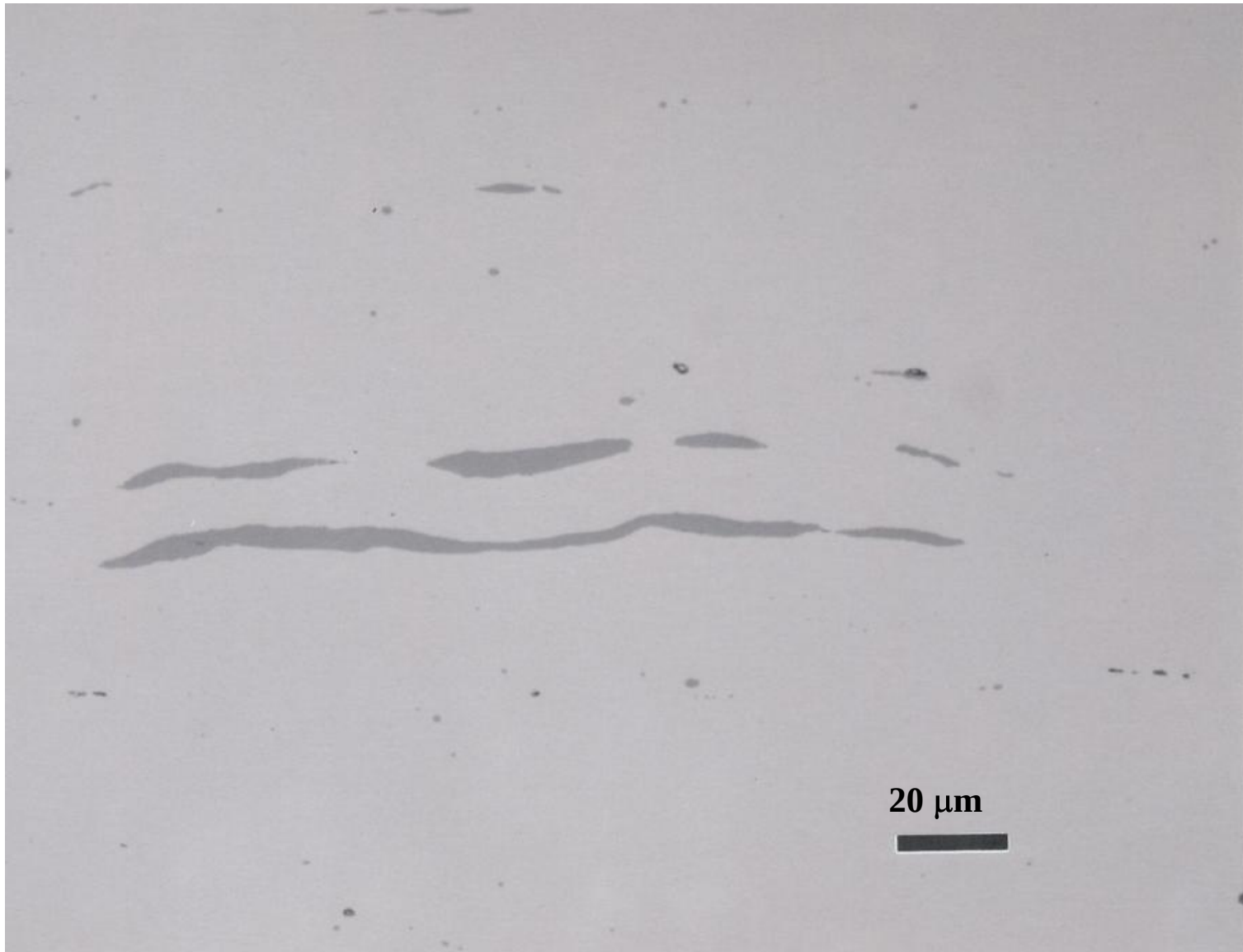
Chi, Laves and Borides

Nonmetallic Inclusions

Particles of oxides or sulfides that precipitate before or during solidification as a result of very limited solid-solubility for oxygen and sulfur in solid iron. These are called indigenous or endogenous) inclusions. Particles from refractory components and other materials in contact with the melt may become entrapped in the solid metal. These are called exogenous inclusions.



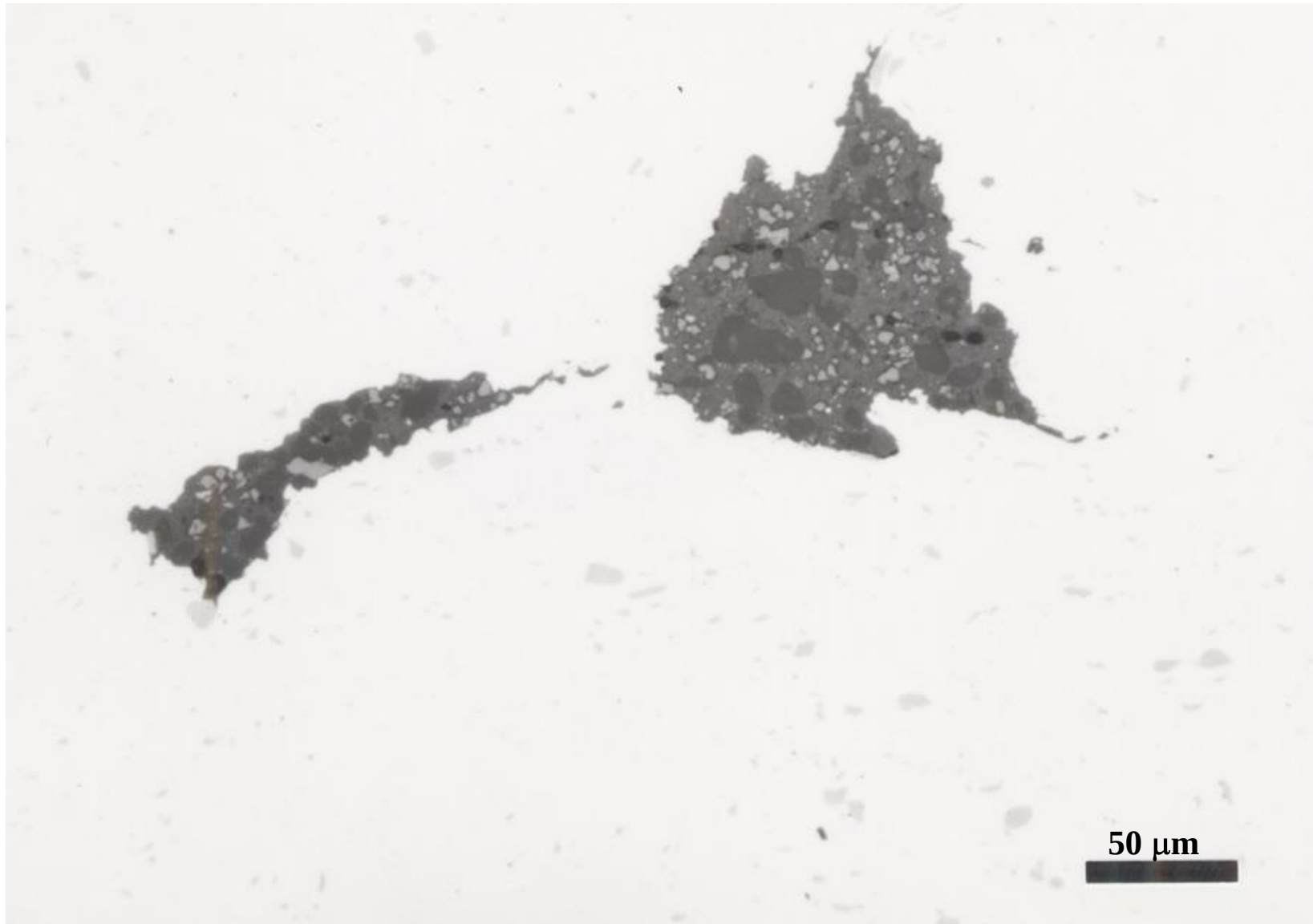
Slag inclusions in wrought iron (Fe – 0.07% C – 0.06% Mn – 0.2% Si – 0.16% P – 0.02% S + several percent slag). Note the two-phase structure of the slag stringers. The matrix is ferrite. Etched with 2% nital to reveal the ferrite grain boundaries.



Manganese sulfides in type 416 stainless steel (Fe - $<0.15\%$ C – 13% Cr - $>0.15\%$ S) in the hot rolled condition. Not etched.



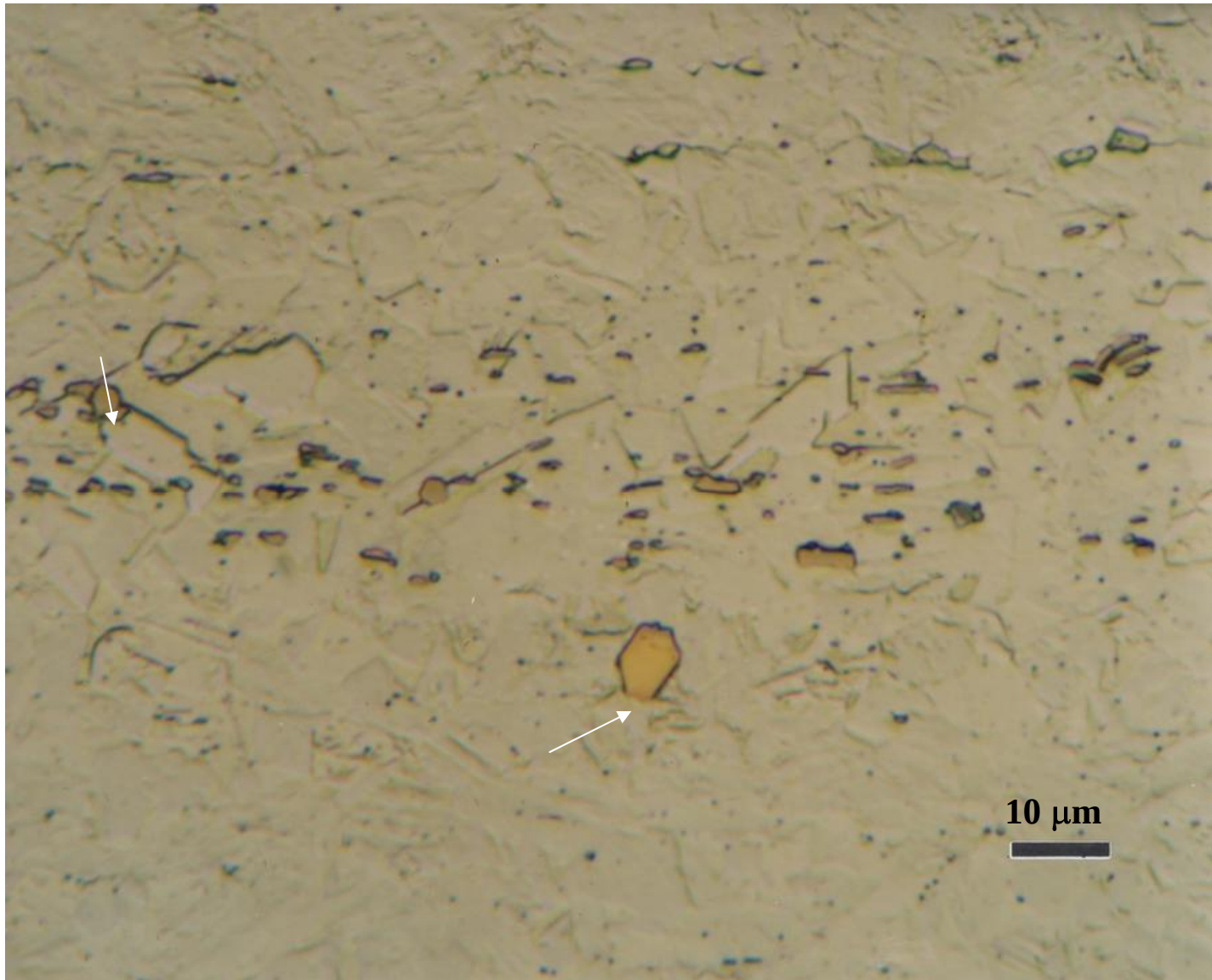
A cluster of aluminate inclusions in as-cast P20 tool steel (Fe – 0.35% C – 0.78% Mn – 1.74% Cr – 0.41% Mo – 0.050% Al – 57 ppm O). Deep etching usually reveals that the particles are interconnected. Hot working with “string” out the particles in a linear manner. Not etched.



Two complex exogenous inclusions detected by immersion ultrasonic inspection. Note that at least three phases are present. Not etched.

Nitrides and Carbonitrides

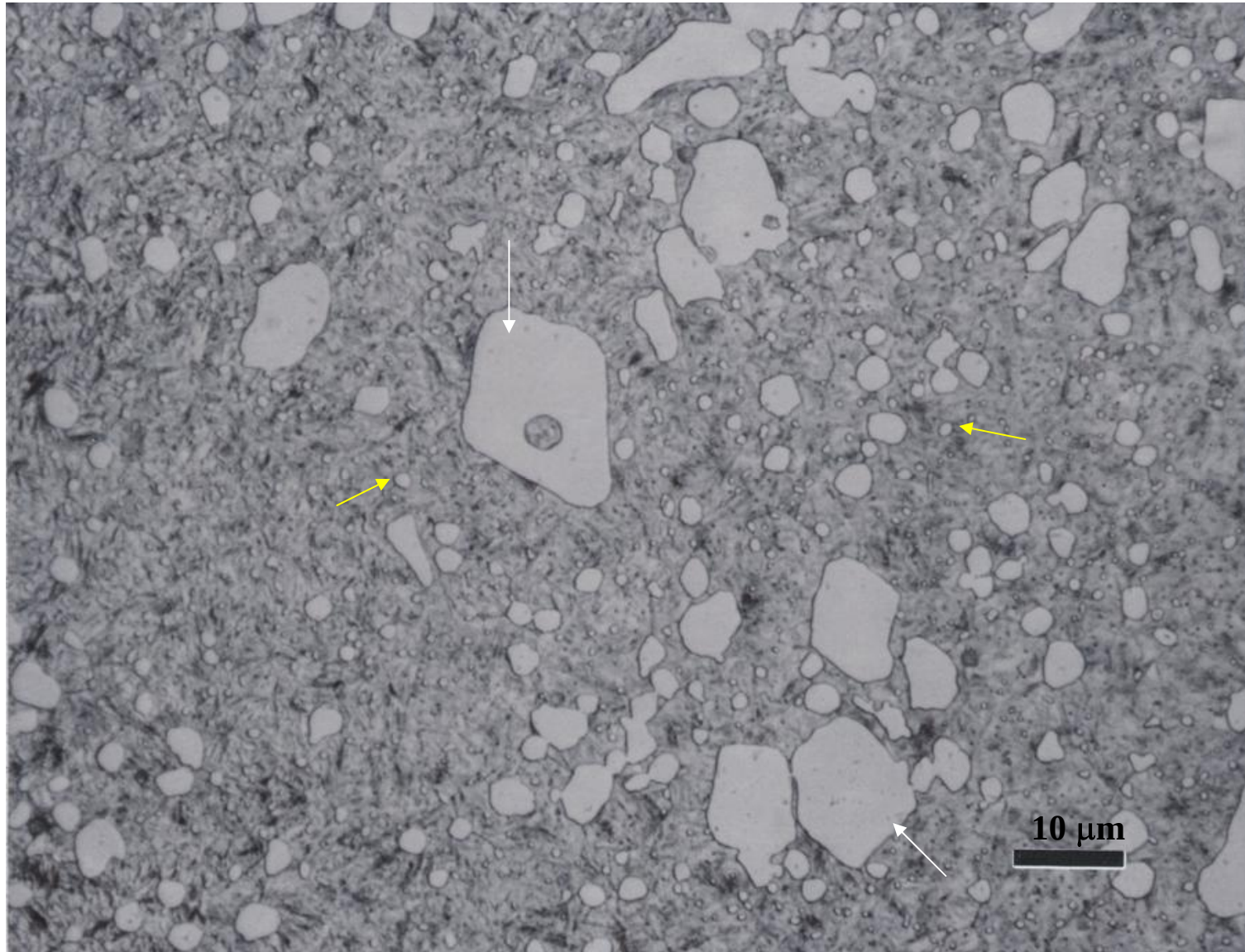
Colored idiomorphic cubic crystals with the general formula $M_x(C,N)_y$ where M may be Ti, Nb, Zr, or Ta. AlN and VN may also be present, but these are normally sub-microscopic in size.



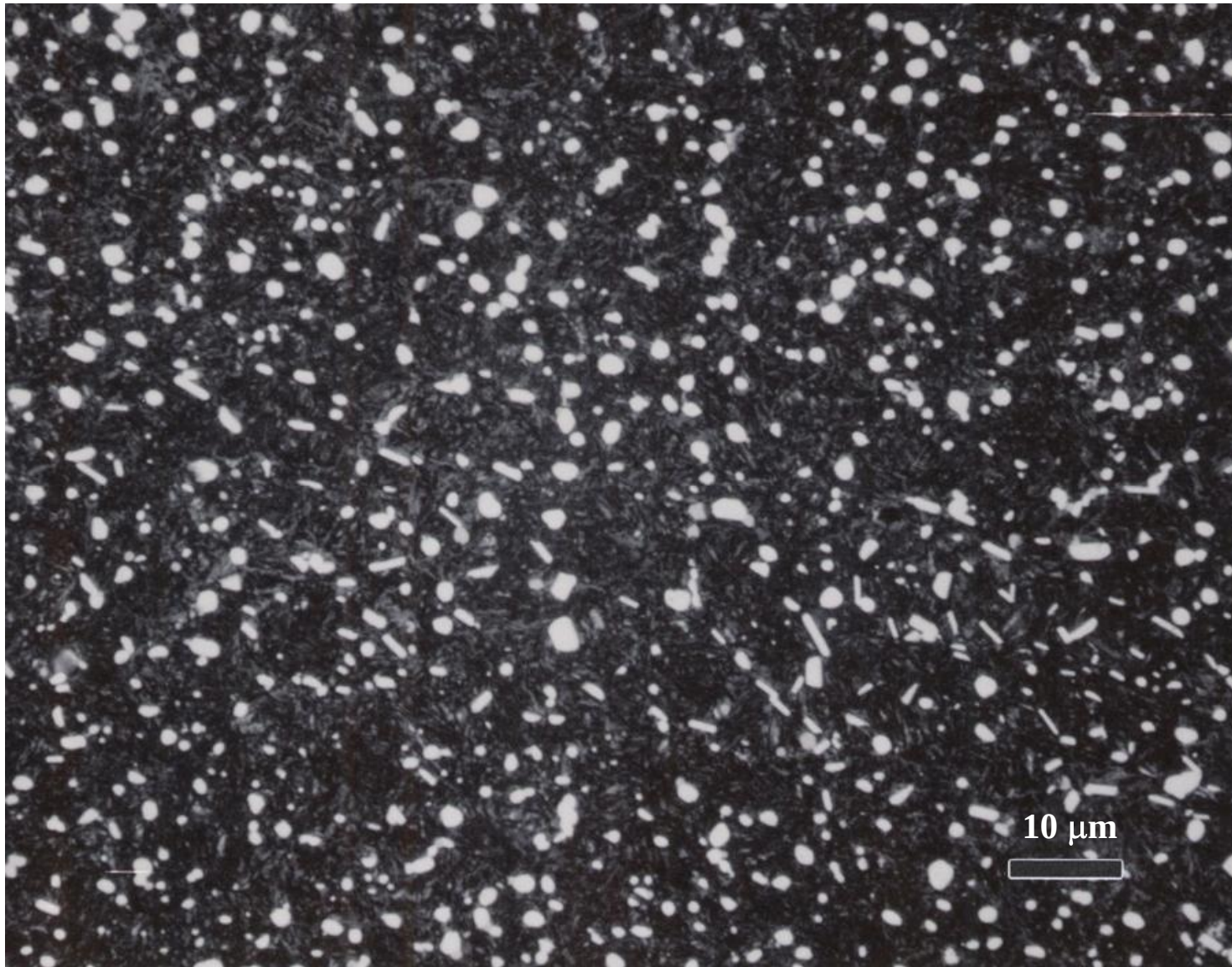
Titanium nitrides (arrows) in a Ti-stabilized austenitic stainless steel (Fe – 0.1% C - 0.8% Mn – 18.2% Cr – 8.2% Ni – 0.42% Ti). The other, smaller particles may also be nitrides, titanium carbides or carbonitrides. Etched with glycereregia.

Alloy Carbides

When amounts of carbide forming elements (such as Cr, V, Mo, W, Ti and Nb) exceed a certain level, an alloy carbide will form rather than cementite. There are numerous examples with formulae such as MC , M_2C , M_6C , M_4C_3 , M_7C_3 and $M_{23}C_6$. These are usually harder than cementite.



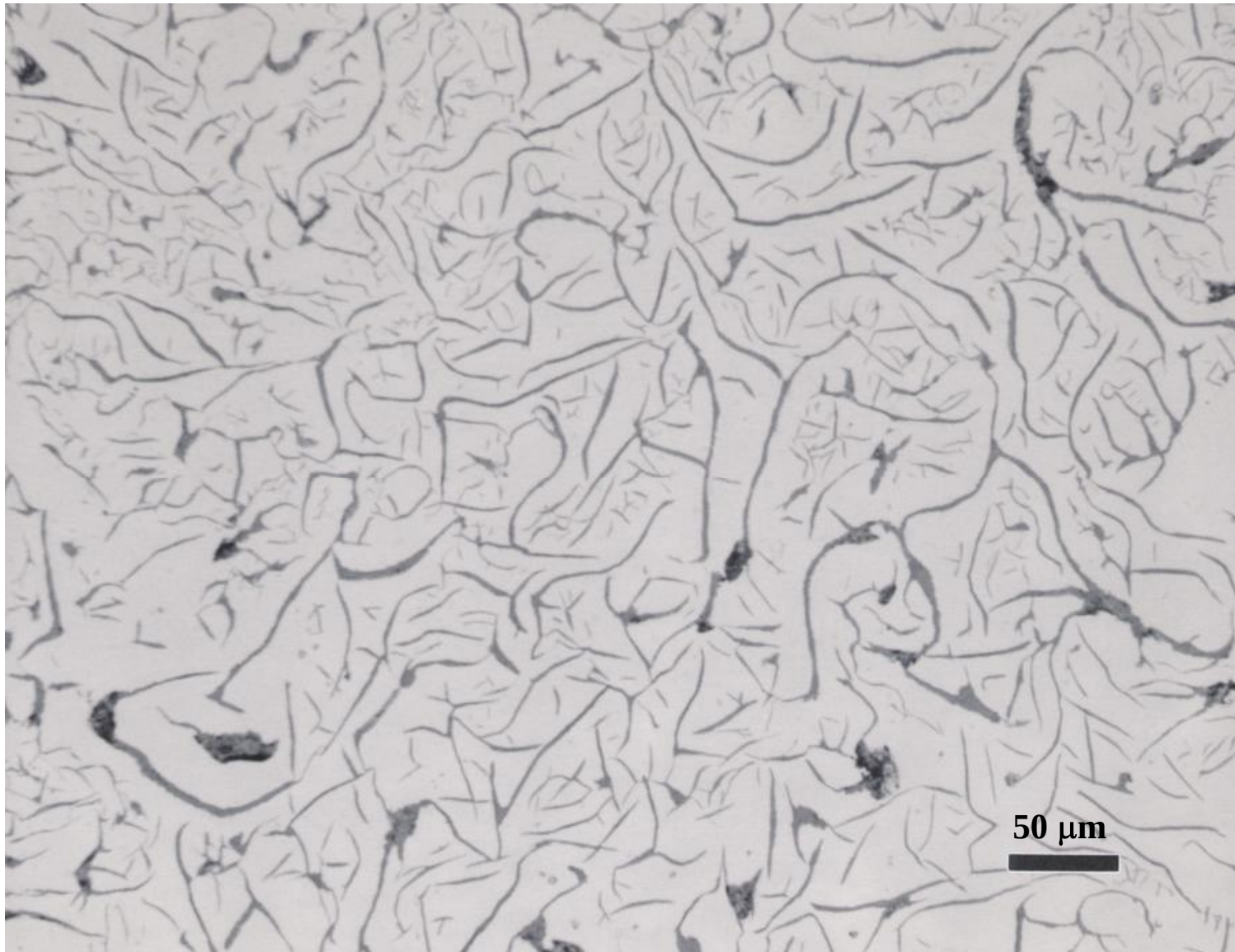
Massive Cr_7C_3 alloy carbides (white arrows) in type D3 tool steel (Fe – 2.1% C – 0.25% Mn – 0.25% Si – 0.5% Ni – 12.5% Cr) in the quenched and tempered condition (fine tempered martensite matrix). The fine carbides are Cr_{23}C_6 (yellow arrows). 10% nital etch.



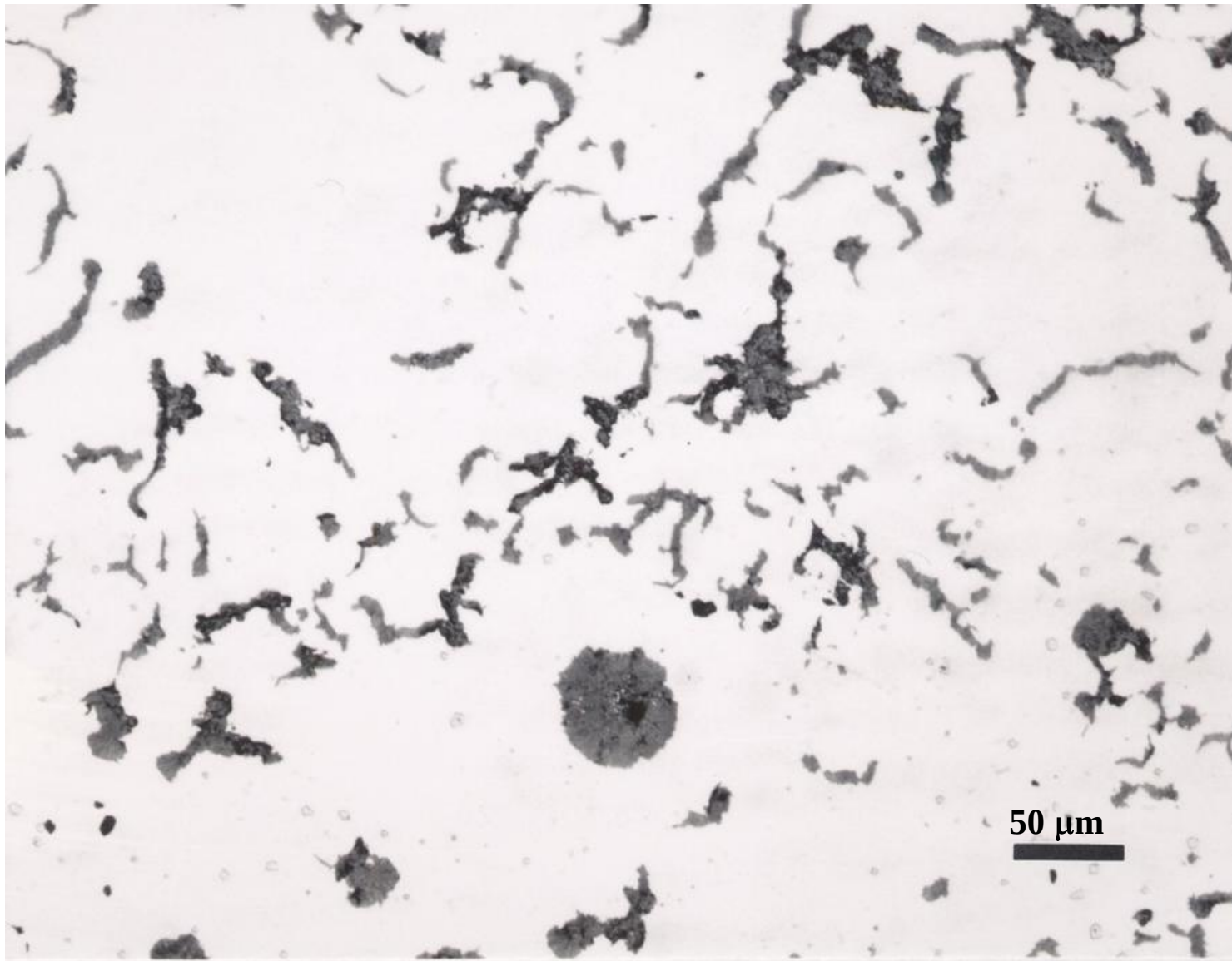
Residual alloy carbide (white particles) and a tempered martensite matrix in P/M M42 high speed steel (Fe – 1.1% C – 8.25% Co – 9.5% Mo – 1.5% W – 3.75% Cr – 1.15% V) at 68 HRC. Etched with Vilella's reagent.

Graphite

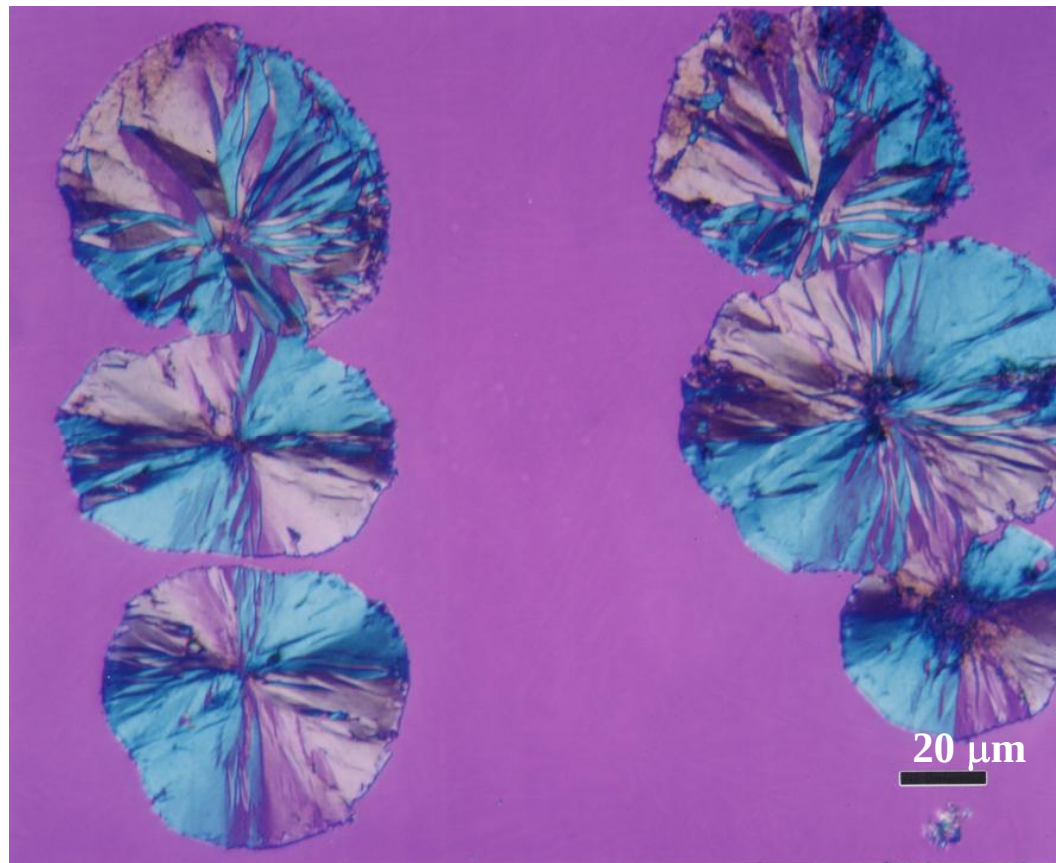
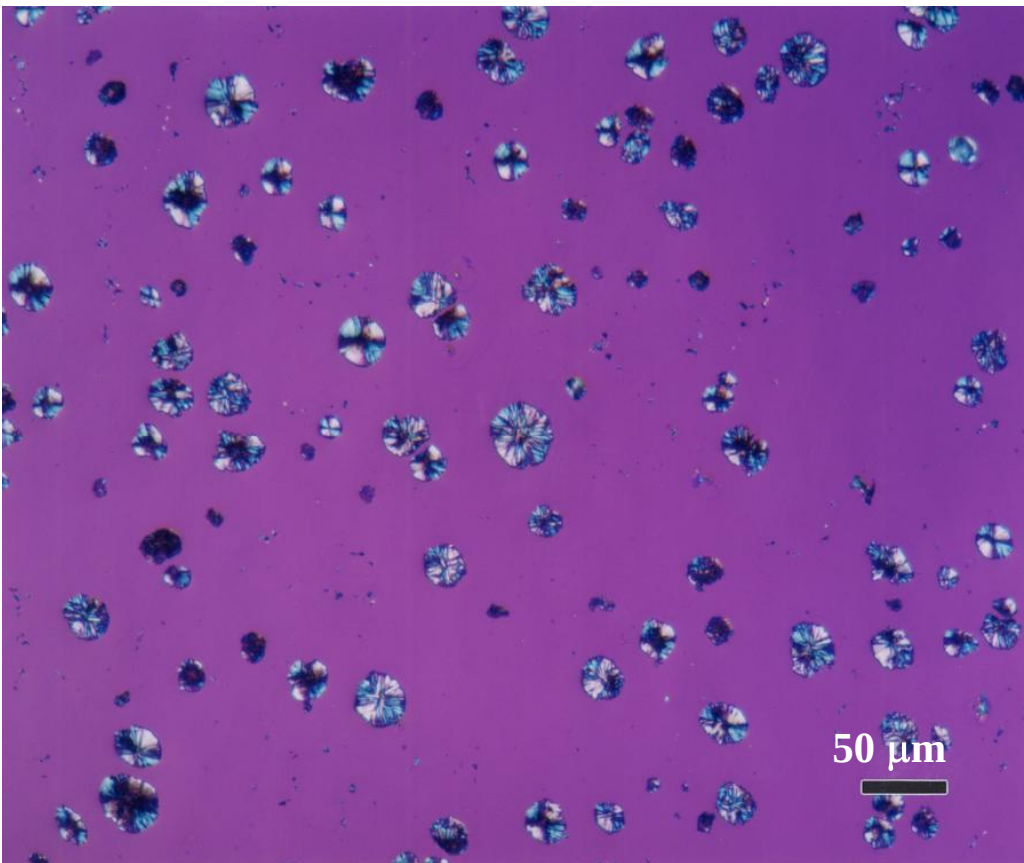
Graphite is the most stable form of carbon in Fe-based alloys, but it is normally present only in cast irons with high carbon and silicon contents. Graphite may be deliberately formed in certain tool steels, such as Type O6, and it has been observed to precipitate in carbon steel pipe held at elevated temperatures for many years. In cast irons, graphite shape may be controlled to produce a variety of shapes from flakes to nodules.



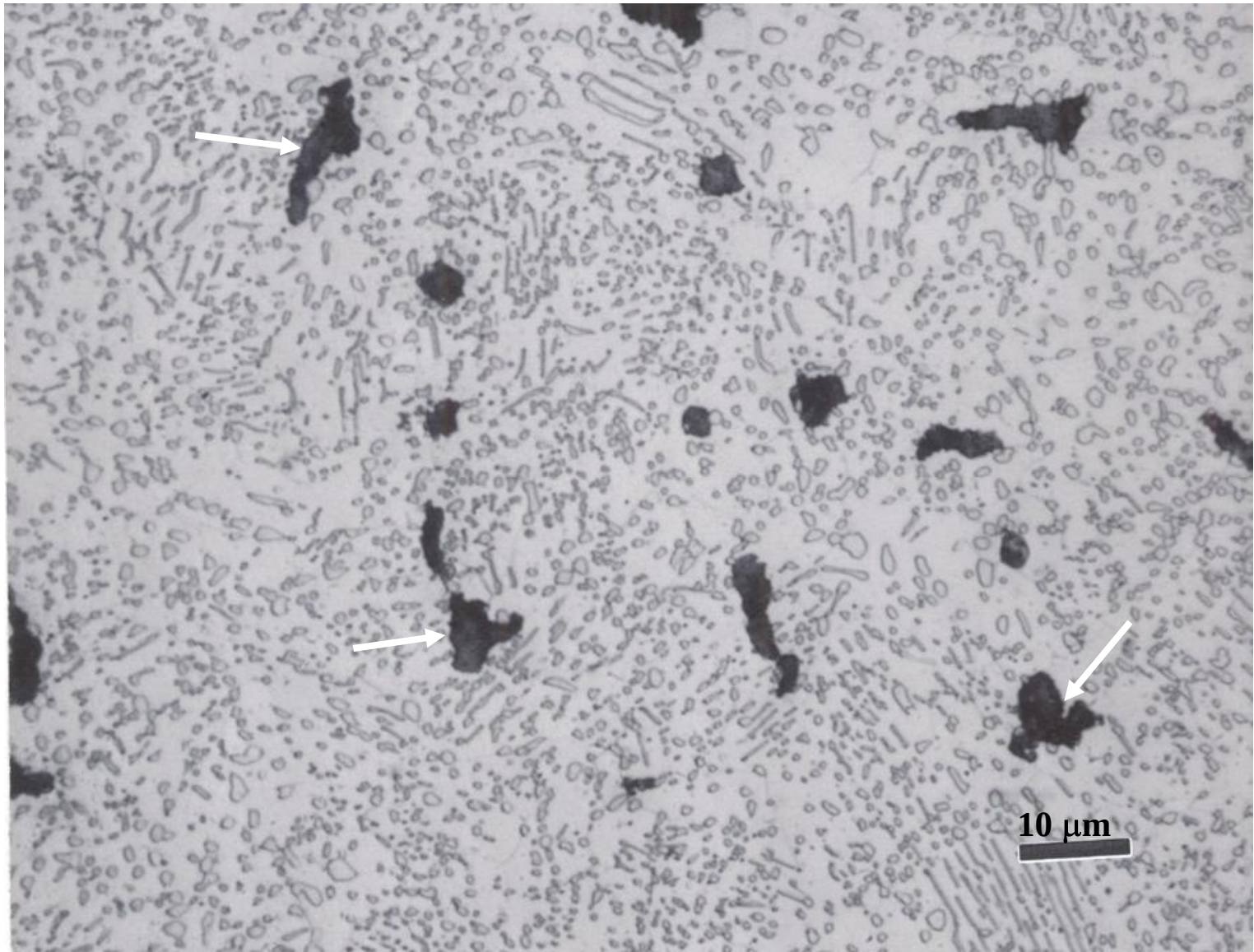
Flake-shaped graphite in gray cast iron. Not etched.



Compacted graphite (and one nodule of graphite) in CG Iron. Not etched.



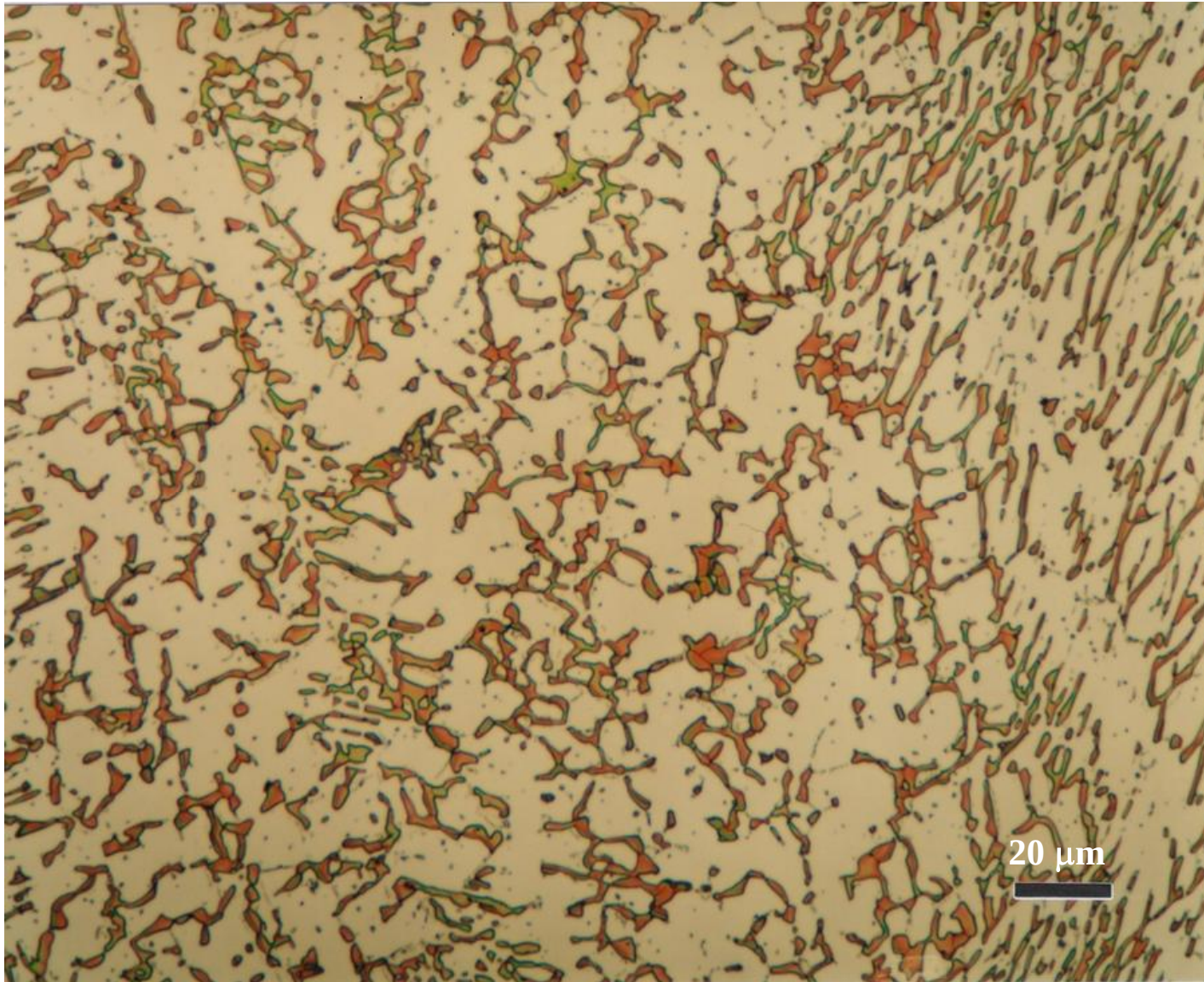
Graphite nodules in ductile iron. Not etched. Viewed with crossed polarized light plus sensitive tint.



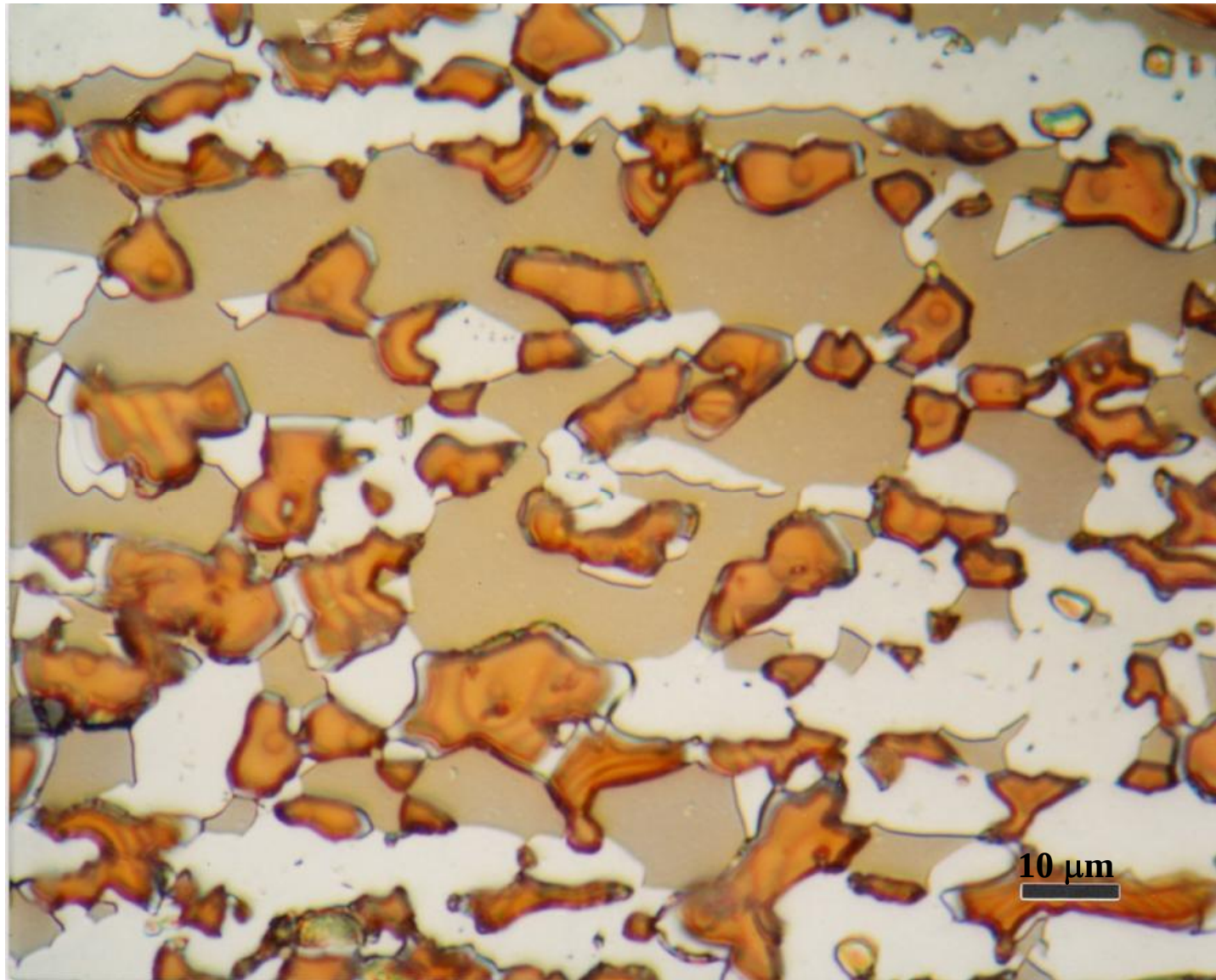
Graphite in wrought, spheroidize annealed type O6 graphitic tool steel (Fe – 1.45% C – 0.8% Mn – 1.15% Si – 0.25% Mo), viewed on a transverse plane (there is some elongation of the graphite in the hot-working direction). The matrix is revealed by etching with 4% picral. Some of the cementite is not spheroidized (note area near magnification bar).

Sigma

A brittle intermetallic phase, first observed in Fe-Cr alloys, with approximately equal amounts of Fe and Cr (with substitution) and a complex body-centered tetragonal crystal structure. Sigma forms slowly at high temperatures in ferrite (and much slower in austenite). Today, any phase with similar crystal structure and composition, regardless of the alloy system, is called sigma phase.



Sigma phase formed in a type 312 stainless steel weld (from the delta ferrite phase) by aging at 816 °C for 160 h. Sigma was colored green and orange by etching with Murakami's reagent (10 g sodium hydroxide, 10 g potassium ferricyanide, 100 mL water) for 60 s at 80 °C.



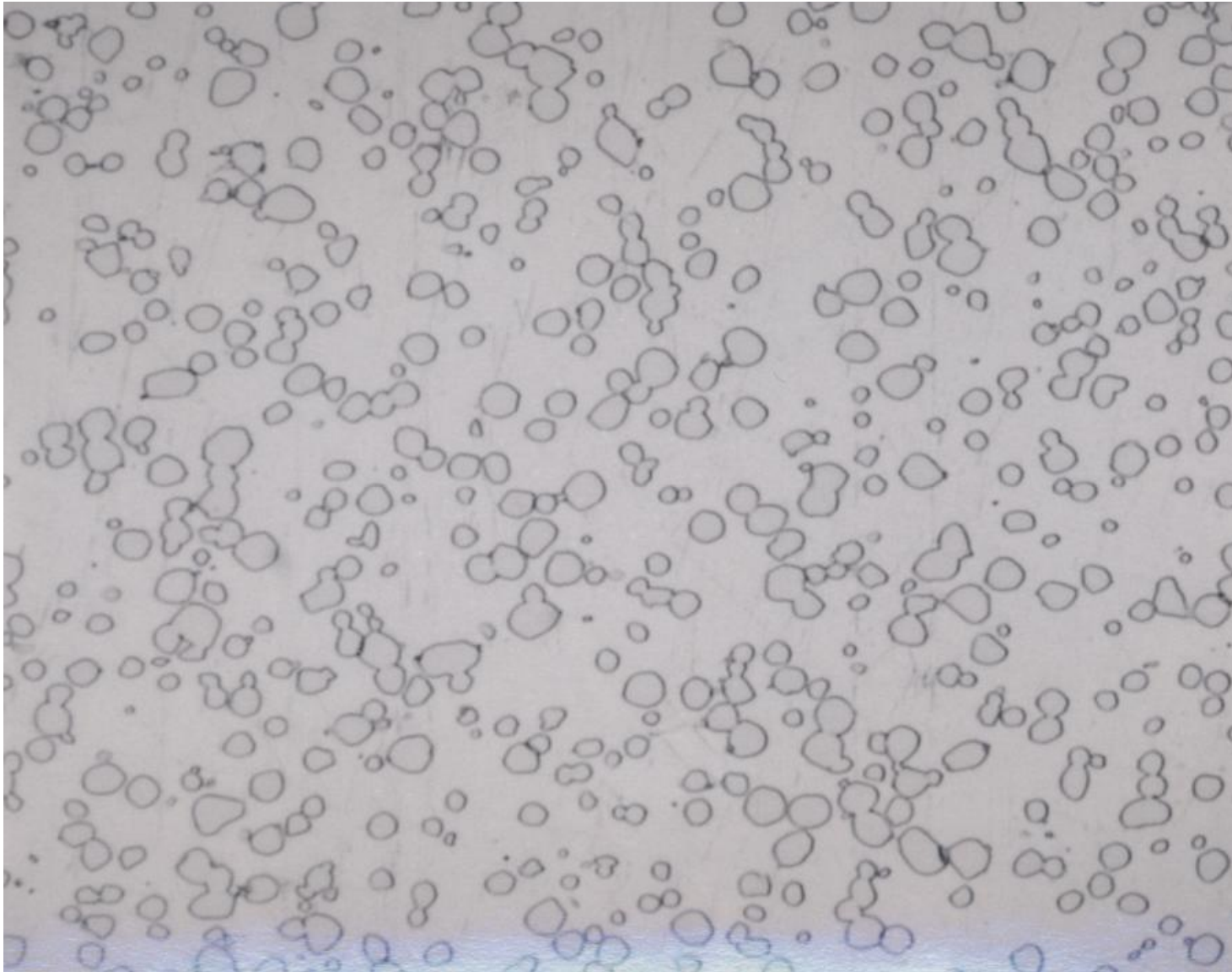
Microstructure of wrought 7-Mo duplex stainless steel (Fe - <0.1% C – 27.5% Cr – 4.5% Ni – 1.5% Mo) solution annealed and then aged 48 h at 816 °C to form sigma. Electrolytic etching with aqueous 20% NaOH (3 V dc, 10 s) revealed the ferrite as tan, the sigma orange, while the austenite was not colored.

Other Intermetallic Phases

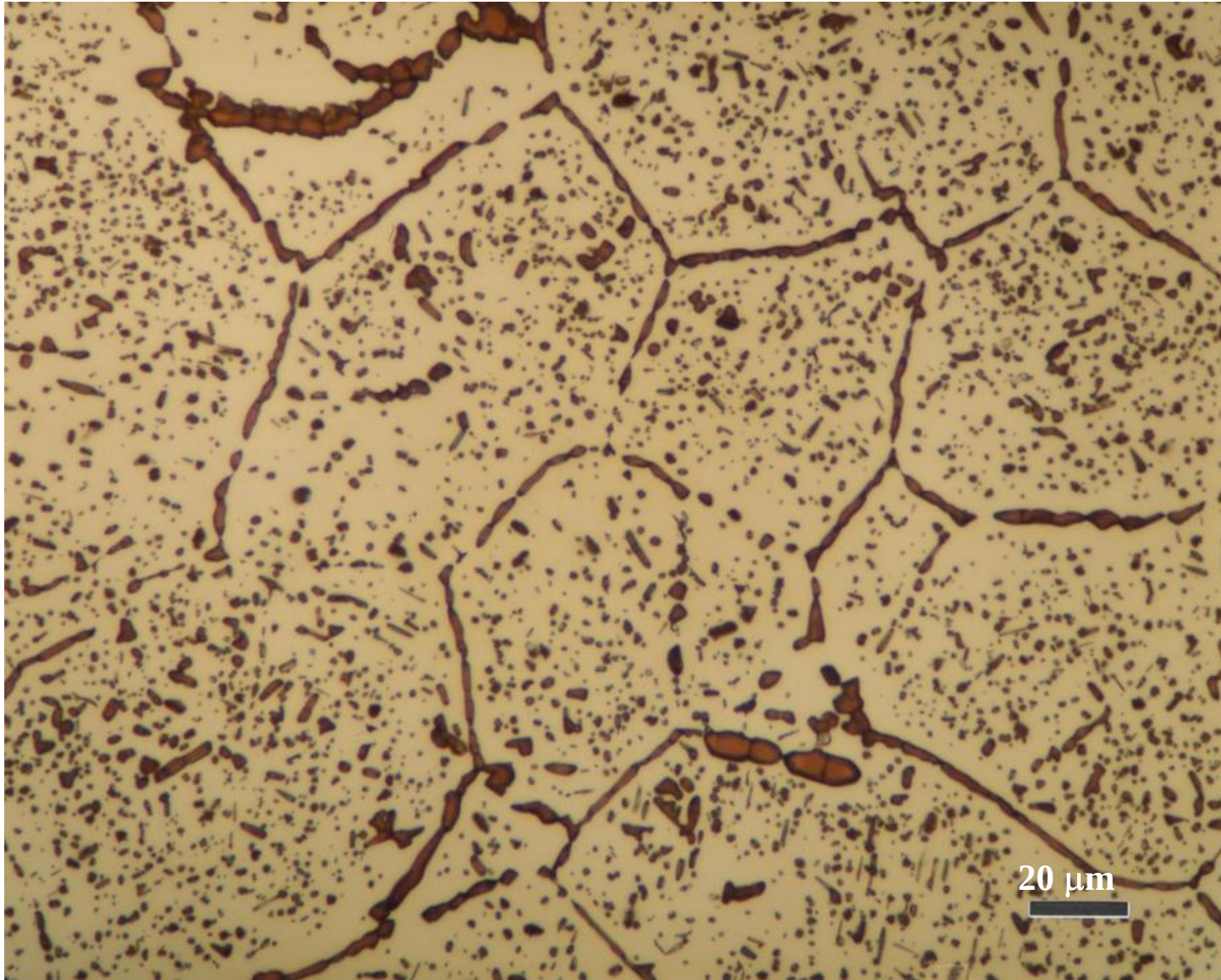
A number of intermetallic phases have been observed in highly alloyed steels, particularly stainless steels after high temperature exposure. High molybdenum content stainless steels may form sigma and chi and at higher levels, chi and Laves. Chi is a complex Fe-Cr-Mo phase with a cubic crystal structure while Laves has the general composition A_2B , as in Fe_2Mo , with a hexagonal crystal structure. Borides are rarely observed in steels, except for boronated stainless steels containing up to 2 wt. % B.



As-cast microstructure of type 304 stainless steel containing 2% B (Cr rebalanced) tint etched with Beraha's reagent (10% HCl plus 1 g/100 mL potassium metabisulfite) to color the austenite. The chromium boride, Cr_2B , is unaffected.



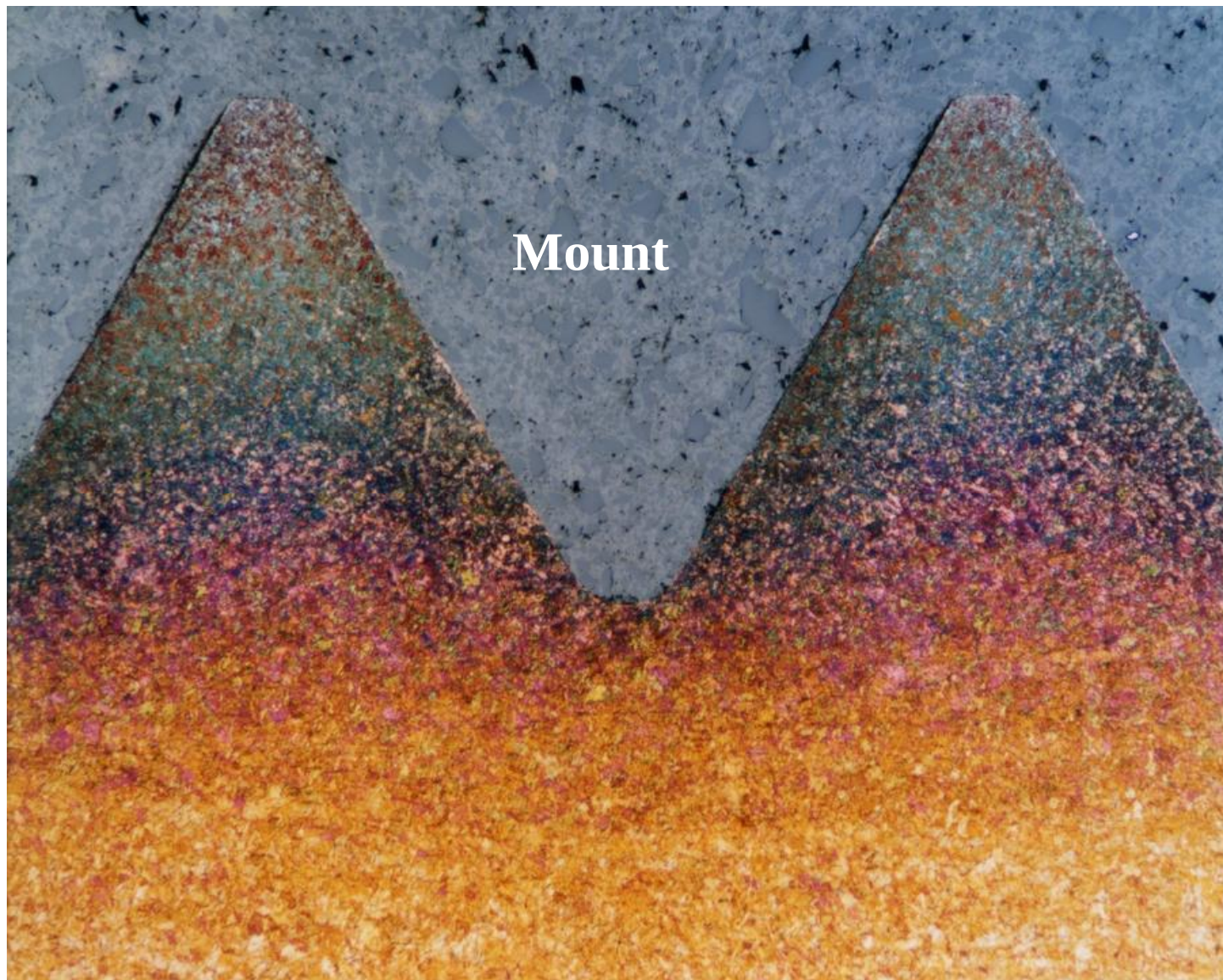
Microstructure of wrought, solution annealed P/M 304 stainless steel with 1.75% B showing chromium boride particles, Cr_2B , in an austenitic matrix after light etching with Kalling's No. 2 reagent. Original at 1000X.



Microstructure of SCF-23 stainless steel (Fe – 0.02% C – 4.27% Mn – 22.43% Cr – 17.78% Ni – 5.54% Mo – 0.40% N) in the solution annealed condition after aging 48 h at 927 °C) forming chi phase in the grain boundaries and within the grains. The chi particles were colored using Murakami's reagent (10 g sodium hydroxide, 10 g potassium ferricyanide, 100 mL water at 80 °C for 30 s).

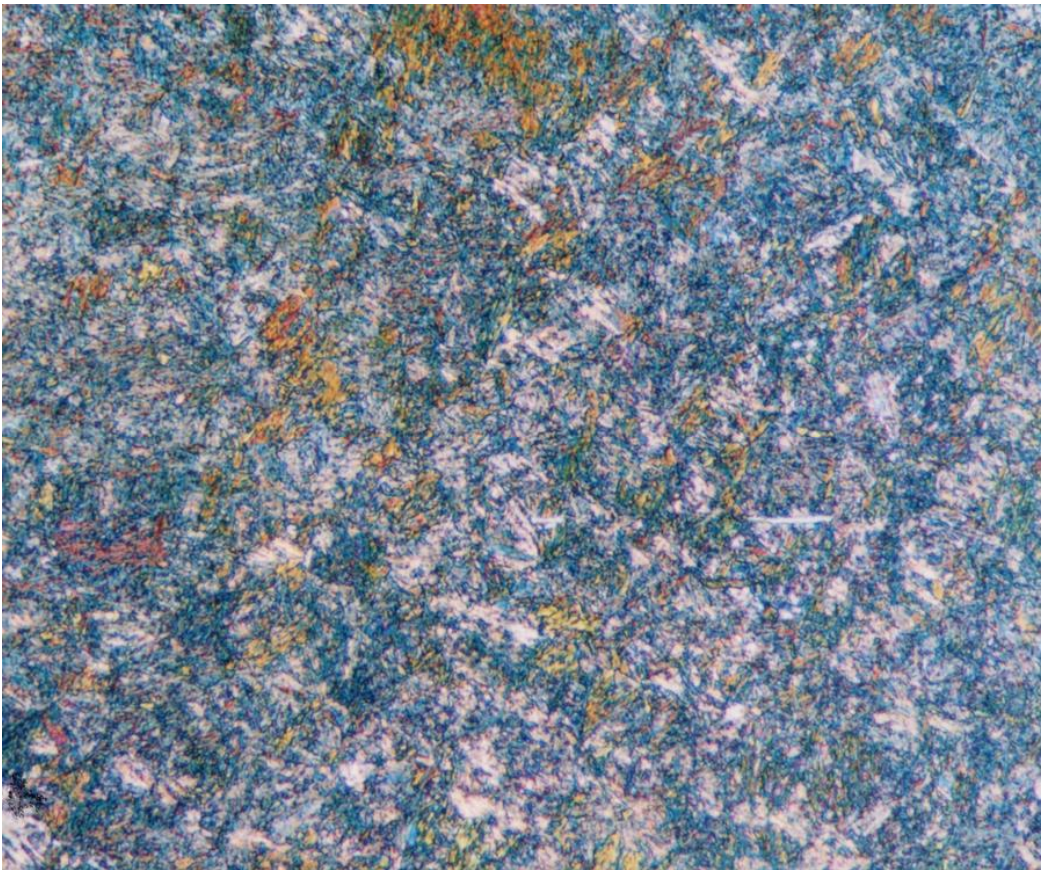
Surface Treatments

The surface chemistry can be modified by diffusion of carbon, nitrogen (carbon and nitrogen) or boron, as in carburizing, nitriding, carbonitriding or boriding, to change the properties of the alloy, aided by heat treatments. Surfaces can also be plated electrolytically or by hot-dipping, or by other processes such as CVD, PVD or ion-implantation.



Carburized gear teeth made with 8620 alloy steel (Fe – 0.2% C – 0.8% Mn – 0.25% Si – 0.55% Ni – 0.5% Cr – 0.2% Mo) with a gradation of carbon and microstructure revealed by tint etching with Beraha's sulfamic acid reagent (100 mL water, 3 g potassium metabisulfite, 2 g sulfamic acid and 0.7 g ammonium bifluoride) and viewed with polarized light plus sensitive tint. Original at 50X.

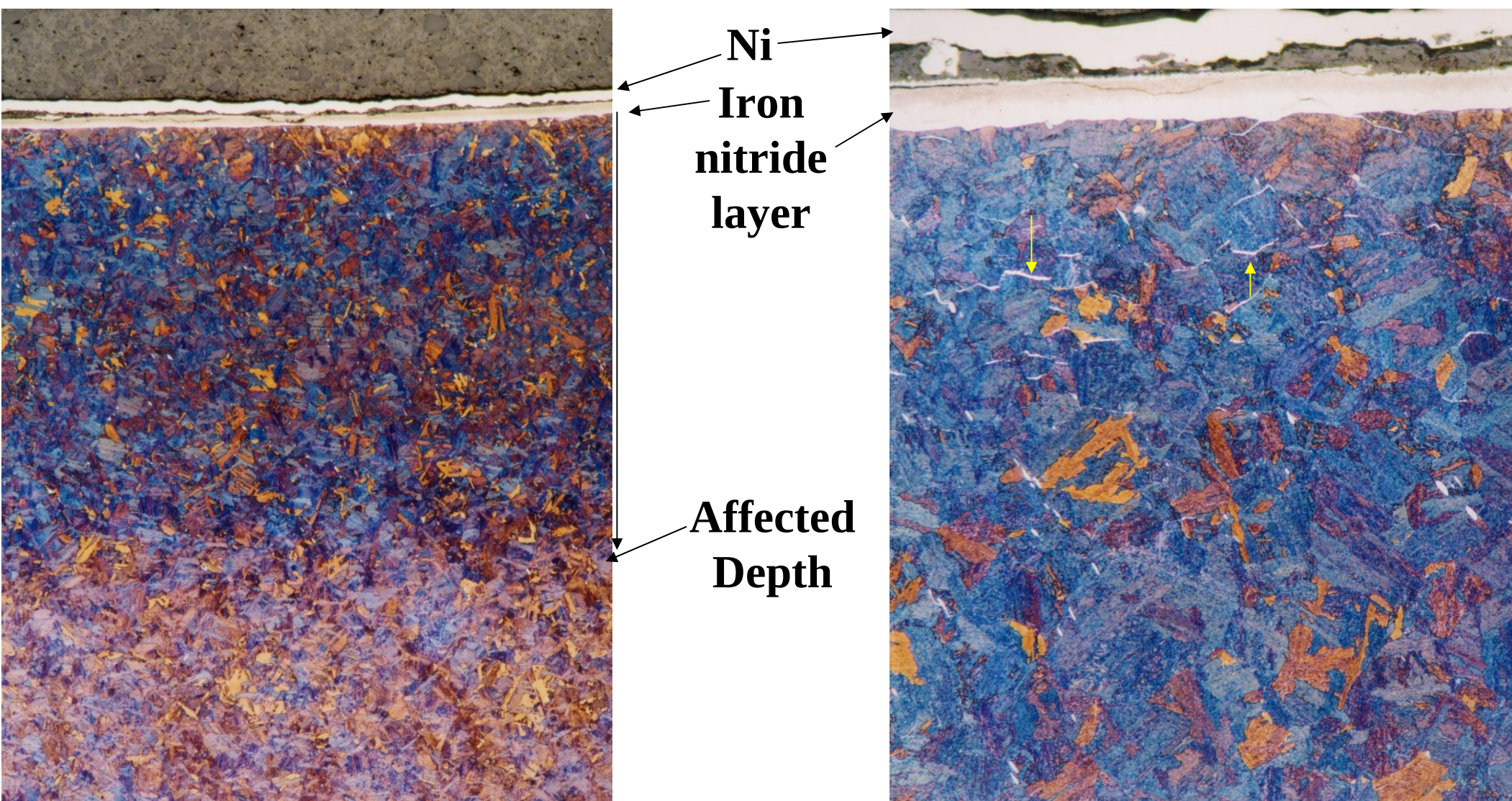
Case



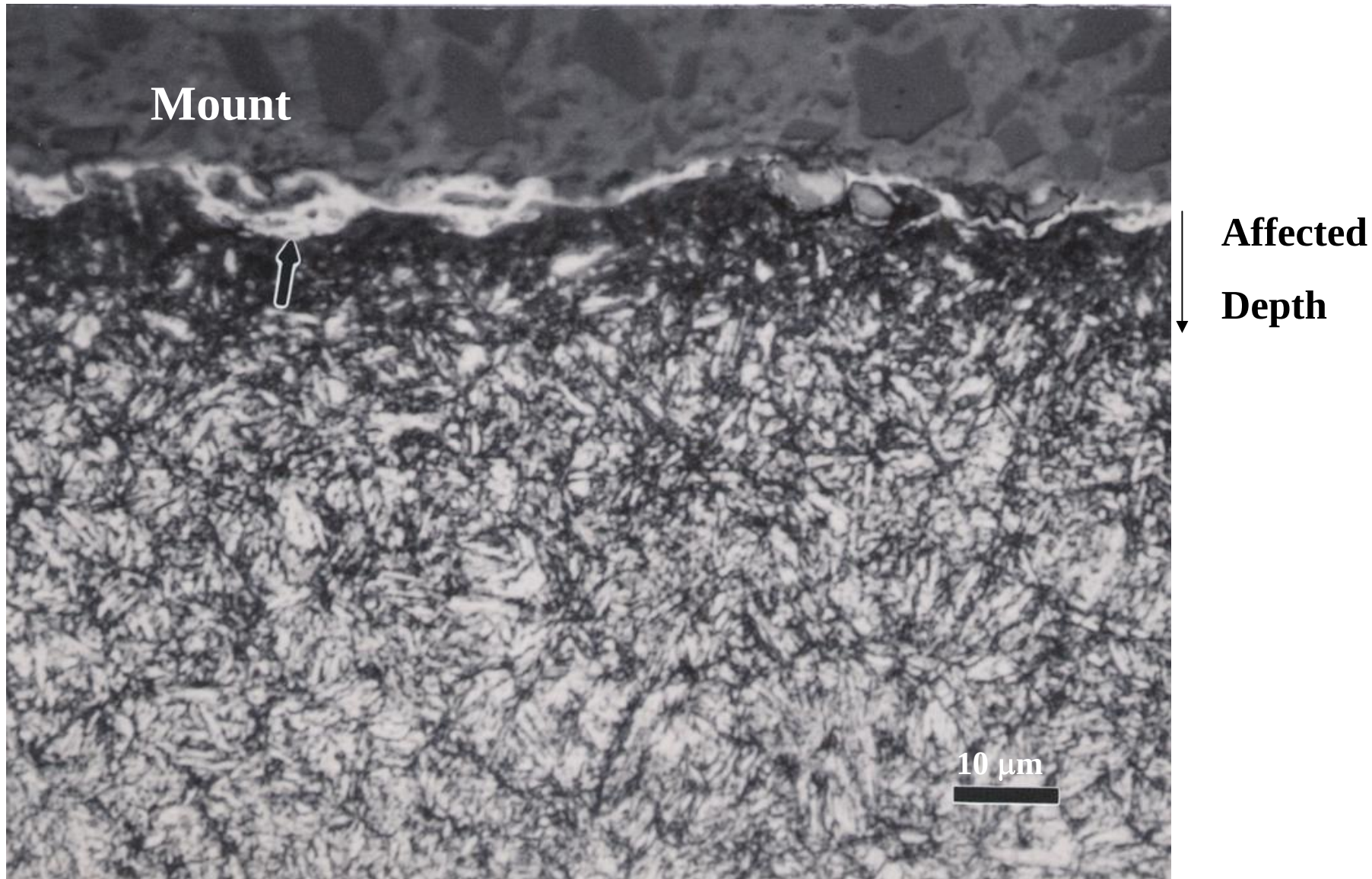
Core



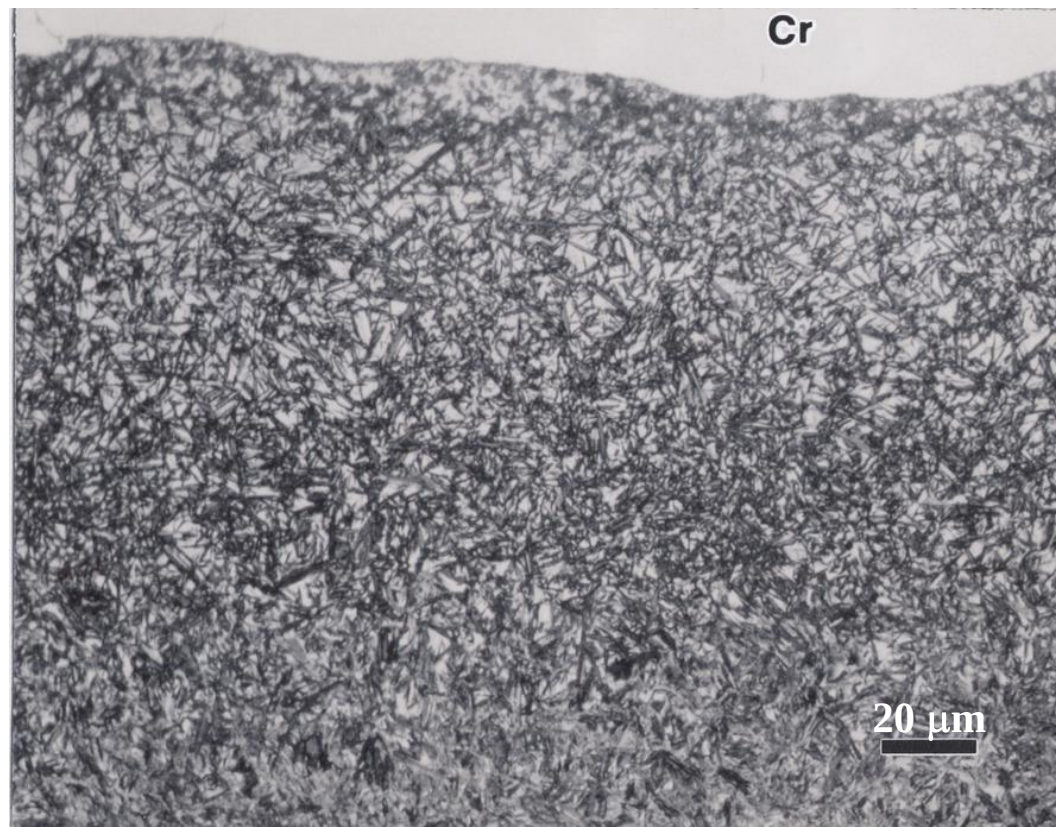
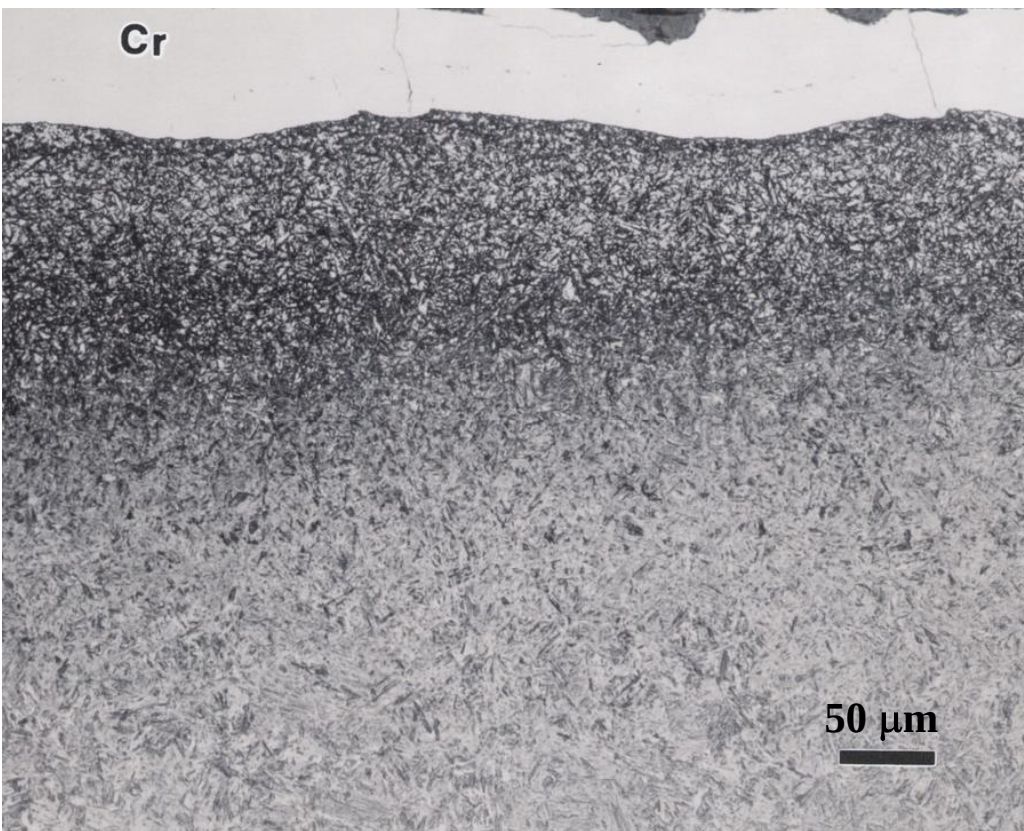
Case and core microstructures of the carburized 8620 alloy steel specimen in the previous slide revealing high-carbon plate martensite and low-carbon lath martensite, respectively, after tint etching and viewing with polarized light and sensitive tint. Originals at 500X.



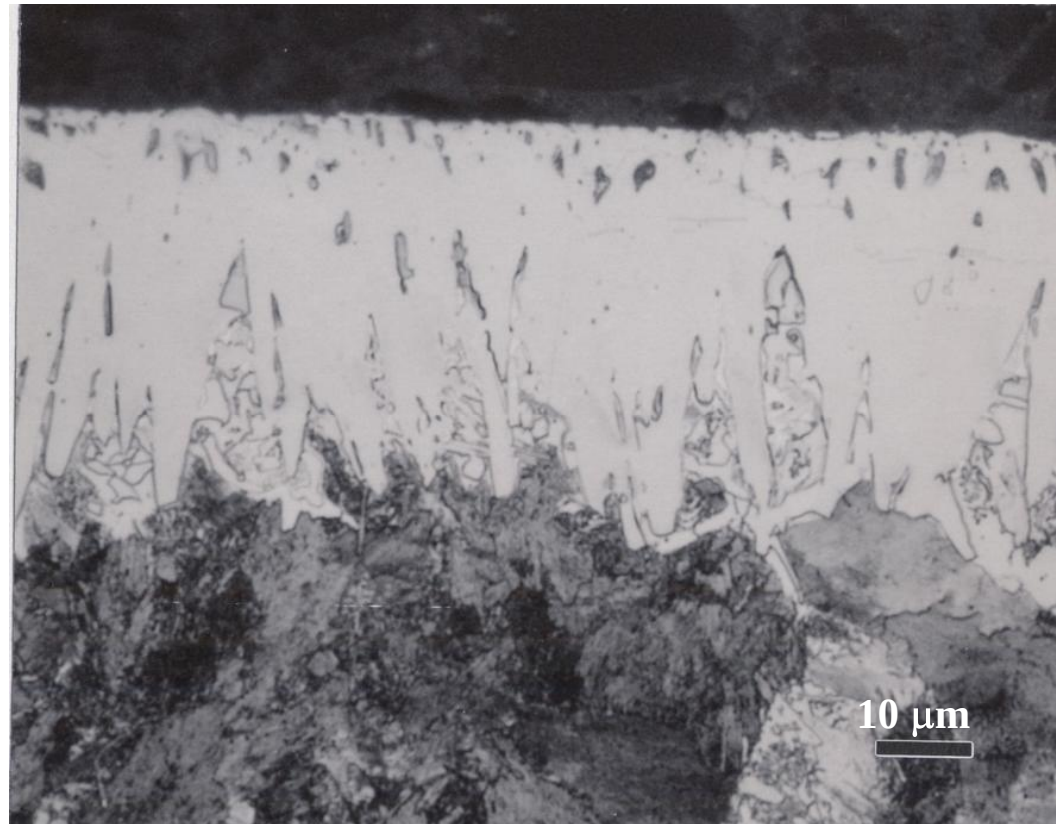
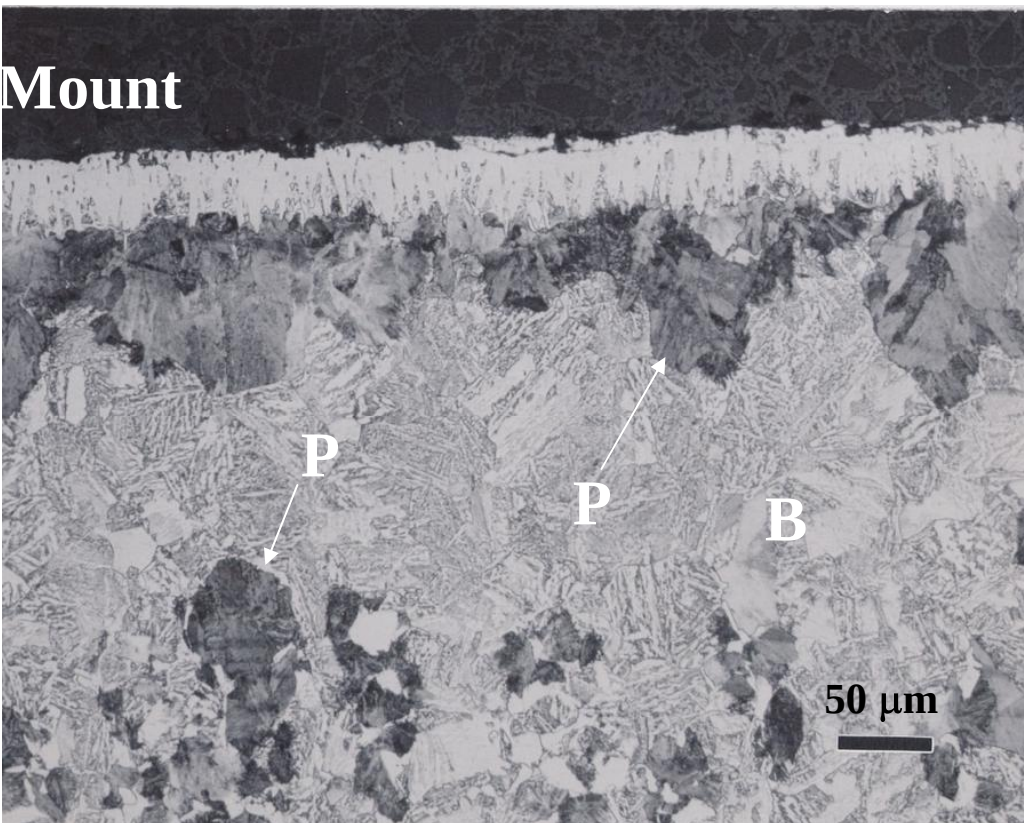
Microstructure of gas nitrided 41B50 alloy steel (Fe – 0.5% C – 0.85% Mn – 0.25% Si – 0.95% Cr – 0.2% Mo – 0.003% B) previously quenched and tempered to 325 HV (tempered martensite). The surface was electroless nickel plated before mounting as the outer edge of the iron nitride “white layer” is considerably softer than the inner layer (very hard and brittle). The left view (original at 50X) reveals the depth of the nitrided case. The right view (original at 200X) reveals some intergranular nitride films (yellow arrows). Tint etched with Beraha's CdS reagent.



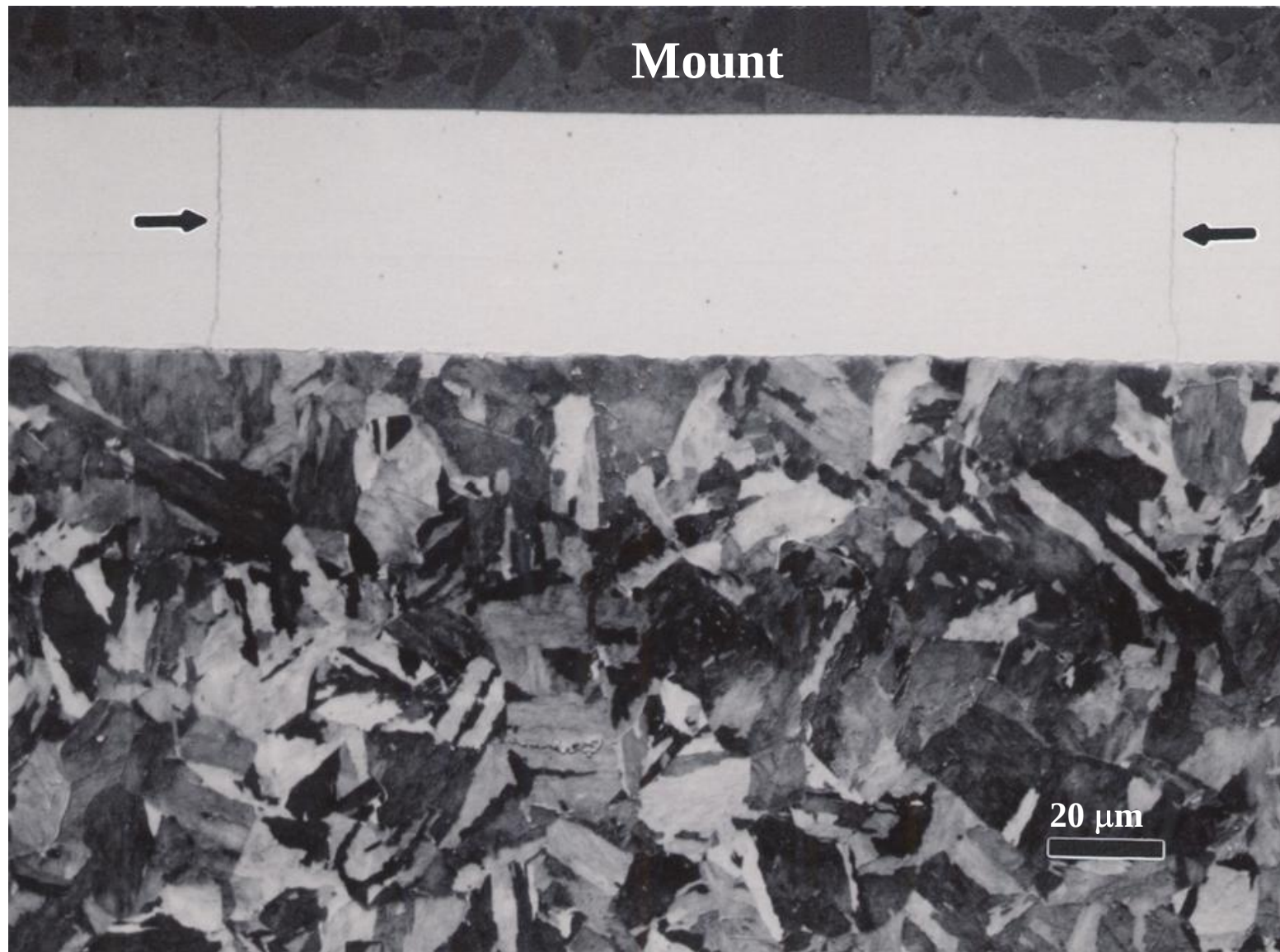
Ion nitrided surface of type H13 tool steel (Fe – 0.4% C – 0.35% Mn – 1% Si – 5% Cr – 0.9% V – 1.35% Mo) previously quenched and tempered. The arrow points to a white, iron nitride layer at the surface. The matrix is tempered martensite. Etch with Vilella's reagent.



Microstructure of carbonitrided type 8720 alloy steel (Fe – 0.2% C – 0.8% Mn – 0.25% Si – 0.55% Ni – 0.5% Cr – 0.25% Mo) after quench and tempering. The surface was plated with Cr for edge protection. There is considerable retained austenite (white patches) between martensite plates (black) in the carbonitrided surface. The specimen was etched with 2% nital.



Microstructure of a borided 42CrMo4 as-rolled alloy steel (Fe – 0.42% C – 0.75% Mn – 0.4% Si – 1.1% Cr – 0.2% Mo). Note the acicular nature of the iron-boride layer. The matrix contains pearlite (P) and bainite (B). Etched with 2% nital.



Microstructure of chromium plated 18Ni300 maraging steel (Fe - $<0.03\%$ C - $<0.1\%$ Mn and Si, 18.5% Ni – 8.75% Co – 4.9% Mo – 0.65% Ti – 0.1% Al) in the solution annealed and aged condition. Note the cracks (arrows) in the Cr plating which are expected. Etched with modified Fry's reagent.

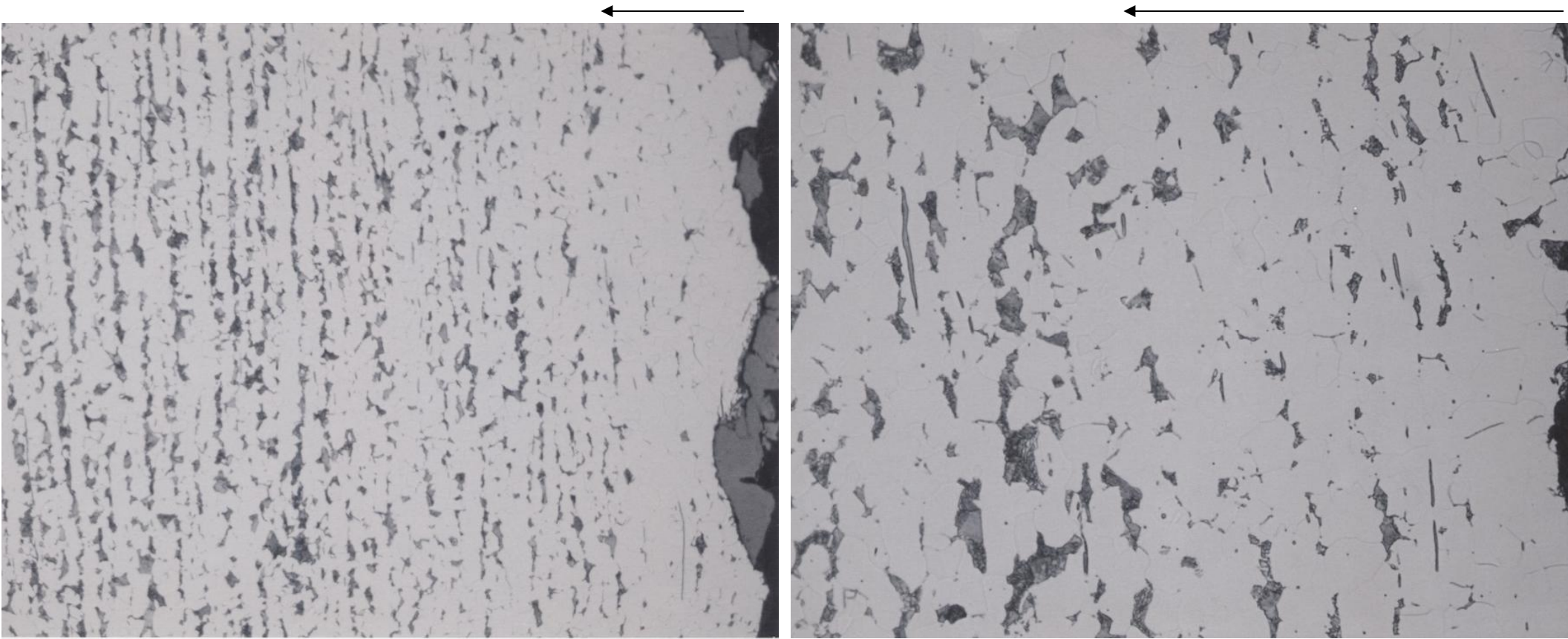
Decarburization

When iron-based alloys are heated in air, carbon will diffuse outward depleting the surface. Decarburization can occur in hot working or in heat treatment. If the surface of a heat treated part is decarburized there will be a loss in hardness which reduces wear resistance and fatigue life.

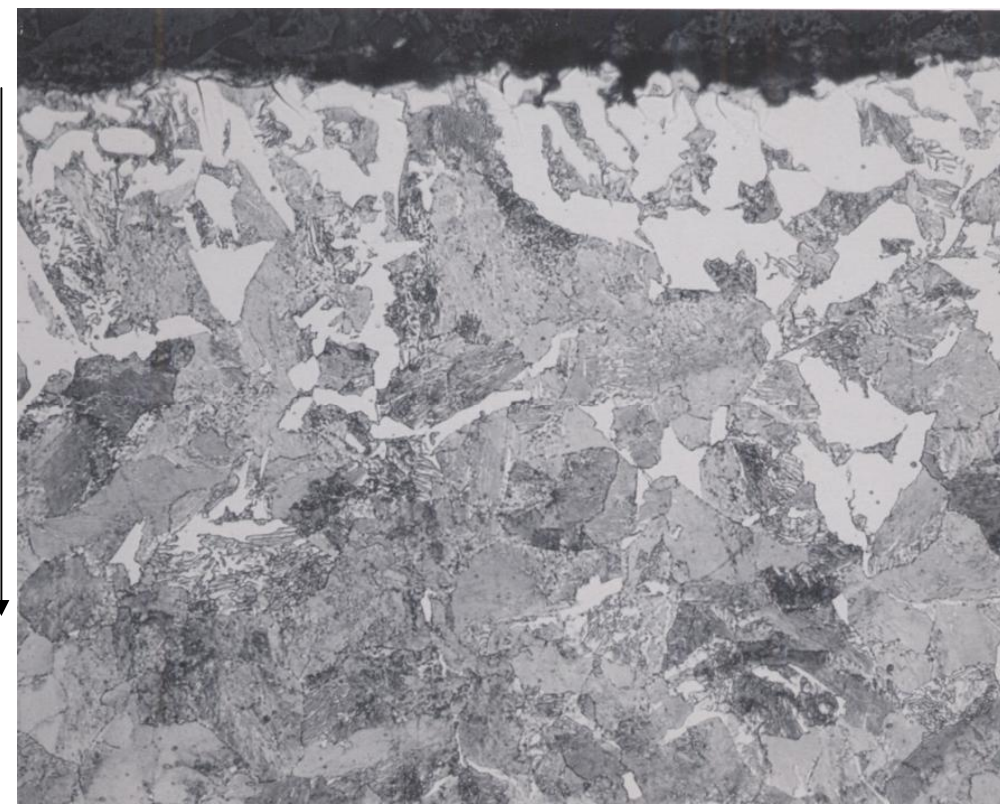
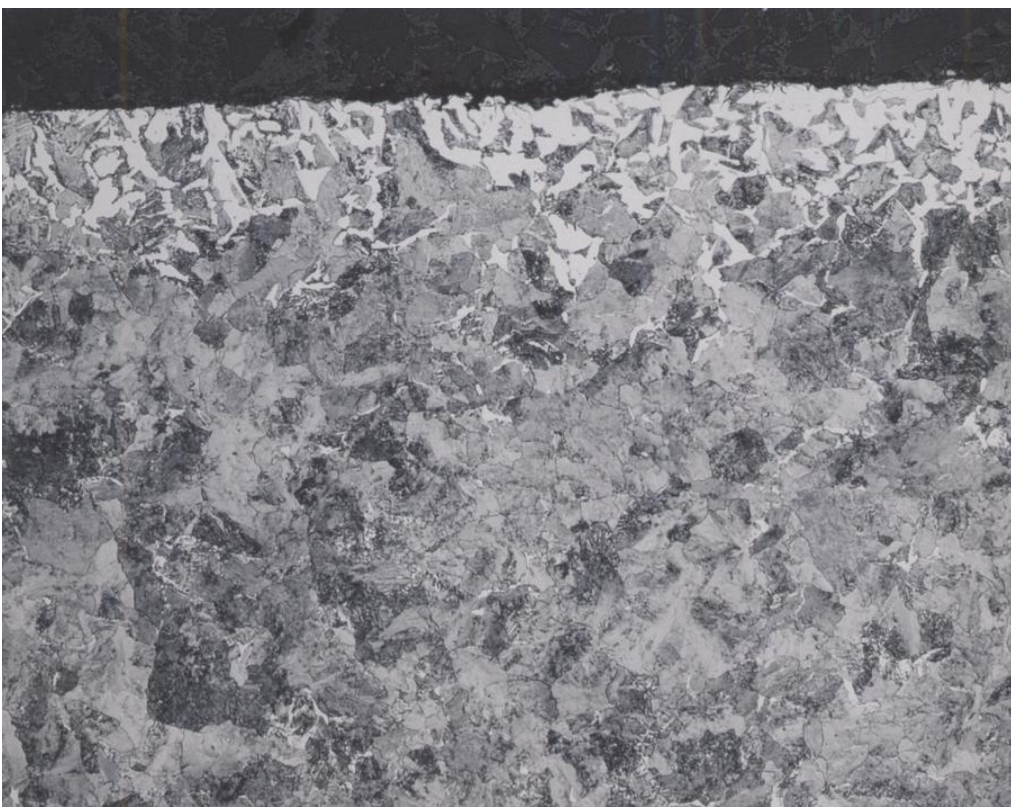


**Maximum
Affected
Depth**

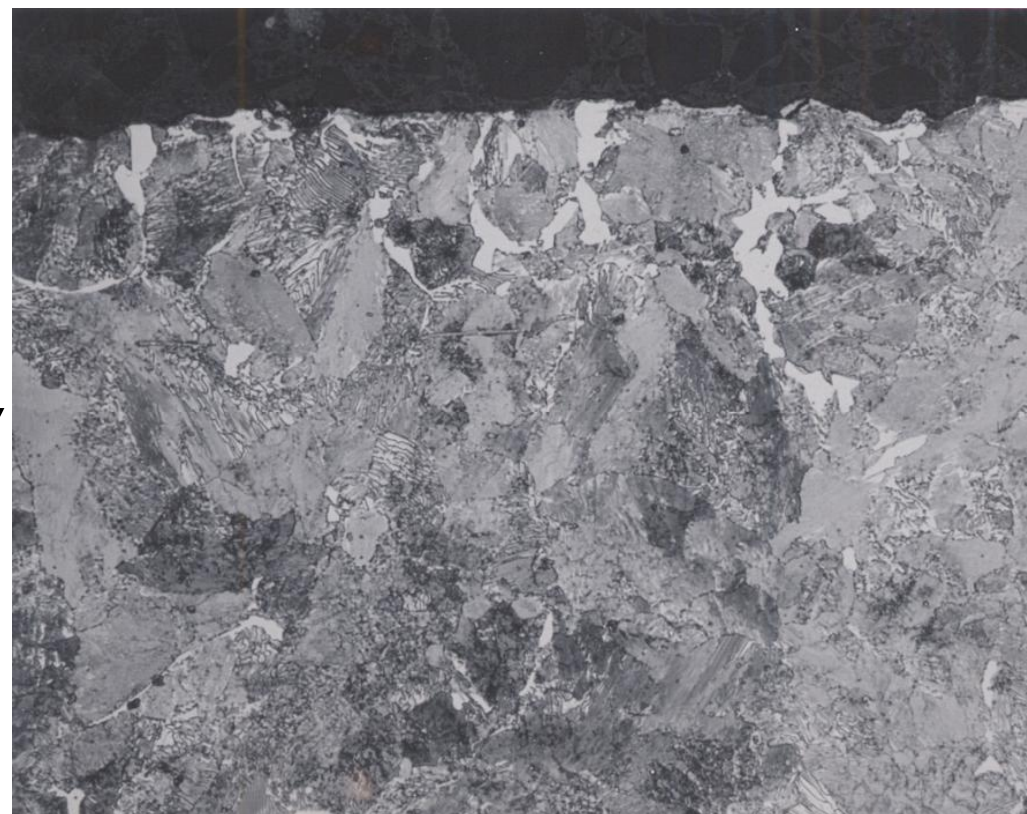
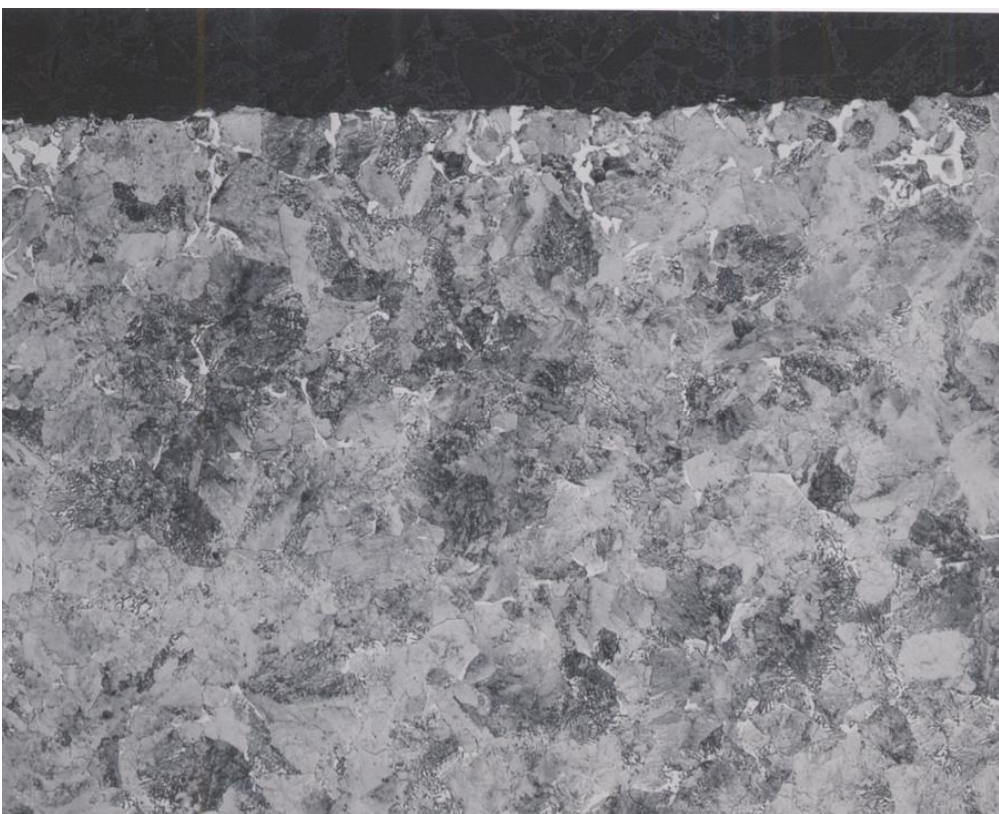
Decarburized surface of an as-rolled carbon steel (Fe – 0.11% C – 0.85% Mn – 0.21% Si) with a ferrite-pearlite microstructure. Note that the amount of pearlite present is reduced in the surface zone due to loss of carbon during the hot working process. Etched with 4% picral. Original is at 200X.



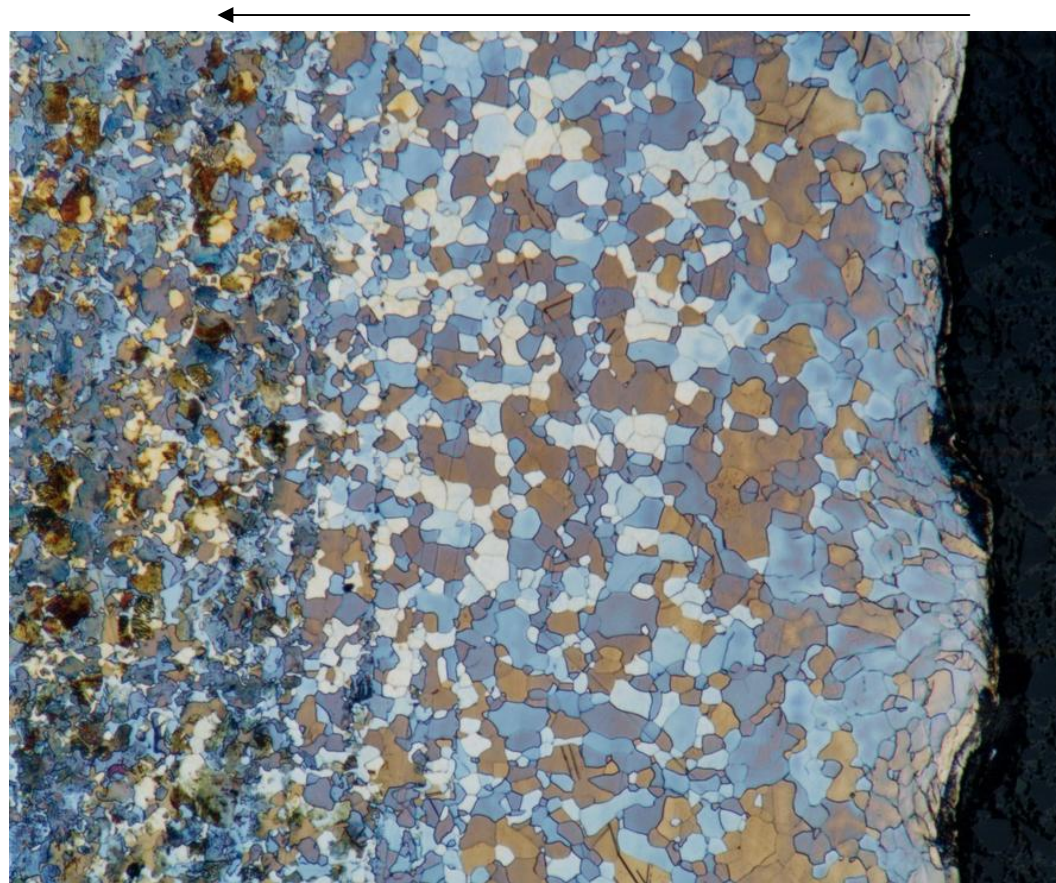
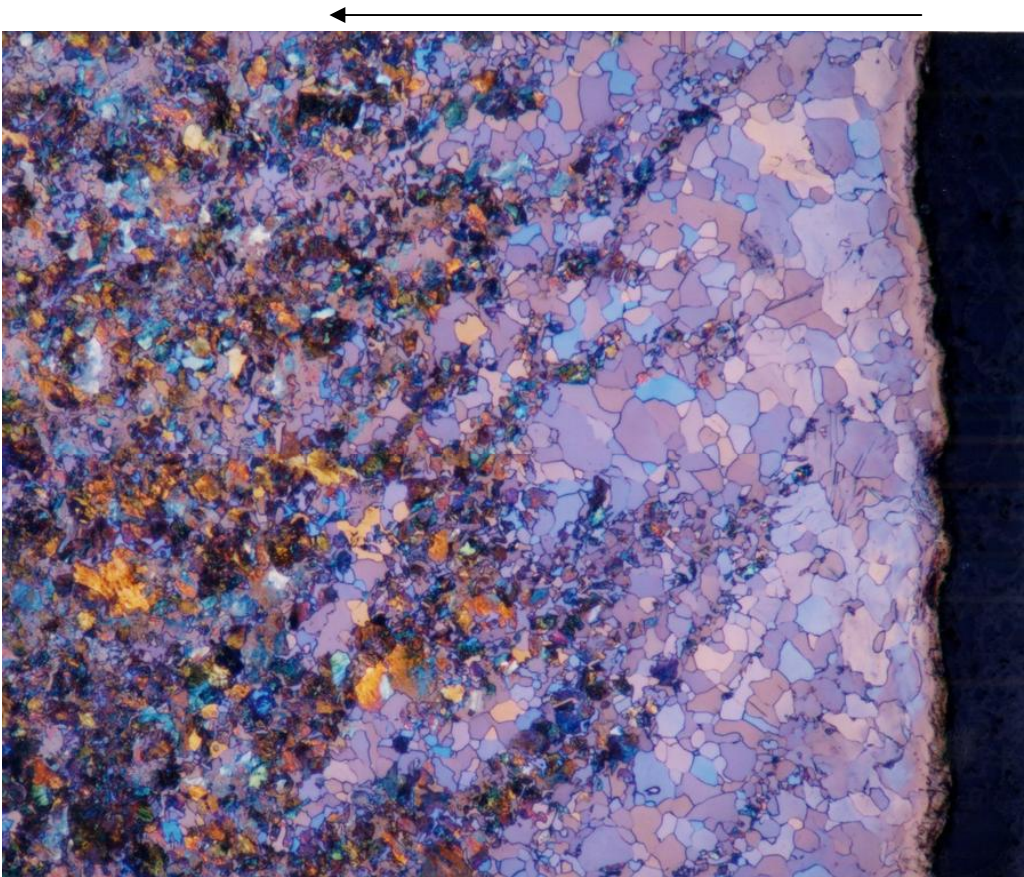
Decarburized surface of a hot rolled carbon steel (Fe – 0.23% C – 0.85% Mn – 0.22% Si) due to loss of carbon during the hot working process. The arrows show the approximate maximum affected depth where the amount of pearlite becomes normal. There is scale (S) on the surface, which is not flat. The specimen was etched with 4% picral. Originals at 200X (left) and 500X (right).



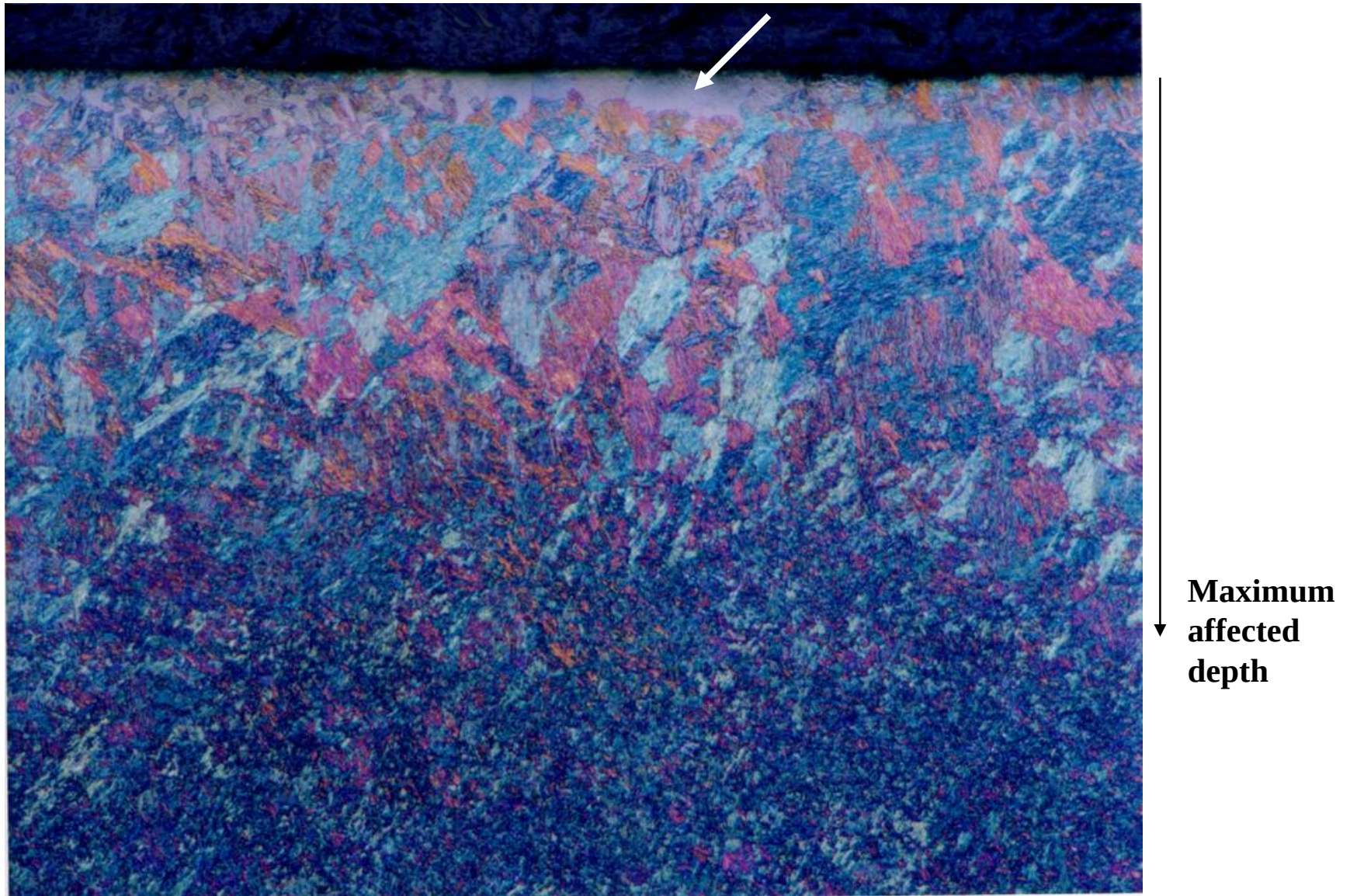
Decarburized surface of an as-rolled, eutectoid carbon steel (Fe – 0.8% C – 0.21% Mn – 0.22% Si). The matrix should be nearly all pearlitic, but there is ferrite at the surface to a considerable depth (arrows). The specimen was etched with 4% picral. The original magnifications were 200X (left) and 500X (right).



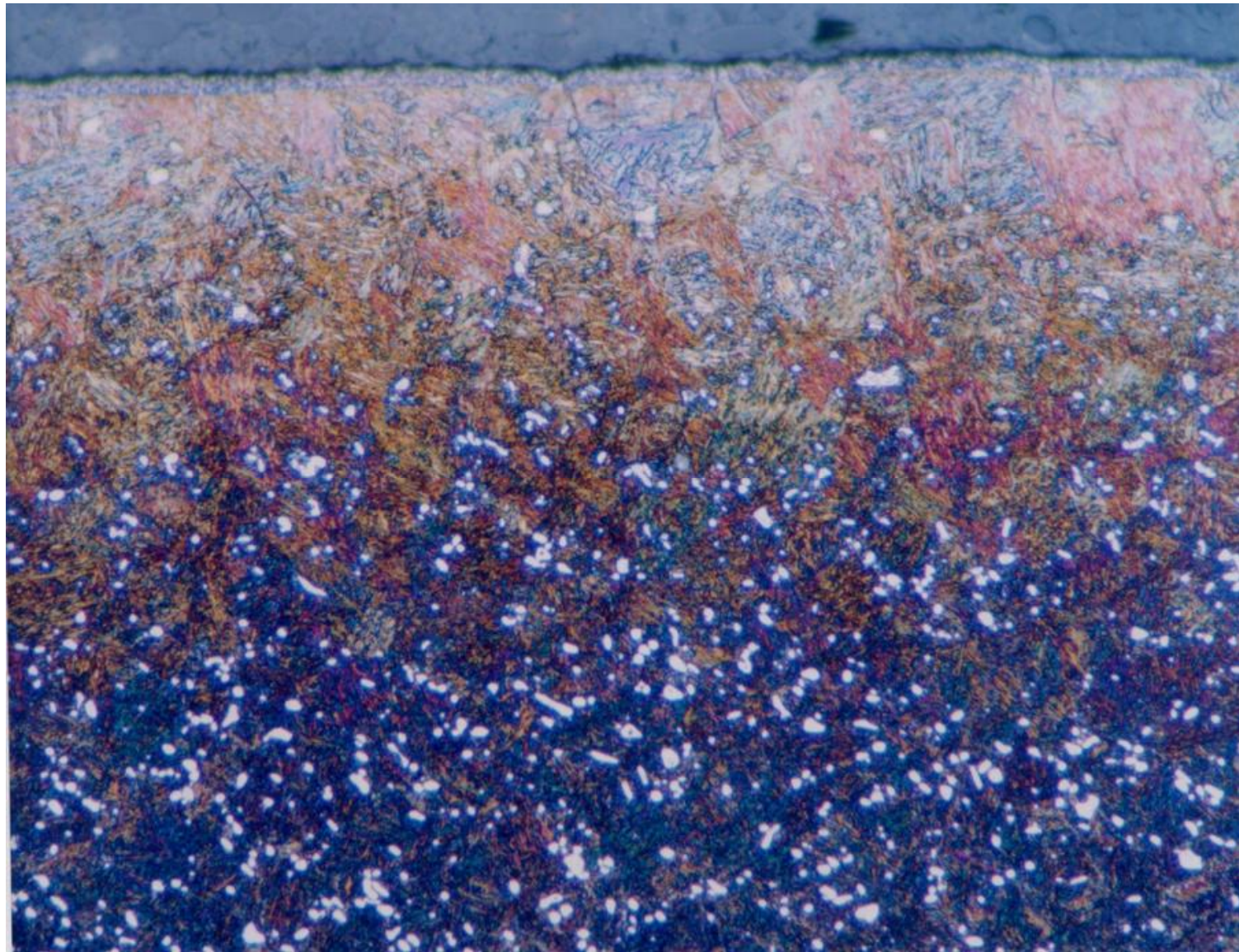
Decarburization in a different location than shown on the previous slide for the as-rolled eutectoid carbon steel. Note that the loss of carbon, as shown by the amount of ferrite present, is less at this location. 4% picral etch. Originals were at 200X (left) and 500X (right).



Decarburized, fully annealed type 4140 alloy steel (Fe – 0.4% C – 0.85% Mn – 0.25% Si 0.95% Cr – 0.25% Mo) color etched with Beraha's sulfamic acid reagent (100 mL water, 3 g potassium metabisulfite, 2 g sulfamic acid). Different areas on the same specimen. Originals were both at 100X magnification but the left image is viewed with polarized light and sensitive tint while the right image is in bright field.

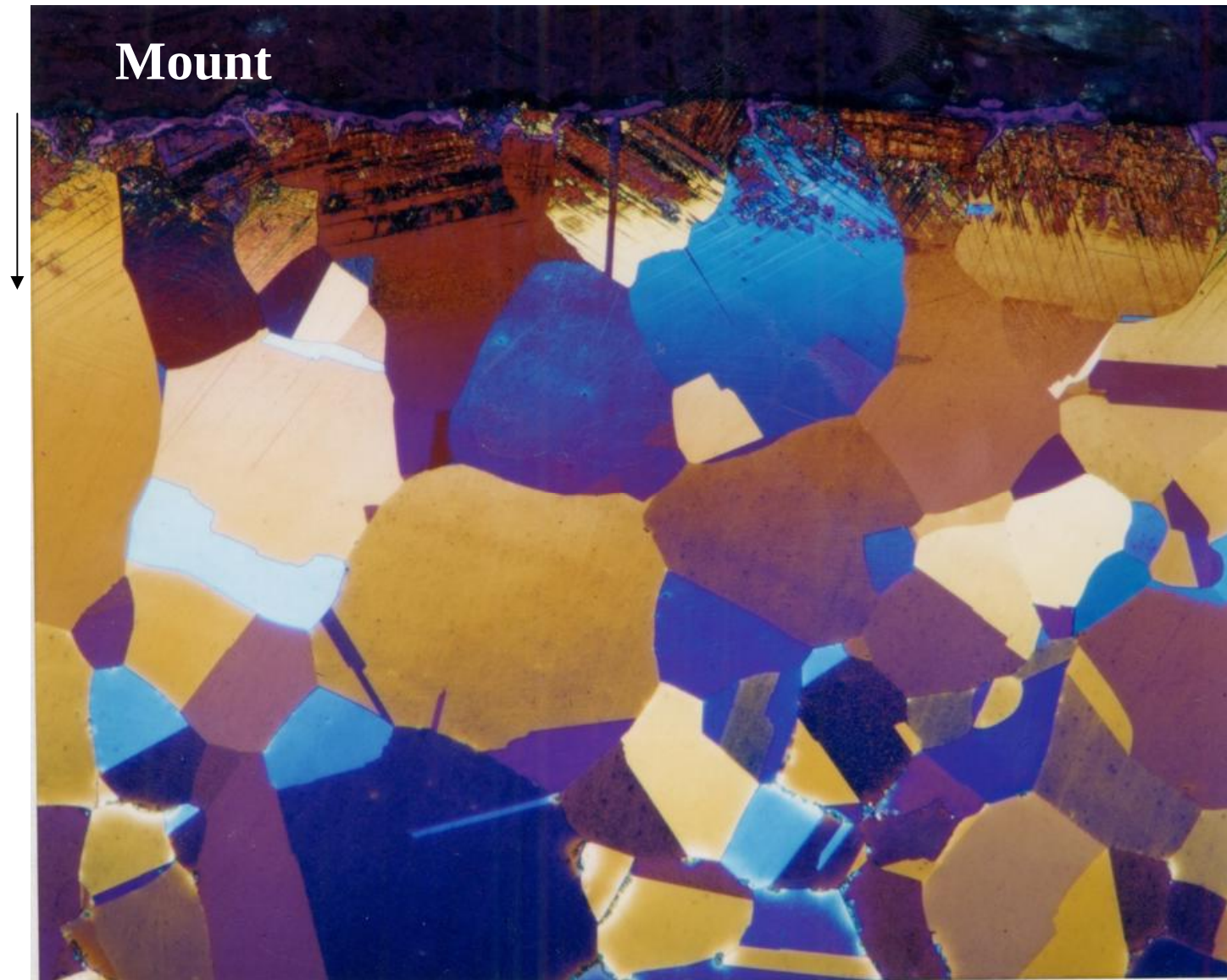


Decarburization in quenched and tempered 420 martensitic stainless steel (Fe – 0.35% C – 13% Cr) revealed by color etching with Beraha's sulfamic acid etch (100 mL water, 3 g $K_2S_2O_5$, 2 g NH_2SO_3H and 0.7 g NH_4FHF) . Note the change in appearance of the martensite, a coarsening of the grain size until free ferrite (arrow) is observed at the surface. Original at 100X magnification. Viewed with polarized light plus sensitive tint.



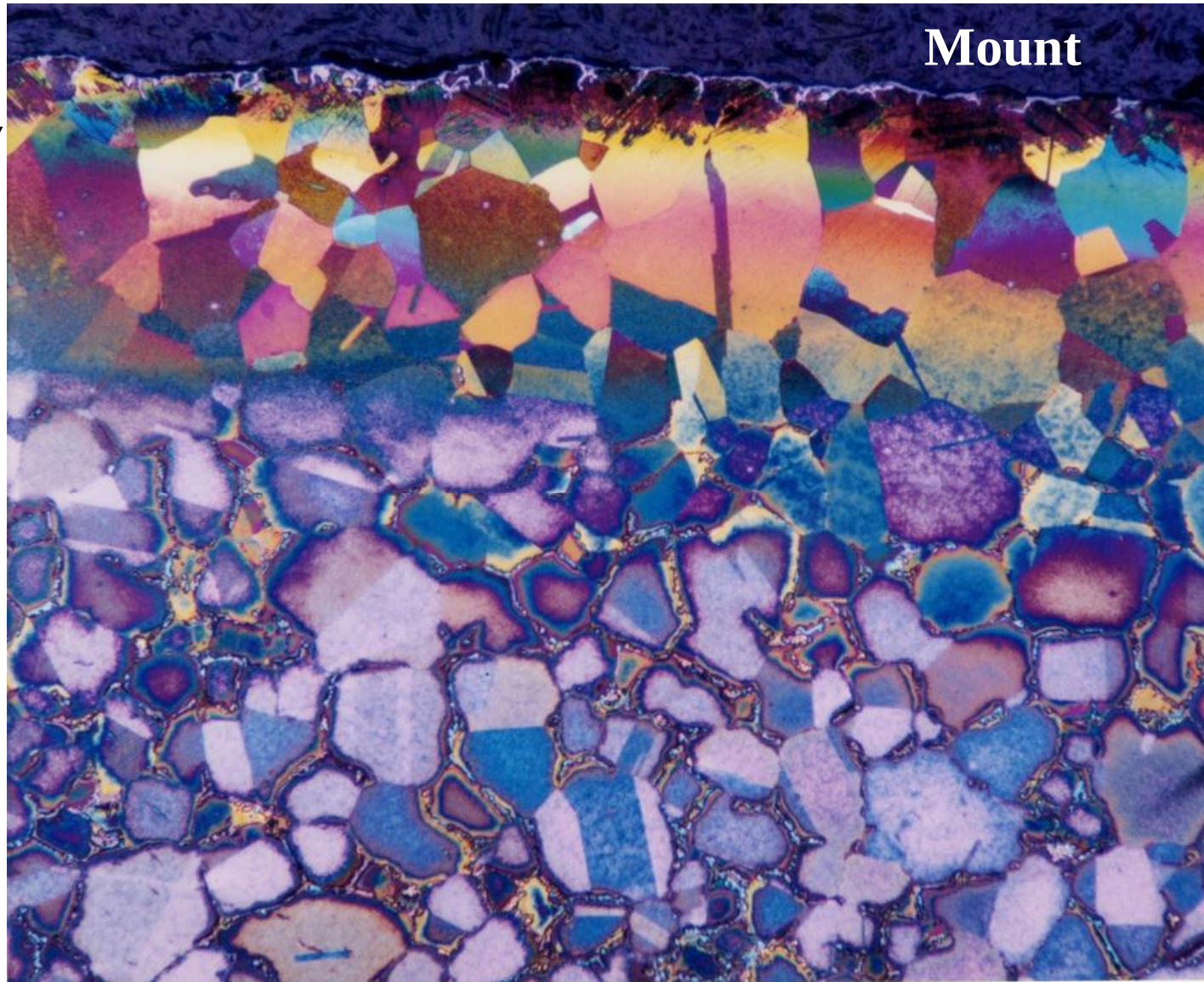
Maximum
affected
depth

Decarburization at the surface of as-quenched (1095 °C – oil quench) type D3 tool steel (Fe – 2.1% C – 12.5% Cr – 0.5% Ni) revealed by color etching with Beraha's sulfamic acid etch (100 mL water, 3 g $K_2S_2O_5$, 2 g NH_2SO_3H and 0.7 g NH_4FHF). Note the gradual loss of primary carbides as the surface is approached and a change in the coloring of the martensite due to a change from high carbon to low carbon martensite. Original magnification 200X.



Decarburized surface of wrought Hadfield manganese steel (Fe – 1.2% C – 13% Mn) color etched with aqueous 10% sodium metabisulfite. Where the carbon is <0.5%, epsilon martensite has formed (arrow). Original at 100X. Viewed with polarized light plus sensitive tint.

**Depth to
0.5% C**



**Depth to
1.16% C**

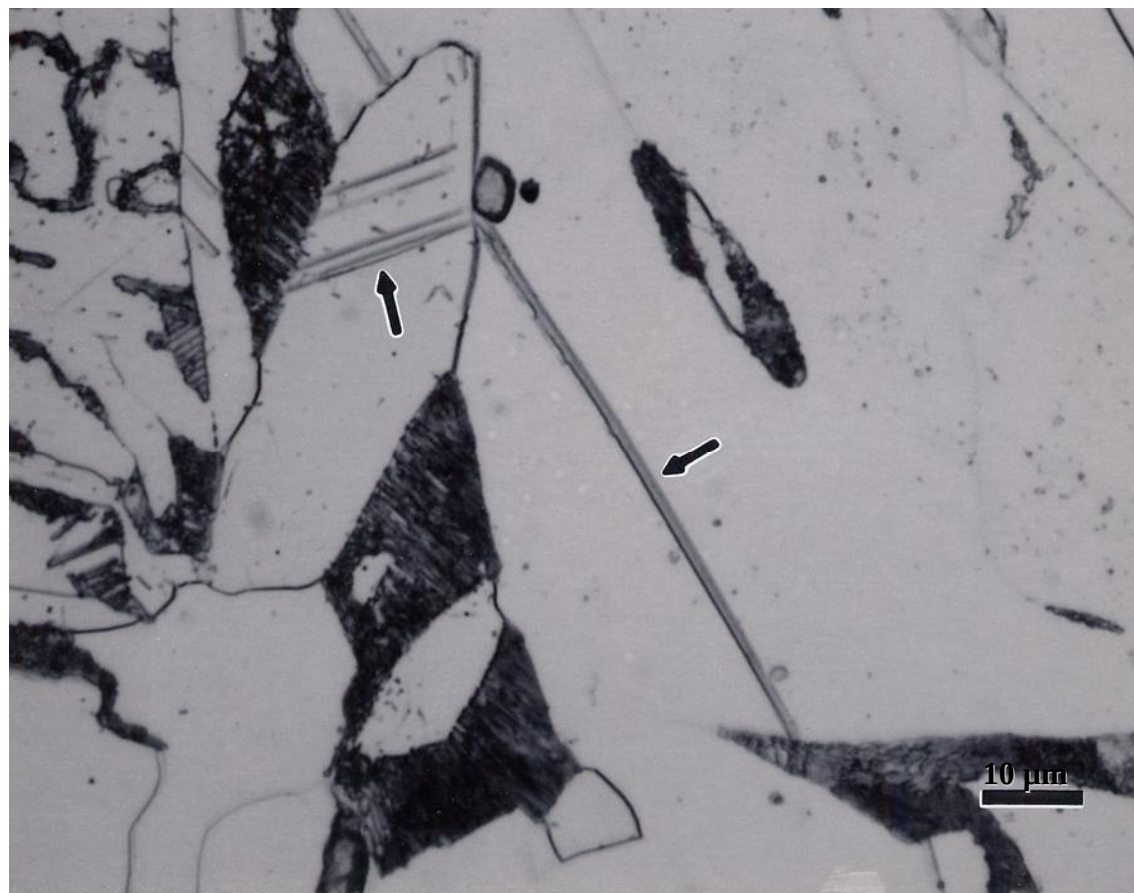
Decarburized surface of wrought, solution annealed and aged Hadfield manganese steel (Fe – 1.2% C – 13% Mn) color etched with aqueous 10% sodium metabisulfite. Where the carbon is <0.5%, epsilon martensite has formed (left arrow). Aging at 575 °C for 1 h precipitates pearlite in the grain boundaries where the carbon is 1.16% or greater. The right arrow shows the approximate depth to 1.16% carbon. Original at 50X. Viewed with polarized light plus sensitive tint.

Deformation by Slip and Twinning

Iron-based alloys deform by slip, that is a movement of atoms across a slip plane. This produces linear marks within grains. Under certain circumstances (high strain rate and low temperatures), mechanical twinning may be observed. These are lenticular-shaped lines within grains.



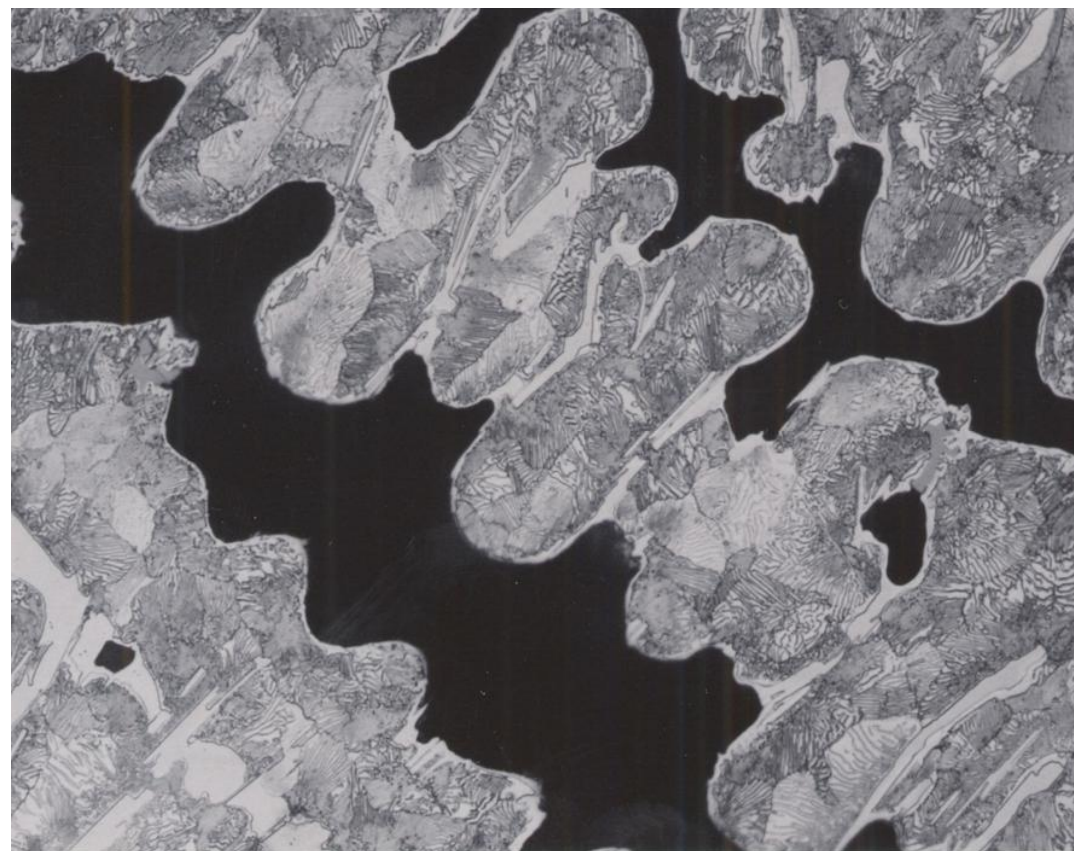
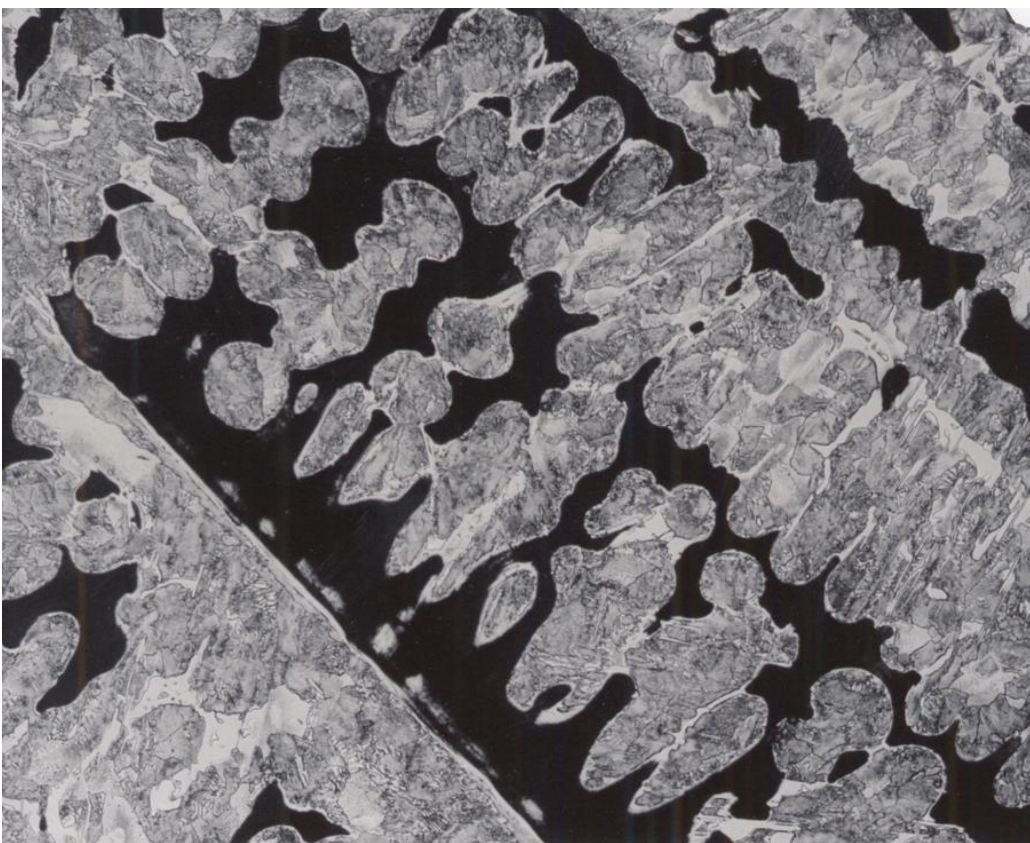
Well-developed slip at the surface of a bar of type 347 austenitic stainless steel (Fe - $<0.8\%$ C – 18% Cr – 11% Ni - $>0.3\%$ Nb) that was reduced in thickness by 30% by cold drawing. Note also that the grains are being elongated in the deformation direction (horizontal). Etched with 60% HNO_3 at 4 V dc. Original at 100X. Slip is developed best in fcc austenitic alloys.



Examples of mechanical twins (arrows) in iron-based alloys: left, twins below the fracture of a carbon steel Charpy V-notch specimen broken at $-173\text{ }^{\circ}\text{C}$ (2% nital etch); right, twins in the shank, above the fracture, in a large Roman nail (Beraha's sulfamic acid tint etch).

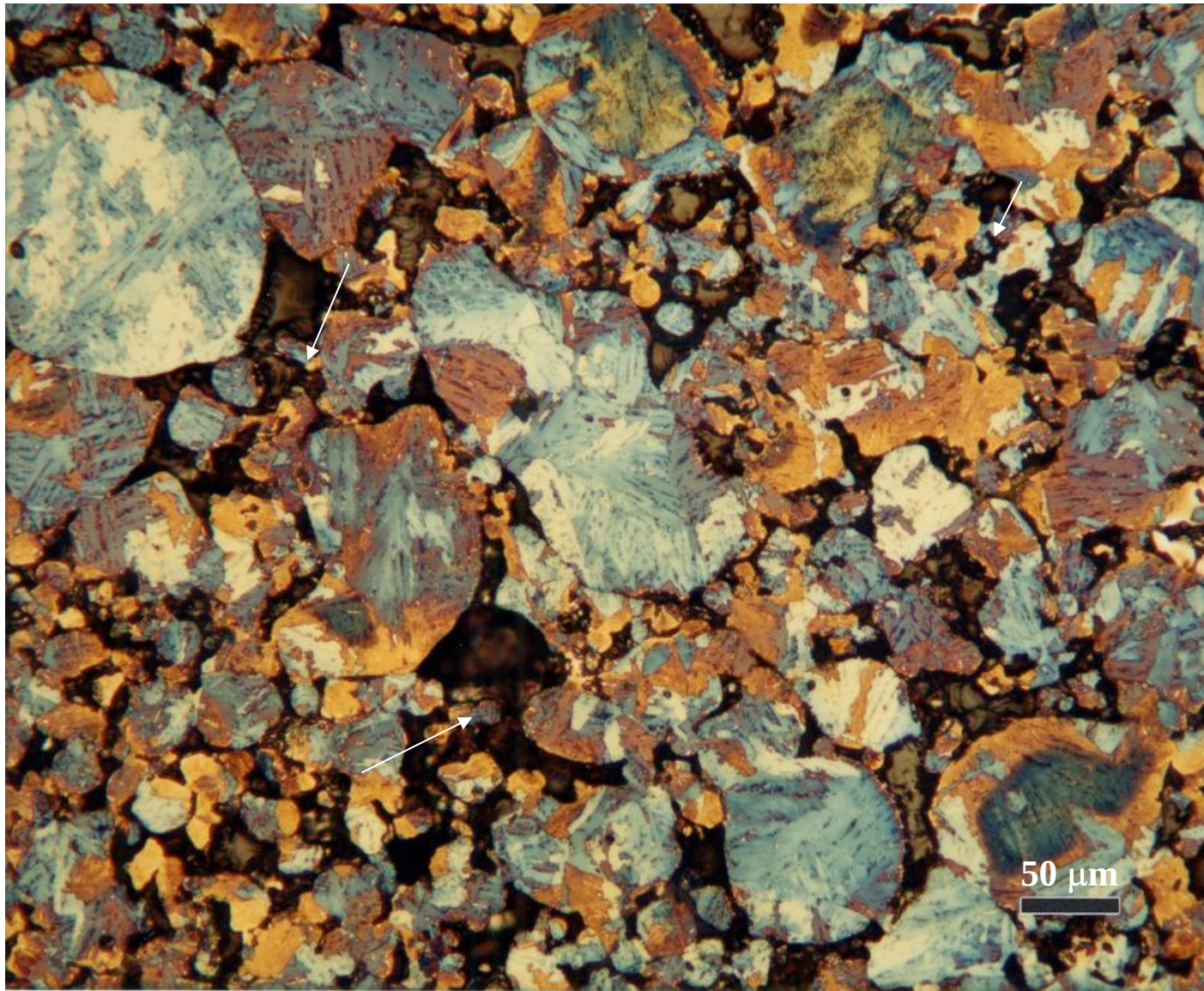
Voids and Prior-Particle Boundaries

Voids of several type can occur in all cast and wrought metals. Porosity historically refers to holes formed by gas evolution during solidification and are round. Shrinkage cavities form due to lack of consolidation during solidification and are irregular in shape. Voids can be found in powder-made products if not processed to full density. Oxides on the surface of the powder may leave behind evidence of the original shape and size of the starting powder.

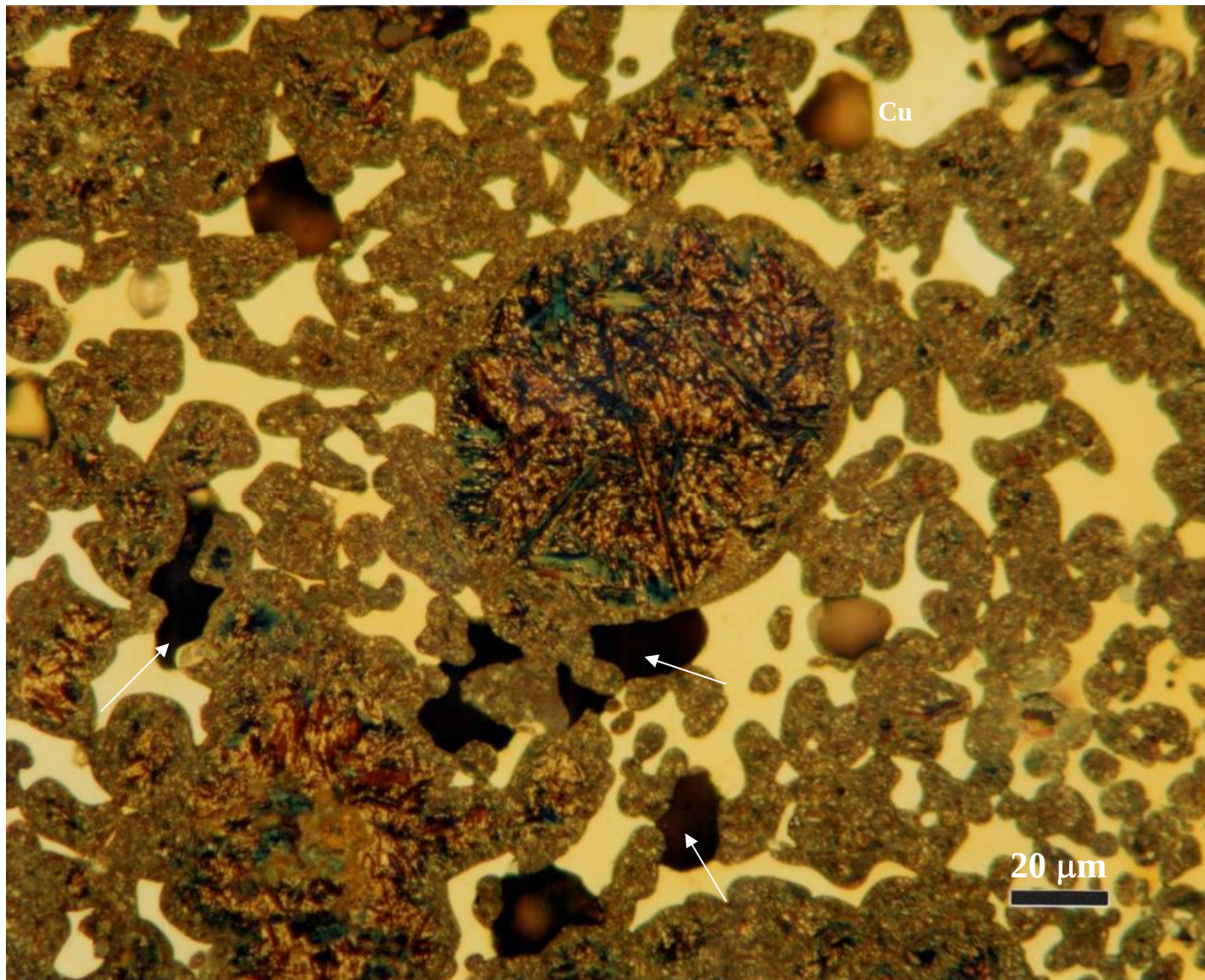


Shrinkage cavities in white cast iron. Note the dendritic appearance as there was not enough liquid metal to fill in the interdendritic region. Original magnifications are 200X (left) and 500X (right). Etched with 2% nital.

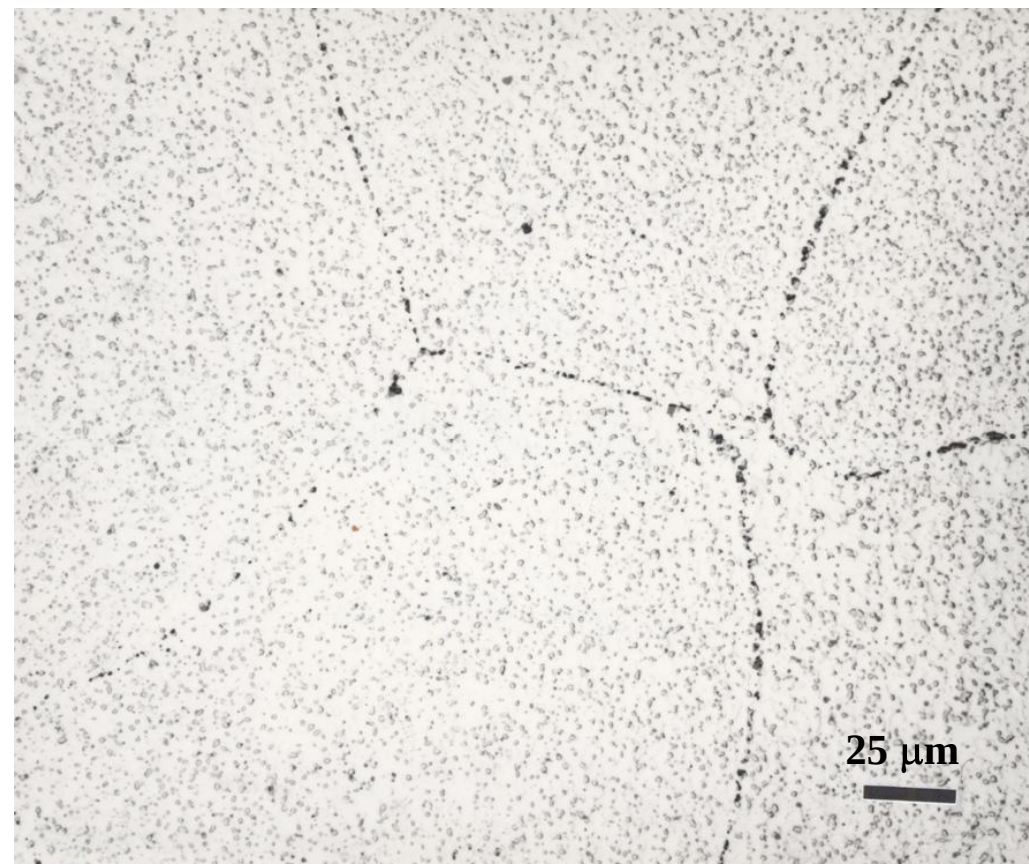
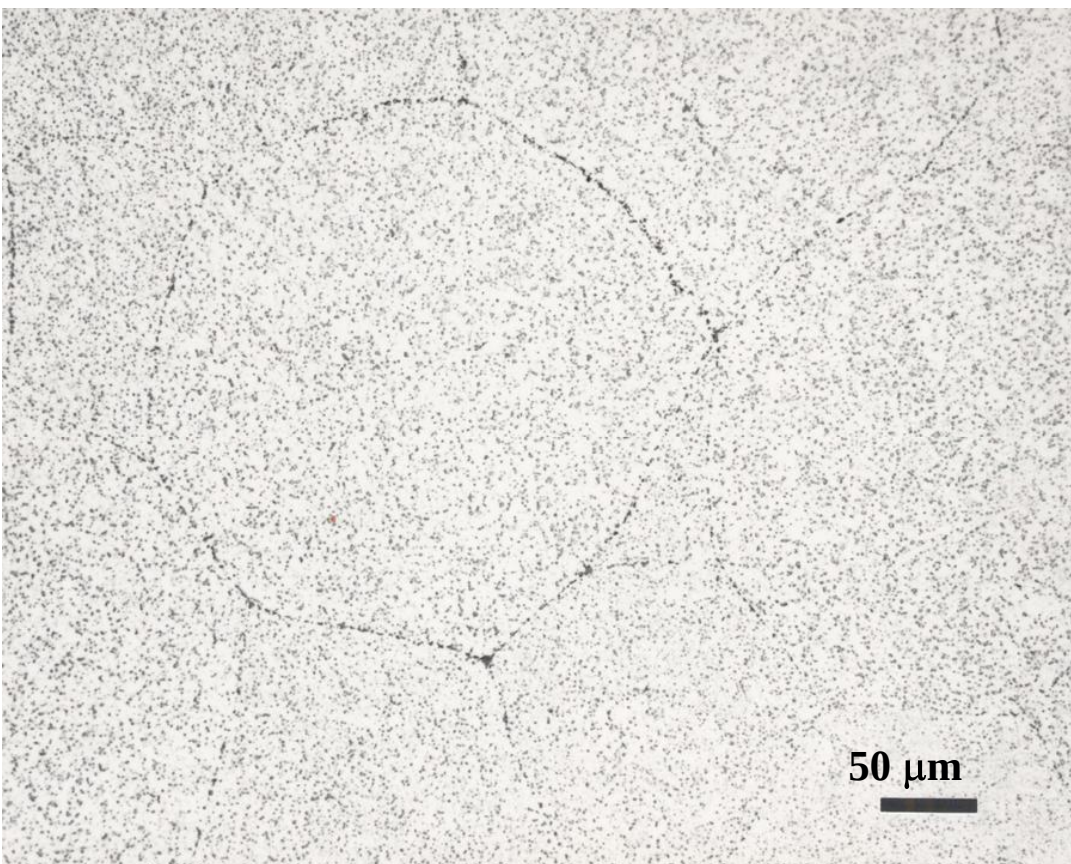
Prior particles
are observed
in the
structure



Microstructure of a low-alloy steel gear made by powder metallurgy but not compacted to full density. The black areas (arrows) are voids. Note the prior-particle shapes visible in the image. The microstructure is low-carbon martensite. Etched with Klemm's I reagent, viewed with polarized light plus sensitive tint.



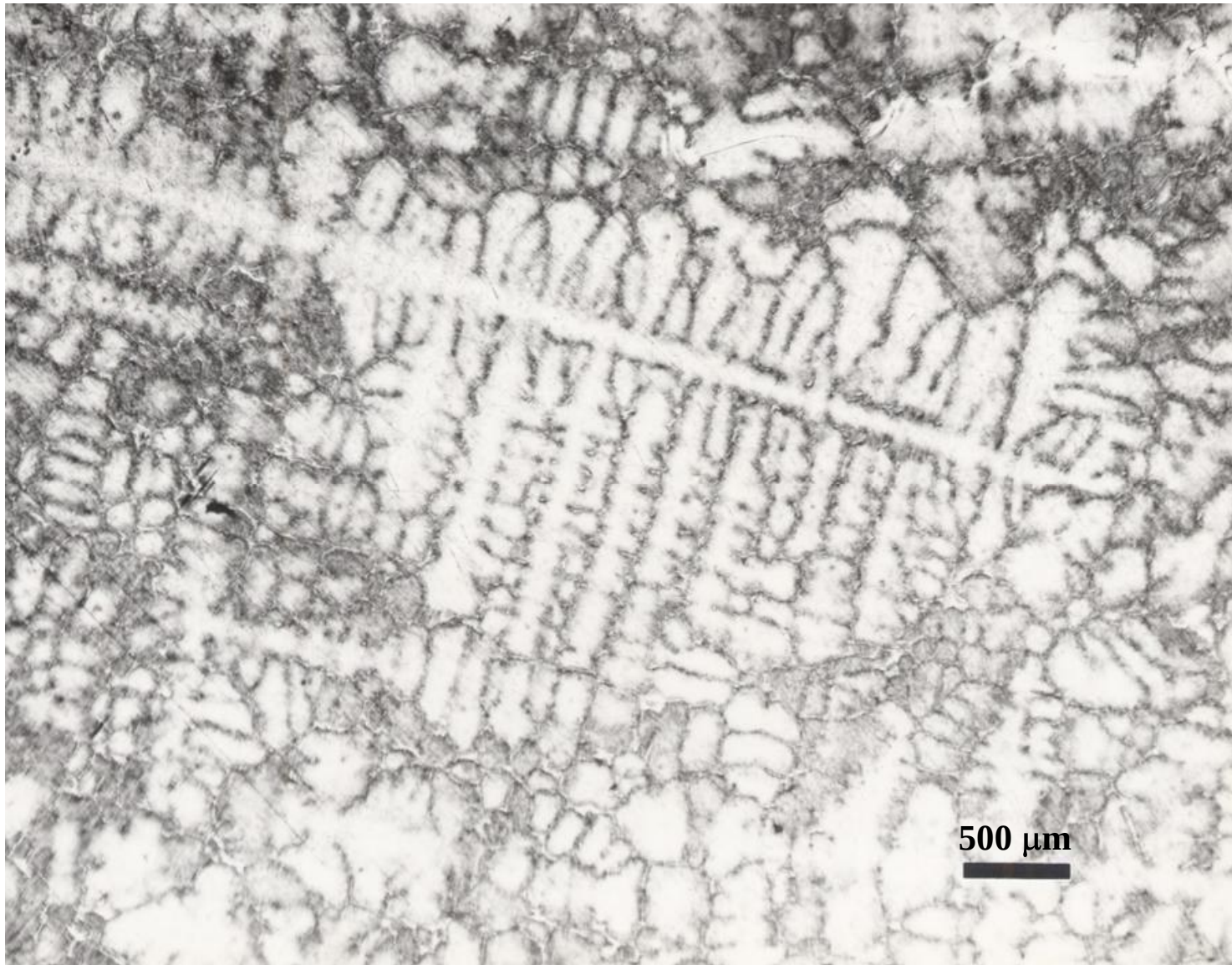
Microstructure of a copper-infiltrated porous powder metallurgy high-carbon steel (after heat treatment which produced coarse martensite and retained austenite. The copper did not completely fill the pores, as some remained (arrows). Color etched with Beraha's reagent and viewed with polarized light plus sensitive tint.



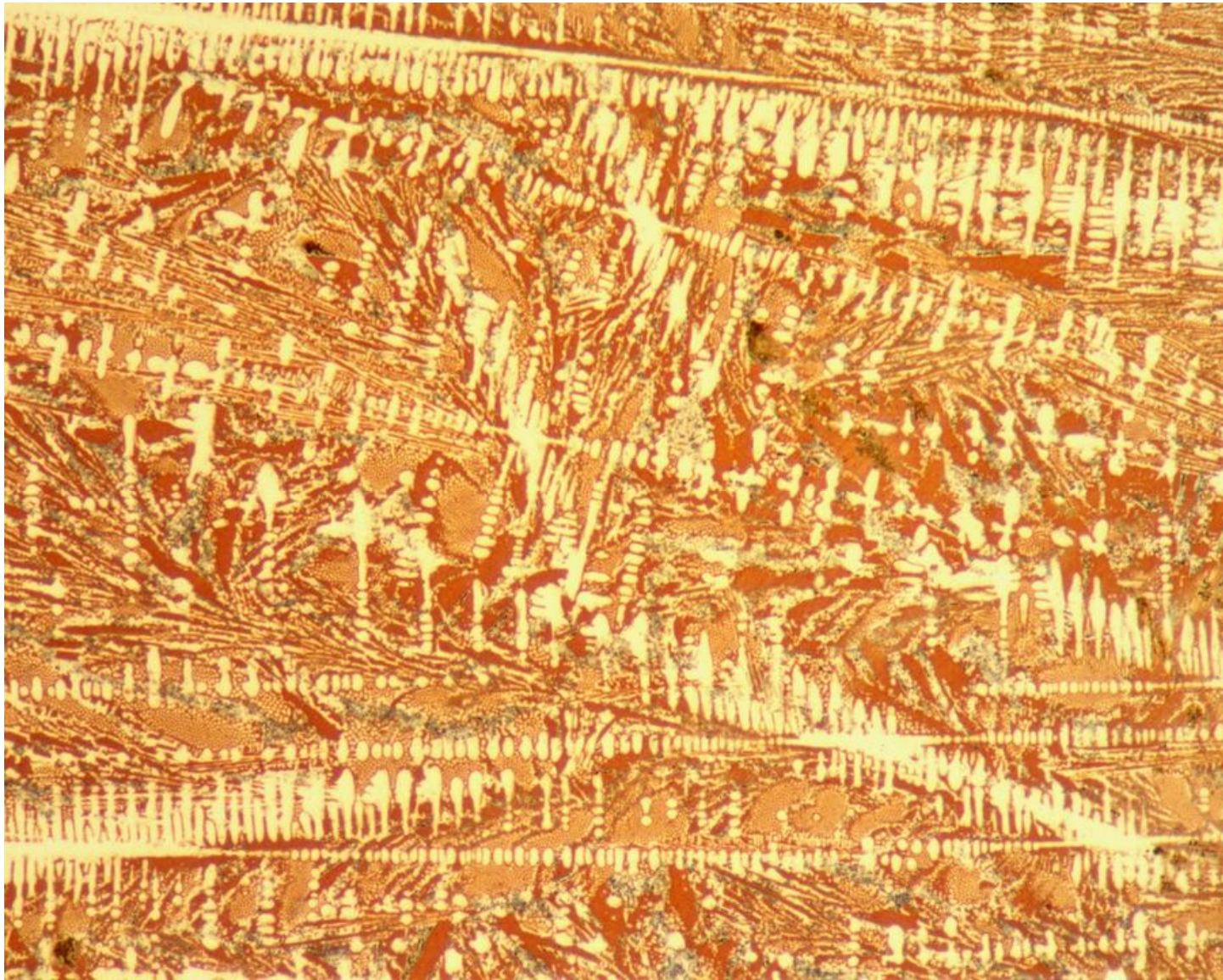
Prior-particle boundaries visible due to the presence of oxides on the powder surface in as-hipped HS-76 P/M high speed steel (Fe – 1.55% C – 4% Cr – 5.25% Mo – 3.1 % V – 10% W – 9% Co). The specimen was not etched.

Dendrites

In the as-cast condition, dendrites may be seen in many steels and cast irons. In some cases, they may be difficult to reveal. The center of the dendrite freezes first and is richest in iron while the interdendritic area solidifies last and is enriched in alloying elements and impurity elements, such as phosphorus and sulfur. Special etchants are often required to reveal the dendrites.



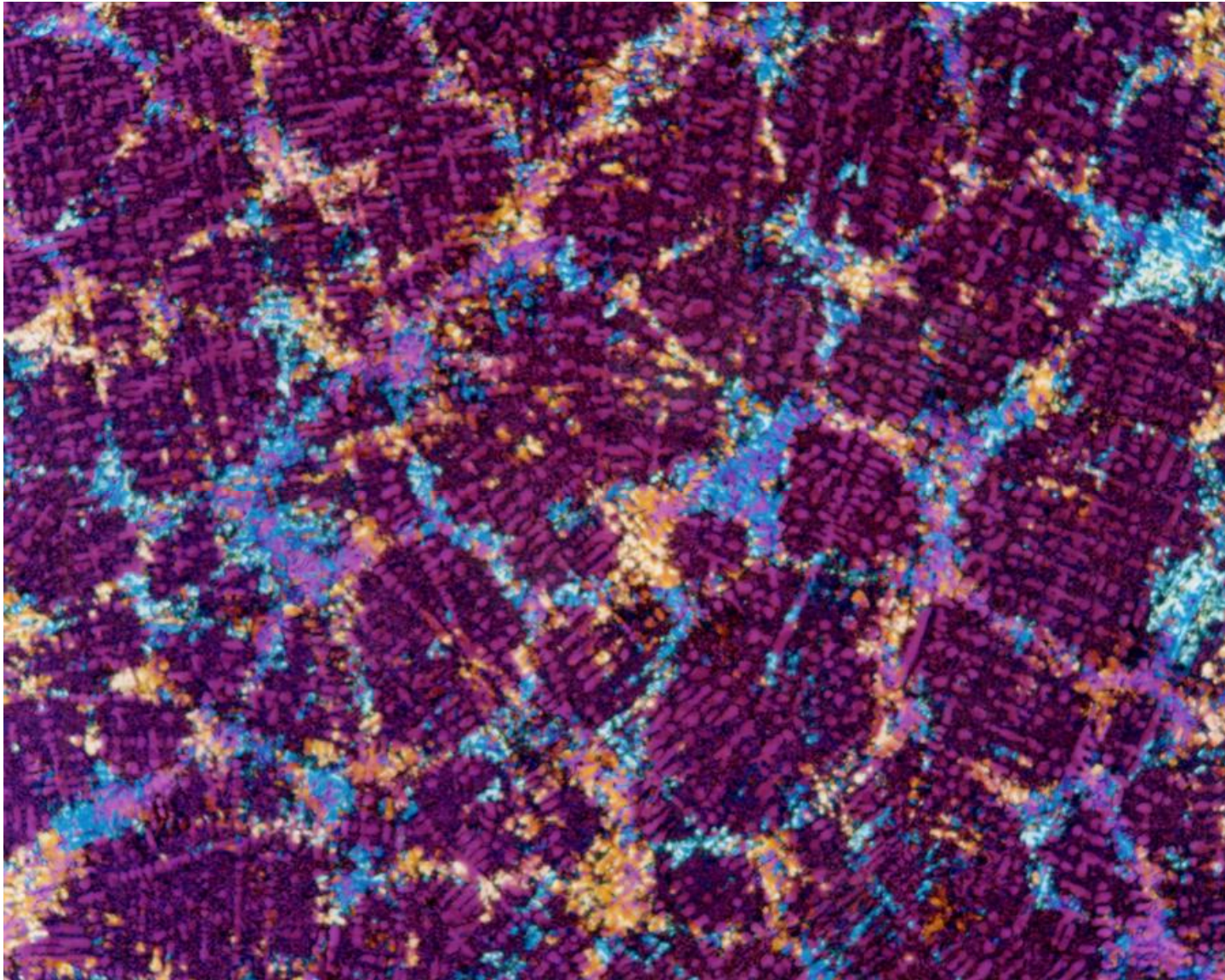
Dendrites in VIM/VAR M-50 bearing steel (Fe – 0.8% C – 4% Cr – 1% V – 4.5% Mo) in the as-cast condition after vacuum-arc remelting. The specimen was tint etched with Beraha's (300 mL water – 1 mL HCl – 0.5 g $K_2S_2O_5$ – 2.5 g NH_4FHF).



Dendrites of alpha ferrite in a chill-cast gray iron specimen etched with Beraha's selenic acid reagent (100 mL ethanol, 2 mL HCl, 1 mL selenic acid) which colored the cementite. Original at 200X magnification. Viewed with bright field.

Eutectic Cells

Eutectic cells can often be seen in cast iron specimens. As the melt solidifies, the alloying elements and impurities are driven into the remaining liquid which freezes last. The cell boundary is formed when the growing eutectic regions impinge and the last remaining liquid freezes.



Eutectic cells in gray cast iron revealed by etching with Klemm's I reagent and enhanced by using polarized light with sensitive tint. Original at 50X magnification.